

Urinary Tract Infection in Adults

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KEY POINTS

- Urinary tract infection is a common problem for all ages, affecting primarily healthy females and individuals with underlying genitourinary abnormalities.
- The optimal approach to management of urinary tract infection requires evaluation of the clinical presentation and critical interpretation of urine culture results, together with an assessment of underlying patient comorbidities and urologic abnormalities.
- The evolution of antimicrobial resistance in common uropathogens, including *Escherichia coli*, requires continuing reassessment of empiric antimicrobial treatment.
- Asymptomatic bacteriuria is common in many populations, but treatment is indicated only for pregnant women or individuals who undergo invasive urologic procedures associated with mucosal bleeding.
- Indwelling devices in the urinary tract become coated with biofilm following insertion, and the presence of the biofilm will have an impact on the urine culture and outcomes with antimicrobial therapy.
- Small renal abscesses (<5 cm) can usually be managed medically with prolonged courses of antimicrobial therapy, but larger abscesses, including most perinephric abscesses, require drainage in addition to antimicrobial therapy.
- Genitourinary tuberculosis is one of the most common forms of extrapulmonary tuberculosis, and usually presents as reactivation with involvement of only one kidney or the ureter.
- *Schistosoma mansoni*, an endemic parasite in many regions of the world, localizes to the bladder wall; acquisition of infection with this parasite is a risk for travelers to these areas.

Urinary tract infection of the bladder, the kidney, or (in men) the prostate is one of the most common human infections. Infecting organisms are usually bacteria; fungi also contribute. Much less frequently, infection is caused by viruses or parasites. Other manifestations of genitourinary tract infection are renal and perinephric abscesses, emphysematous cystitis and pyelonephritis, xanthogranulomatous pyelonephritis, and pyocystitis. Disseminated viral infections (e.g., mumps, cytomegalovirus, and other herpes viruses) and fungal infections (e.g., blastomycosis, histoplasmosis) may also involve the urinary tract but are not discussed in this chapter. Polyomavirus BK infection in kidney transplant recipients and urinary

tract infection in children, including vesicoureteral reflux, are addressed in Chapters 70 and 72, respectively.

DEFINITIONS

Urinary tract infection results from the presence of bacteria or other microorganisms in the urine or genitourinary tissues. The term “bacteriuria” describes isolation of any bacteria in the urine, although in practice it usually refers to isolation of organisms in concentrations that meet specific quantitative criteria. Infection is asymptomatic when the urine culture

Table 36.1 Abnormalities of the Urinary Tract That May Be Associated With Complicated Urinary Tract Infection

Abnormality	Example(s)
Obstruction	Pelviccalyceal junction obstruction, ureteric or urethral strictures, prostate hypertrophy, urolithiasis, tumor, extrinsic compression, female genital mutilation
Neurologic impairment	Neurogenic bladder
Urologic devices	Indwelling catheter, ureteric stent, nephrostomy tube
Urologic abnormalities	Vesicoureteral reflux, bladder diverticuli, cystoceles, urologic procedures, ileal conduit, augmented bladder, neobladder
Metabolic/congenital diseases	Nephrocalcinosis, medullary sponge kidney, urethral valves, polycystic kidneys
Immunologic impairment	Renal transplantation

result meets quantitative criteria for bacteriuria without signs or symptoms attributable to infection. Symptomatic urinary tract infection may manifest as bladder infection (cystitis or lower tract infection), kidney infection (pyelonephritis or upper tract infection), or prostate infection (acute or chronic bacterial prostatitis). Acute uncomplicated urinary tract infection occurs in women with a normal genitourinary tract, usually manifesting as cystitis.^{1–3} Pyelonephritis, also referred to as “acute nonobstructive pyelonephritis” or “acute uncomplicated pyelonephritis,” also occurs in these women but much less frequently.² Complicated urinary tract infection occurs in individuals with functional or structural abnormalities of the genitourinary tract (Table 36.1).^{4,5} In healthy postmenopausal women without genitourinary abnormalities and diabetic women without nephropathy or neurologic bladder impairment, urinary tract infection should be considered uncomplicated. Acute uncomplicated urinary tract infection rarely occurs in men. A urinary tract infection in a man should be considered complicated until underlying abnormalities have been ruled out.

Urinary tract infection commonly recurs. “Reinfection” is infection that recurs after entry of an organism into the genitourinary tract, usually from the periurethral flora. Reinfection characteristically occurs with a different organism. However, when periurethral colonization with a potential uropathogen persists, the same strain may be isolated from reinfection. “Relapse” occurs when the infecting organism persists in the urinary tract despite antimicrobial therapy; the same organism is isolated from recurrent infection after therapy.

GENERAL CONCEPTS

HOST DEFENSES OF THE NORMAL URINARY TRACT

Using standard microbiologic methods, the urine and genitourinary tract are normally sterile apart from the distal

Table 36.2 Some Host Defenses Other Than Voiding That Contribute to Maintaining Sterility of Urine

Defense	Example(s)
Normal flora	Urethra, possibly bladder microbiome
Urine characteristics	pH, osmolality, concentrations of organic acids
Urine proteins	Tamm–Horsfall protein (uromodulin), secretory immunoglobulins, lactoferrin, lipocalin, cationic peptides (defensins, cathelicidins), siderophores
Innate immune response	Toll-like receptors, polymorphonuclear leukocytes, chemokines/cytokine production, antimicrobial peptides, myeloid dendritic cells, macrophages, mast cells
Uroepithelium	Mucopolysaccharide layer, glycoprotein plaque (uroplakins), epithelial barrier, shedding of superficial epithelial cell layer
Prostate secretions	Chemokines, immunoglobulins

urethra. The normal flora of the distal urethra plays an important role in host defense by preventing colonization at this site by potential uropathogens. The flora includes aerobic bacteria that are common skin commensals, such as coagulase-negative staphylococci, viridans group streptococci, and *Corynebacterium* species.^{6,7} There is also a large and complex anaerobic flora.⁷ Recent studies using 16S ribosomal RNA sequence analysis of bladder urine collected by suprapubic aspiration or urethral catheterization to minimize contamination also describe a urine microbiome of predominantly nonpathogenic gram-positive organisms, with some other species present.^{8,9} This microbiome shows substantial heterogeneity among women and variation with age. The role of the microbiome is not clear, but a protective role based on biofilm barrier, nutrient competition, or priming of host defenses has been suggested.

Urine is a good nutrient source for most bacterial species, and common uropathogens grow well in urine. The most important host defense that maintains sterility of the urine is normal, unobstructed voiding. A complex array of urine and uroepithelial cell components also contributes to maintenance of sterile urine in the normal genitourinary tract (Table 36.2).^{10–12} Inhibitors of bacterial adherence to uroepithelial cells prevent persistence of bacteria once they have entered the urinary tract. Tamm–Horsfall protein, the most abundant protein in the urine, appears to have an important role in this regard.¹¹ This protein prevents attachment of *Escherichia coli* to uroepithelial cell receptors by binding to the type 1 fimbria adhesin (FimH), and removes some other uropathogens, such as *Klebsiella pneumoniae* and *Staphylococcus saprophyticus*.¹³ It also may have an immunomodulatory role through activation of the innate immune response by a Toll-like receptor (TLR)-4-dependent mechanism.¹¹

Adherence of bacteria to uroepithelial cells is also prevented by the surface mucopolysaccharide–glycosaminoglycan layer of the uroepithelium, by urine immunoglobulin G (IgG) and secretory immunoglobulin A (IgA), and by some low-

molecular-weight oligosaccharides present in the urine. The relative in vivo importance of any of these is not yet established. Despite the many components contributing to maintaining sterility of the urine, bacteriuria is readily established once normal voiding is impaired. In the complicated urinary tract, infection occurs through increased entry of organisms into the bladder or kidney, which may be attributed to the use of urologic devices, turbulent urine flow, or ureteric reflux. Organisms may then persist despite other host defenses when infected urine is retained if voiding is incomplete, or in biofilm on urologic devices.

IMMUNE AND INFLAMMATORY RESPONSES TO URINARY TRACT INFECTION

The innate immune system provides the major immunologic response to urinary tract infection.¹² The intensity of response is determined by the interactions of microbial pathogenicity, individual genetic regulation, and the site of infection.^{14,15} Unique *E. coli* strains have a variable capacity to stimulate or evade activation of the innate immune response. Uropathogenic strains that cause symptomatic infection induce a strong innate immune response, whereas strains isolated from asymptomatic bacteriuria evoke a limited response.^{16,17} Strains that successfully evade immune activation may have a pathogenetic advantage for establishing bladder colonization and persistent infection.^{18,19} Host genetic polymorphisms affecting the innate immune response predispose to acute pyelonephritis or asymptomatic bacteriuria.^{15,20}

Infecting organisms that gain access to the bladder adhere to uroepithelial cells, and stimulation by bacterial lipopolysaccharide leads to activation of these cells.²¹ Recognition is through TLRs (TLR-4, TLR-2, TLR-5, TLR-11) and activation promotes cytokine production, particularly interleukin (IL)-1, IL-6, and IL-8 (chemokine [C-X-C motif] ligand 1 [CXCL-1]). These cytokines recruit neutrophils and other immunocompetent cells to the kidney and bladder.^{11,12,15} The chemotactic cytokine IL-8 is released at the mucosal site and induces a rapid influx of neutrophils into the bladder, with subsequent phagocytosis and clearance of bacteria. Macrophages and mast cells resident in the submucosa are also recruited. Organisms may also infect superficial bladder epithelial cells, with subsequent shedding of these cells. This innate immune response rapidly clears most uropathogenic *E. coli* organisms from the bladder, but it does not produce a sterilizing immunity in murine models.²² In humans, bacteriuria often persists despite marked pyuria.

Urine and serum IL-6 concentrations are correlated with the severity of infection. The highest levels occur in patients with pyelonephritis. Systemic elaboration of IL-1 β and IL-6 produces fever and activation of the acute-phase response. The acute inflammatory infiltrate of polymorphonuclear leukocytes that develops in renal tissue during pyelonephritis limits bacterial spread and persistence within the kidney but also contributes to tissue damage and renal scarring.

IgA-producing plasma cells are found in higher numbers in the bladder submucosa of patients with bacterial cystitis than in healthy controls. However, acute cystitis is associated with a reduced or undetectable serologic response, presumably reflecting the superficial nature of the infection.¹² The local immune response is of short duration and is reactivated for each infection. This limited immunologic response to bladder

infection may explain why early reinfection with the same *E. coli* strain is observed in some women with acute cystitis. However, animal studies have reported some protection against same-strain reinfection mediated by systemic and local antibodies.²²

A vigorous local and systemic humoral immune response occurs in patients with pyelonephritis.^{10,16} The antibody response is directed against surface antigens of the infecting bacteria, including O antigens, and surface proteins such as the type 1 (FimH) and P fimbriae, which are major adhesins of *E. coli*.^{22,23} IgM antibodies dominate the systemic humoral response in the first episode of infection in the upper urinary tract, but subsequent episodes are characterized by an IgG response. In pyelonephritis, elevations of IgG antibodies to lipid A are correlated with severity of renal infection and parenchymal destruction. There is also a substantial urinary IgG and secretory IgA antibody response. Despite this robust response, the protective role, if any, of the antibody response in pyelonephritis is not clear. Bacteria often persist in the renal parenchyma despite very high levels of specific antibodies. In addition, the frequency of urinary tract infection is not increased in women who do not produce secretory IgA.

Cell-mediated immunity appears to have a limited role in the host defense against urinary tract infection. A small number of mucosal T lymphocytes are present throughout the urinary tract, and both CD4⁺ and CD8⁺ T cells can be found in the submucosa and lamina propria of the bladder and urethra. Recruitment of B and T lymphocytes to the bladder wall is observed with secondary infections. T-cell-derived proinflammatory cytokines also stimulate renal tubular epithelial cells to produce IL-6, which may increase IgA secretion of committed B cells.²² However, women infected with human immunodeficiency virus (HIV) who have very low CD4⁺ counts do not have an increased susceptibility to or severity of urinary tract infection²⁴; this finding suggests that cell-mediated immunity is not an essential defense against such infection.

URINE CULTURE

The definitive diagnosis and appropriate management of urinary tract infection usually require microbiologic confirmation by urine culture. Urine specimens for culture should always be obtained before antimicrobial therapy is initiated because urinary excretion of antimicrobial agents rapidly sterilizes urine. Once collected, the specimen should be forwarded promptly to the laboratory. Organisms present in small quantitative counts (i.e., contaminants) grow readily in urine at room temperature and reach high quantitative counts within a few hours. If the specimen is delayed in reaching the laboratory, it should be refrigerated at 4°C until transported.

A urine specimen for culture must be collected with a method that minimizes contamination. A clean-catch voided specimen without additional periurethral cleaning is usually appropriate. When patients cannot cooperate for the collection of a voided specimen, urine may be collected by an in-and-out catheter. For men, a specimen may be obtained in an external condom catheter after application of a clean condom catheter and collecting bag.²⁵ Urine samples may also be collected by suprapubic aspiration or directly from the renal pelvis when percutaneous drainage of an obstructed

urinary tract is necessary. Specimens obtained from patients with short-term indwelling catheters should be collected by puncture of the catheter port. For a long-term indwelling catheter, two to five organisms are present in the catheter biofilm at any time, so urine collected through the catheter will be contaminated by organisms present in the biofilm.²⁶ The long-term indwelling catheter should be removed and replaced by a new catheter, and a specimen of bladder urine obtained through the newly placed catheter.^{25,26}

The standard quantitative criterion for diagnosis of urinary tract infection with voided specimens is an organism count of $\geq 10^5$ colony-forming units (CFU)/mL of a potential uropathogen. Women usually have contaminating organisms from vaginal or periurethral flora isolated from voided specimens, and the quantitative criterion distinguishes bacteriuria from contamination. Application of this quantitative standard is always appropriate for the diagnosis of asymptomatic bacteriuria, but for symptomatic cases the quantitative urine culture results must be interpreted in the context of the clinical presentation and with consideration of the method of specimen collection (Table 36.3). Bacteria require several hours of incubation in bladder urine to achieve a concentration $\geq 10^5$ CFU/mL. Some patients with frequency or diuresis may not retain urine in the bladder for a sufficient time to achieve the concentration of $\geq 10^5$ CFU/mL. Quantitative counts may also be lower when infection is caused by some fastidious organisms or if the patient is receiving a urinary antiseptic. For symptomatic men, a single urine specimen in which $\geq 10^3$ CFU/mL of a uropathogen is isolated is diagnostic for bladder bacteriuria, on the basis of paired comparisons of voided specimens and suprapubic aspirates.

Table 36.3 Quantitative Counts of Bacteria in the Urine for Microbiologic Diagnosis of Urinary Tract Infection in Patients Not Receiving Antimicrobial Therapy

Collection Method	Quantitative Criteria (CFU/mL)
Voided Specimen	
Asymptomatic women or men	$\geq 10^{5a}$
Women: acute uncomplicated	
Cystitis	$\geq 10^3$
Pyelonephritis	$\geq 10^{4b}$
Men: Symptomatic	$\geq 10^3$
Men: External condom collection	$\geq 10^5$
Catheter	
In-and-out	$\geq 10^2$
Indwelling ^c	
Asymptomatic	$\geq 10^5$
Symptomatic	$\geq 10^2$
Suprapubic or percutaneous aspiration	Any growth

^aTwo consecutive specimens are recommended for women.

^bIn 95% of cases, values $\geq 10^5$ colony-forming units (CFU) per milliliter.

^cA long-term catheter should be replaced and the specimen collected through a new catheter.

A urine specimen obtained by suprapubic aspiration or other percutaneous collection method such as renal pelvis drainage is assumed to be a sterile specimen, and any quantitative count of an organism represents true bacteriuria. However, in specimens collected by an in-and-out catheter, contaminating organisms are introduced from the periurethral area, and a quantitative criterion of $\geq 10^2$ CFU/mL is recommended. Other relevant considerations in interpreting a urine culture result include the number and type of organisms isolated. A single infecting organism is usual, but in patients with complicated urinary tract infection, particularly those with indwelling urinary devices, more than one organism is frequently present. Commensal bacteria of the normal skin flora, such as *Lactobacillus* spp. and coagulase-negative staphylococci, should usually be considered contaminants when they are isolated from voided urine specimens. In young healthy women, group B streptococci and *Enterococcus* species isolated in any quantitative count are also usually contaminants.²⁷

PHARMACOKINETIC AND PHARMACODYNAMIC CONSIDERATIONS FOR TREATMENT

Therapeutic success in the treatment of cystitis depends on antimicrobial levels in the urine. Antimicrobial levels in renal tissue, which are correlated with serum levels, determine outcome for pyelonephritis.²⁷ Treatment of urinary tract infection is unique in some respects because of the exceptionally high urine concentrations achieved by many antimicrobial agents excreted into the urine (Table 36.4). The urine concentration is determined by the interplay of glomerular filtration, active tubular secretion, and tubular reabsorption, all influenced by urine pH, protein binding, and the molecular structure of the drug. Cystitis and pyelonephritis may be successfully treated with antimicrobial agents at minimum inhibitory concentrations (MICs) to which the infecting organism would not usually be considered susceptible. The “intermediate” susceptibility designation reported by the clinical microbiology laboratory implies clinical efficacy in body sites where antimicrobial agents are physiologically concentrated, such as the urine, and is relevant to treatment of urinary tract infection. Thus when an organism isolated from the urine is reported to have intermediate susceptibility to an antimicrobial agent, the drug is usually appropriate for treatment of urinary tract infection with that organism. The urine bactericidal activity of some antimicrobial agents is modified by the urine pH. Penicillins, tetracyclines, and nitrofurantoin are more active in acidic urine, and aminoglycosides, fluoroquinolones, and erythromycin are more active in alkaline urine. This pH variability has not, however, been shown to be relevant for therapeutic outcomes, with the exception of methenamine salts, for which an acidic pH is necessary to release formaldehyde, the active component.

The prostate is a unique compartment for consideration of antimicrobial efficacy. There are no active antibiotic transport mechanisms for the gland and most antibiotics penetrate poorly into prostate tissue and fluid.^{28,29} The interior of the gland is an acidic environment. Drug entry and activity are determined by concentration gradient, protein binding, lipid solubility, molecular size, local pH, and pK_a of the antimicrobial agent. Alkaline drugs such as trimethoprim

Table 36.4 Urinary Excretion of Antimicrobial Agents in Persons With Normal Renal Function

Antimicrobial Agent	% of Absorbed Drug Excreted Renally as Parent Metabolites (Active Metabolites) ^a	Usual Dosage for Normal Renal Function ^b
Penicillins		
Penicillin G	80	1–2 million U IV q4–6h
Amoxicillin	90	500 mg PO tid
Amoxicillin/clavulanic acid	Clavulanate: 20–60	500 mg PO tid or 875 mg PO bid
Ampicillin	90 (10)	1–2 g IV q6h
Cloxacillin	35–50	1–2 g IV q4–6h
Piperacillin	50–80	200–300 mg/kg/day IV qid
Piperacillin/tazobactam	Tazobactam: 60–80	3.375 g IV q6h
Pivmecillinam	45	200–400 mg bid or tid
Cephalosporins		
Cephalexin	>80 (18)	500 mg PO qid
Cefazolin	>80	1 g IV q8h
Cefuroxime	>80	250–500 mg PO bid
Cefotaxime	50–60 (30)	1 g IV q8h
Ceftriaxone	50	1–2 g IV q24h
Cefepime	85	1–2 g PO q12h
Cefixime	15–20	400 mg PO qd
Cefpodoxime	20–35	100–200 mg PO q12h
Cefprozil	60	250–500 mg PO q12h
Ceftazidime	80–90	1–2 g IV q8–12h
Ceftaroline	50	600 mg IV q12h
Ceftolozane/tazobactam	95/80	1 g/500 mg q8h
Ceftazidime/avibactam	80–90/97	2 g/500 mg q8h
Macrolides, Lincosamides		
Erythromycin	5–15	500 mg PO qid or 1 g IV q6h
Clindamycin	≤6 (some active metabolites)	150–300 mg PO tid or 600–900 mg IV q6–8h
Clarithromycin	20–30 (10–15)	250–500 mg PO q12h
Azithromycin	6	500 mg PO qd
Aminoglycosides		
Gentamicin	99	5 mg/kg/day IV in 1–3 divided doses
Tobramycin	99	5 mg/kg/day IV in 1–3 divided doses
Amikacin	99	15 mg/kg/day IV in 1–3 divided doses
Carbapenems		
Imipenem/cilastatin	70–76	500 mg IV q6h
Meropenem	70–80	500 mg to 1 g IV q6–8h
Ertapenem	40	1 g IV q24h
Doripenem	70 (15)	500 mg q8h
Fluoroquinolones		
Norfloxacin	25–40 (10–20)	400 mg bid
Ciprofloxacin	40 (10–20)	250–750 mg PO or 400 mg IV bid
Levofloxacin	70–80	250–750 mg PO or IV q24h
Moxifloxacin	20	400 mg PO or IV q24h
Other Antibacterials		
Vancomycin	>90	1 g IV q12h
Teicoplanin	>90	6–12 mg/kg IV q12h
Dalbavancin	42	1 g IV
Daptomycin	54	4 mg/kg IV q24h
Linezolid	35	600 mg PO or IV q12h
Tigecycline	32	250–500 mg IV q6h div q12h
Colistin	64–70	1.5–2.5 mg/kg/day
Trimethoprim	66–95	100 mg PO q12h

Continued on following page

Table 36.4 Urinary Excretion of Antimicrobial Agents in Persons With Normal Renal Function (Cont'd)

Antimicrobial Agent	% of Absorbed Drug Excreted Renally as Parent Metabolites (Active Metabolites) ^a	Usual Dosage for Normal Renal Function ^b
Sulfamethoxazole	20–40	
Trimethoprim/sulfamethoxazole		180/800 mg PO bid
Nitrofurantoin monohydrate/macrocrystals	40–60	50–100 mg PO q6h
Fosfomycin tromethamine ^a	30–60	
Uncomplicated UTI		3 g, one dose;
Complicated UTI		3 g q48–72h
Doxycycline	20–30	100 mg PO bid
Aztreonam	66	1–2 g IV q6–8h
Metronidazole	15 (30–60)	500 mg PO or IV tid
Rifampin	<10/50	600 mg PO qd
Antifungals		
Amphotericin B deoxycholate	<10	0.5–1 mg/kg IV qd
Amphotericin B lipid formulations	<1	1–5 mg/kg/day IV
5-Flucytosine	90	100–150 mg/kg/day PO in four divided doses
Ketoconazole	<10	400 mg PO qd
Fluconazole	80	100–400 mg PO or IV q24h
Itraconazole	<1	200 mg bid PO or IV × 2 days, then 200 mg PO qd
Voriconazole	<1	6 mg/kg IV × 1 dose, then 200 mg IV bid; 200-mg PO bid × 1 day, then 100 mg bid
Posaconazole	<1	400 mg PO bid
Caspofungin	<1	70-mg loading dose, then 50 mg IV q24h
Micafungin	<1	50–100 mg IV q24h
Anidulafungin	<1	100- to 200-mg IV loading dose, then 50–100 mg IV q24h

^aExcept where noted, values are the proportion of dose renally excreted unchanged.

^bNot all antimicrobial agents have an indication for urinary tract infection.

bid, Twice a day; IV, intravenous; PO, by mouth; qd, once a day; qid, four times a day; tid, twice a day; UTI, urinary tract infection.

(TMP) diffuse into the prostate and are trapped, with high concentrations achieved, but the drug remains in an inactive, ionized form. Fluoroquinolones and macrolides, however, penetrate well and remain active.

Current pharmacodynamic models for antimicrobial treatment of infection distinguish between time-dependent and concentration-dependent bacterial killing. Bacterial killing by β -lactam antimicrobial agents is time dependent, as the therapeutic efficacy depends on how long the concentration of the antimicrobial agent remains above the MIC of the infecting organism. Bacterial killing by fluoroquinolones and aminoglycosides is concentration dependent; the therapeutic efficacy is measured by the ratio of peak antimicrobial concentration to MIC, or the ratio of the area under the curve to MIC. A pharmacodynamic model in which the urine bactericidal titer replaces the MIC has been applied to predict optimal dosing regimens for antimicrobial treatment of urinary tract infection.^{27,30} The validity of these models for urinary tract infection, however, still requires confirmation in clinical trials.^{31,32} In particular, the relevance to treatment of complicated urinary tract infection is uncertain because impaired renal function, voiding abnormalities, and the presence of biofilms on indwelling devices introduces variability that will affect antimicrobial efficacy.³⁰

USUAL PRESENTATIONS OF URINARY TRACT INFECTION

ACUTE UNCOMPLICATED URINARY TRACT INFECTION: CYSTITIS

EPIDEMIOLOGY

Acute uncomplicated urinary tract infection manifesting as acute cystitis is a common syndrome that affects otherwise healthy women.^{1,3} About 10% of young, sexually active, premenopausal women experience a urinary tract infection each year, and 60% of all women have one or more such infections in their lifetimes. From 2% to 5% of women experience frequent recurrent infection for at least some period. After a first episode of cystitis, 21% of female college students in one study reported a second infection within 6 months.³³ Among postmenopausal women aged 55 to 75 years who were enrolled in a Seattle health group, the incidence was seven infections per 100 patient-years; in 24 months, 7% of women had one infection, 1.6% had two infections, and 1% had three or more infections.³⁴ Acute cystitis is associated with considerable short-term morbidity.³⁵ Female college students reported that symptoms persisted

for an average of 6.1 days,² and, for ambulatory women, a mean duration of symptoms was 4.9 days, with 63% of patients reporting their usual activities were compromised by the infection.³⁶ However, although a large number of women are affected and many have frequent recurrence of infection, there is no long-term morbidity. Acute uncomplicated urinary tract infection is uncommon in healthy young men, with an estimated incidence of <0.1% per year.

PATHOGENESIS

Microbiology

Acute uncomplicated urinary tract infection is primarily a disease of extraintestinal pathogenic *E. coli*, also referred to as uropathogenic *E. coli*. These organisms are isolated in 80% to 85% of episodes of acute cystitis.^{1,2,34} Infection occurs via the ascending route after bacterial strains that originate in the gut flora colonize the vagina or periurethral area.⁷ Although urethral colonization with a potential uropathogen appears to be a prerequisite for infection, cystitis does not subsequently develop in most women with periurethral colonization.⁷ Strains of *E. coli* that colonize the periurethral area and subsequently cause urinary tract infection belong to a restricted number of phylogenetic *E. coli* groups and more frequently express diverse virulence factors than do periurethral strains that do not cause symptomatic infection.^{37,38} A necessary characteristic for establishing bladder infection is production of FimH, an adhesin attaching to receptors on uroepithelial cells.³⁹ This surface protein, however, is common on *E. coli* strains, regardless of whether they cause infection. Other potential urovirulence characteristics include other adhesins, iron sequestration systems, toxin secretion, and motility.³⁸ The putative virulence factors produced by strains isolated from symptomatic infection overlap with those isolated from asymptomatic bacteriuria, and no single characteristic is uniquely correlated with symptomatic infection.⁴⁰ Uropathogenic *E. coli* strains may be acquired during travel, from ingestion of food or water, from other household members, including pets, and from sexual partners.^{41,42} Clonal outbreaks may occur with transmission of a single strain within a community or larger geographic area.^{43,44}

S. saprophyticus, a coagulase-negative staphylococcal species, is an organism virtually unique to acute cystitis. It is the second most frequently isolated species (in 5%–10% of episodes), and there is a seasonal variation of infection, with isolation more common in late summer or fall. Genetic elements described in the *S. saprophyticus* genome that may promote urovirulence include adhesins, transport systems to support growth in urine, and urease production.⁴⁵ Other Enterobacteriaceae, most commonly *K. pneumoniae*, are isolated in fewer than 5% of premenopausal women but in 10% to 15% of postmenopausal women.^{34,46} Gram-positive organisms such as *Enterococcus* species and group B streptococci are uncommon pathogens in premenopausal women.⁴⁷ *Salmonella* species and bacteria associated with sexually transmitted infections, such as *Ureaplasma urealyticum*, *Gardnerella vaginalis*, and *Mycoplasma hominis*, are occasionally isolated.⁴⁸

Acute uncomplicated urinary tract infection recurs in many women and is characteristically reinfection. In as many as 30% of early reinfections—those occurring within 1 month of treatment of an episode of acute cystitis—an *E. coli* strain similar to the pretherapy strain is isolated. This finding is

assumed to be a consequence of failure of the antimicrobial therapy to eliminate virulent strains from the gut or vaginal flora reservoirs.² Intracellular persistence of *E. coli* in uroepithelial cells is an alternative mechanism proposed to explain same-strain recurrence, on the basis of observations in animal studies.^{49,50} However, prospective studies in women document periurethral colonization prior to bladder infection onset for most episodes,⁵¹ and the contribution of an intracellular reservoir to recurrent acute uncomplicated cystitis in humans remains uncertain.

Host Factors

Acute, uncomplicated urinary tract infection is a consequence of the interaction of a virulent organism with host genetic susceptibility and behavioral variables.^{10,14} The notion of genetic propensity is supported by two consistent observations: (1) an increased frequency of urinary tract infection in first-degree female relatives of women with recurrent infection^{1,3,52} and (2) the fact that infection at a younger age is a major risk factor for recurrent cystitis in women of any age.^{1,34,52} One well-characterized genetic association is being a nonsecretor of the ABO blood-group antigens.^{52,53} Women with recurrent urinary tract infection are at least three times more likely to be nonsecretors than are those without recurrent infection. Nonsecretors express cell-surface glycosphingolipids on the vaginal epithelium, and presumably urethral mucosa, that differ from those expressed by secretors and bind uropathogenic *E. coli* more avidly.¹⁴ Other potential genetic determinants include genetic polymorphisms of the IL-8 receptor CXCR1, TLRs, and the tumor necrosis factor promoter.¹⁴

The most important behavioral association of urinary tract infection in premenopausal women is sexual intercourse.^{1–3,54} In young, sexually active women, 75% to 90% of episodes are attributable to intercourse, and there is a correlation between frequency of intercourse and frequency of infection.⁵³ Intercourse appears to promote infection by facilitating ascension of organisms from the periurethral area into the bladder. Spermicide use for birth control is another independent behavioral risk factor for acute cystitis in premenopausal women. The frequency of recurrent infection is at least twice as high among women who use spermicides as among women who do not.⁵³ Spermicides are bactericidal for the hydrogen peroxide-producing lactobacilli of the normal vaginal flora, which maintain the acidic pH. When these bacteria are not present, the vaginal pH increases and this facilitates colonization with potential uropathogens such as *E. coli*. Case-control studies have consistently demonstrated that behavioral variables popularly identified as risks for cystitis—such as type of underwear, bathing rather than showering, postcoital voiding, frequency of voiding, perineal hygiene practices, vaginal douching, and tampon use—are not associated with an increased risk of infection.^{1,2}

A history of prior urinary tract infection at a younger age is the strongest association of recurrent acute cystitis in postmenopausal women.^{55–57} Sexual intercourse is not an important contributor.^{57,58} Estrogen deficiency has been proposed to promote recurrent urinary tract infection in these women through alterations in vaginal flora, including replacement of lactobacilli by potential uropathogens. However, prospective cohort studies and case-control studies uniformly demonstrate no association of oral or topical

estrogen use with recurrent urinary tract infection, regardless of restoration of vaginal lactobacilli and acid pH.⁵⁹ Acute uncomplicated urinary tract infection in men is uncommon, but reported risk factors have included intercourse with a female partner with recurrent urinary tract infection, not being circumcised, and anal intercourse.⁶⁰

DIAGNOSIS

The classic clinical manifestation of symptomatic lower urinary tract infection is the acute onset of one or more irritative bladder symptoms, such as urgency, frequency, dysuria, stranguria, and hesitancy.^{1,3} Gross hematuria is also common. The most important differential diagnoses to exclude are sexually transmitted infections⁶¹; vulvovaginal candidiasis and noninfectious syndromes such as interstitial cystitis should also be considered. The combination of new-onset frequency, dysuria, and urgency, together with the absence of vaginal discharge and pain, has a positive predictive value for acute cystitis of 90%.⁶² Women who experience recurrent infection also have >90% accuracy in self-diagnosis on the basis of symptoms.²

A urine culture is not recommended routinely for women with a clinical presentation consistent with acute uncomplicated cystitis.^{3,48,63} The utility of the urine culture is limited by the reliability of the clinical diagnosis, predictable microbiology, and prompt clinical response with short-course empirical antimicrobial therapy. Final culture results are often not available until therapy is completed, and quantitative bacterial counts <10⁵ CFU/mL are isolated in as many as 30% of women with acute cystitis. Therefore interpretation of culture results may be problematic.^{1,47} In fact, urine culture results are negative in as many as 10% of women with a characteristic clinical presentation, and these women have a clinical response to antimicrobial therapy similar to that of women with positive cultures.⁶⁴ These women with low quantitative counts of organisms isolated in the urine culture may have urethritis, rather than cystitis. However, both urinary frequency and increased fluid intake are characteristic of patients with acute cystitis. Thus the limited dwell time of urine in the bladder seems the likely explanation for the observation of a high frequency of urine cultures with lower quantitative counts. Culture results may be negative because organisms are present in quantitative counts below the level of detection by standard laboratory procedures (usually <10³ CFU/mL) or because fastidious organisms may not be identified by routine laboratory procedures used in processing urine specimens.

A urine specimen for culture should be obtained, however, from selected women with possible acute uncomplicated urinary tract infection. When the clinical presentation is not characteristic, a urine culture may help confirm or exclude the diagnosis of urinary tract infection. Any quantitative count of an Enterobacteriaceae or *S. saprophyticus* is considered positive. *Enterococcus* spp. or group B streptococcus in any quantitative count should be interpreted as contamination.⁴⁷ Failure to respond to appropriate empirical antimicrobial therapy or an early (<1 month) symptomatic recurrence after therapy is suggestive of infection with a resistant organism. In these situations, a urine culture should be obtained to confirm whether antimicrobial resistance is present and to facilitate selection of an effective alternative regimen.

The presence of pyuria, identified by routine urinalysis or leukocyte esterase dipstick testing, is a consistent accom-

paniment of acute cystitis.^{1,63} The absence of pyuria is suggestive of an alternative diagnosis but does not rule out urinary tract infection in women with a consistent clinical presentation.^{1,64–66} Thus routine screening for pyuria is not recommended in the management of women presenting with presumed acute cystitis.^{1,63} A urine nitrite dipstick test screens for the presence of bacteria, rather than leukocytes, and results are usually positive in women with infection.⁶⁶ False-negative nitrite test results may occur when there is infection with bacteria that do not reduce nitrate, such as *Enterococcus* species, or when urine has not been retained in the bladder a sufficient time to allow bacteria to convert nitrate to nitrite. Nitrite tests uncommonly have false-positive results, but these may occur when blood, urobilinogen, or some dyes are present in the urine.

TREATMENT

For many women, the natural history of acute uncomplicated cystitis is spontaneous clinical and microbiologic resolution within a few days or weeks. In a clinical trial in which subjects were randomly assigned to receive antibiotic therapy or placebo, 28% of 277 women who received placebo were asymptomatic by 1 week, and 45% had negative culture results by 6 weeks.⁶⁷ In another study, 54% of women who received placebo were asymptomatic by 3 days and 52% at 7 days.⁶⁸ However, antimicrobial treatment is associated with a significantly shorter duration of symptoms. The rates of clinical cure were 77% with nitrofurantoin in comparison with 54% with placebo at 3 days, and 88% and 52%, respectively, at 7 days.⁶⁸ After initiation of antimicrobial therapy, 54% of women reported symptom improvement by 6 hours, 87% by 24 hours, and 91% by 48 hours. In another case series, 72% of women reported complete symptom resolution by the fourth day of effective treatment.³⁶ In a trial of antiinflammatory therapy with ibuprofen alone compared with empiric fosfomycin antimicrobial therapy for women with mild to moderate symptoms, antimicrobial therapy was associated with more rapid resolution of symptoms and fewer cases of pyelonephritis.⁶⁹

Many antimicrobial agents are effective for treatment of acute cystitis (Table 36.5). The anticipated cure rate for recommended first-line empirical regimens is 80% to 95%.^{3,70} TMP/sulfamethoxazole (TMP/SMX) has been a mainstay of empirical treatment of acute cystitis for decades and remains highly effective against susceptible organisms.^{70,71} However, increasing rates of TMP/SMX resistance in community *E. coli* isolates compromise the use of this agent as first-line empirical therapy.⁷² Some antimicrobial agents—nitrofurantoin, pivmecillinam, and fosfomycin—have indications largely restricted to acute cystitis. These drugs do not induce cross-resistance with other classes of antimicrobial agents and, to date, limited resistance has been observed in community uropathogens.^{3,72} Thus they are ecologically attractive for treatment of acute cystitis.^{35,70,73} Pivmecillinam, however, is not available in all countries. Fosfomycin is now also used both orally and parenterally for infections with some resistant organisms, and resistance to this agent is emerging in some settings.⁷⁴ The fluoroquinolones with urinary excretion—norfloxacin, ciprofloxacin, and levofloxacin—are not generally recommended as first-line therapy because widespread use promotes the emergence of resistance.⁷³ β -Lactam antimicrobial agents, including amoxicillin, amoxicillin/clavulanic acid,

Table 36.5 Preferred Antimicrobial Regimens for Treatment or Prevention of Acute Uncomplicated Urinary Cystitis in Women With Normal Renal Function

First-Line Therapy	Other Therapy
Acute Cystitis	
TMP/SMX, 160/800 mg bid × 3 days	Norfloxacin, 400 mg bid × 3 days
Nitrofurantoin, 50–100 mg qid, or monohydrate/macrocrystals, 100 mg bid × 5 days ^a	Ciprofloxacin, 250 mg bid, or extended-release preparation, 500 mg qd × 3 days
Pivmecillinam, 400 mg bid × 5 days or 200 mg bid × 7 days ^a	Levofloxacin, 250–500 mg qd × 3 days
Fosfomycin trometamol, 3-g single dose ^a	Trimethoprim, 100 mg bid × 3 days
	Amoxicillin 500 mg tid × 7 days ^a
	Amoxicillin/clavulanic acid, 500 mg tid or 875 mg bid × 7 days ^a
	Cephalexin, 250–500 mg qid × 7 days ^a
	Cefpodoxime proxetil, 100 mg bid × 3 days
	Cefuroxime axetil, 500 mg bid × 7 days ^a
	Cefixime, 400 mg qd × 7 days ^a
	Doxycycline, 100 mg bid × 7 days
Prophylaxis	
Long-Term Low-Dose Regimens (at Bedtime)	
Nitrofurantoin, 50 mg qd, or monohydrate/macrocrystals, 100 mg qd	Cephalexin, 500 mg qd ^a
TMP/SMX, 40/200 mg od or every other day	Norfloxacin, 200 mg every other day
	Ciprofloxacin, 125 mg qd
	Trimethoprim, 100 mg qd
Postcoital (Single-Dose) Regimen	
TMP/SMX, 40/200 mg or 80/400 mg	Cephalexin, 250 mg ^a
Trimethoprim, 100 mg	Norfloxacin, 200 mg
Nitrofurantoin, 50 or 100 mg ^a	Ciprofloxacin, 125 mg

^aRecommended for pregnant women.

bid, Twice a day; qd, once a day; qid, four times a day; tid, twice a day; TMP/SMX, trimethoprim/sulfamethoxazole.

and cephalosporins, are reported to be 10% to 15% less effective than first-line agents.^{70,75} The β -lactam agents are useful, however, for treatment in pregnant women because they are safe for the fetus.⁴⁸

To limit adverse effects, cost, and emergence of bacterial resistance, the shortest effective duration of antimicrobial therapy should be prescribed. For TMP/SMX and the fluoroquinolones, 3 days is optimum.^{3,70} The minimum duration of nitrofurantoin therapy is 5 days.^{76,77} For β -lactam antimicrobial agents, 7 days is recommended; shorter courses are less effective. Fosfomycin is given as a single dose. For other agents a single dose is generally 5% to 10% less effective than the recommended longer regimens.^{48,70}

The antimicrobial susceptibility of uropathogenic *E. coli* strains acquired in the community evolves continually in response to antimicrobial pressure.^{72,78–80} Resistance in community isolates has compromised the efficacy of ampicillin, cephalosporins, and TMP/SMX for use as empirical therapy and there is increasing resistance to fluoroquinolones.^{3,72} Recent prior antimicrobial therapy is most strongly associated with isolation of a resistant organism.⁸⁰ Treatment of cystitis with an antimicrobial agent to which the infecting organism is resistant is associated with a high failure rate, despite very high urinary levels of the drug.^{81,82} In fact, cure rates when the organism is resistant to the antimicrobial agent are similar to those reported with placebo: about 50%. If the local prevalence of resistance to an antimicrobial agent in community *E. coli* strains exceeds 20%, that agent should not be used for first-line empirical therapy.⁷⁰ The current increase in extended-spectrum β -lactamase (ESBL)–producing *E. coli* in community-acquired infections, attributed to global expansion of the *E. coli* clone ST131,⁴⁴ is of particular concern because these strains are usually also resistant to TMP/SMX and fluoroquinolones.^{80,83} Optimal treatment of infection with ESBL-producing *E. coli* is not yet well defined. Nitrofurantoin, fosfomycin, and pivmecillinam currently remain effective for most of these strains.^{80,84}

OTHER INVESTIGATIONS

Young women with a characteristic clinical presentation and prompt response following appropriate empirical therapy do not require further diagnostic imaging or urologic investigation.⁸⁵ Fewer than 5% of these women have urologic abnormalities, and the few women in whom abnormalities are identified usually do not require further intervention. However, investigation may be appropriate to rule out alternative pathologic processes when the diagnosis is uncertain or clinical presentation is atypical.

RECURRENT INFECTION

Episodes of acute cystitis recur frequently in many women. Effective control can be achieved with low-dose prophylactic antimicrobial therapy given either daily or every other day at bedtime or after intercourse (Table 36.5).^{2,86,87} This strategy is recommended for women who experience more than two episodes in 6 months. The initial course of prophylaxis lasts 6 or 12 months. Antimicrobial prophylactic therapy decreases recurrent symptomatic episodes by about 95% while the agent is being taken, but it does not alter the frequency of recurrent infection once prophylaxis is discontinued. About 50% of women experience reinfection within 3 months of discontinuing prophylaxis. Reinstitution of prophylaxis for as long as 2 years may be considered for these women. Self-treatment is another effective strategy for managing recurrent infections; this approach is often preferred by women who are traveling or who experience less frequent recurrences.² A 3-day course of TMP/SMX or ciprofloxacin has been shown to be effective for empirical self-treatment in clinical trials, but other regimens are also probably effective.

The only feasible behavioral intervention to prevent recurrent urinary tract infection is to avoid spermicide use. Other proposed nonantimicrobial approaches for prevention include daily intake of cranberry products, oral or vaginal probiotics to reestablish normal vaginal flora, and estrogen replacement for postmenopausal women.^{88,89} Initial studies reported that

daily intake of cranberry juice or tablets decreased episodes of recurrent cystitis by 30% in comparison with placebo.⁹⁰ The proposed mechanism for this effect is inhibition of the P fimbria-mediated adherence of *E. coli* to uroepithelial cells by the proanthocyanidins present in cranberry products and excreted in the urine. However, more recent clinical trials reported no benefit of daily cranberry juice in comparison with placebo,⁹¹ and cranberry capsules were less effective than TMP/SMX prophylaxis.⁹² Trials of oral or vaginal *Lactobacillus* probiotics do not support efficacy of these products in preventing urinary tract infection.^{93–95} The role of estrogen replacement to prevent urinary tract infection in postmenopausal women is controversial.^{2,96} In prospective clinical trials, researchers have uniformly reported no benefit of systemic estrogen over placebo, despite restoration of an acidic vaginal pH and increased vaginal colonization with lactobacilli in subjects who received estrogen.⁵⁹ Two small clinical trials comparing vaginal estrogen with placebo in postmenopausal women with frequent, recurrent infection demonstrated a decreased frequency of symptomatic episodes in women treated with topical estrogen therapy, but in a comparative trial, an estrogen-containing pessary was substantially less effective than nitrofurantoin prophylaxis.^{59,97} Currently, topical vaginal estrogen should likely only be considered for use to prevent infection in selected women with a very high frequency of recurrent infection.

Several other potential nonantimicrobial approaches for prevention of recurrent urinary tract infection are under investigation.^{49,98,99} These include strategies to reinstate the glucosaminoglycan layer,^{100,101} to block bacterial FimH adhesion with D-mannose,⁹⁹ using vitamin D to enhance antimicrobial peptide production,¹⁰² vaccination using FimH or iron receptors,⁹⁹ and immune stimulation with heat-killed whole bacteria.⁹⁹ A recent trial with one of the whole bacterial immunostimulants (OM-89S), however, showed no benefit.¹⁰³ The clinical efficacy of any of these proposed interventions remains uncertain.

ACUTE NONOBSTRUCTIVE PYELONEPHRITIS

EPIDEMIOLOGY

Pyelonephritis is a less common manifestation of acute uncomplicated urinary tract infection than cystitis. The ratio of pyelonephritis to cystitis episodes is reported to be between 1:18 and 1:29 in women with recurrent infection.² The highest incidence is among women aged 20 to 30 years. Pyelonephritis is associated with substantial morbidity; hospitalization is required for as many as 20% of affected nonpregnant women.¹⁰⁴ Severe manifestations such as sepsis syndrome are, however, uncommon. Acute pyelonephritis complicates 1% to 2% of pregnancies. When this complication occurs at the end of the second trimester or early in the third trimester, preterm labor and delivery may occur and lead to poor fetal outcomes, as with any febrile illness in later pregnancy.¹⁰⁵

Acute nonobstructive pyelonephritis is rarely a direct cause of renal failure. In the few reports of renal failure attributed to pyelonephritis, the patients were elderly¹⁰⁶ or had comorbid conditions such as diabetes and HIV infection.¹⁰⁷ Renal scarring is a complication of pyelonephritis in some women with more severe clinical presentations. An Italian study reported renal scars identified by computed tomography (CT) or magnetic resonance imaging (MRI) at a 6-month

follow-up in 29% of women who required hospitalization for pyelonephritis.¹⁰⁸ In an Israeli study, 30% of a cohort of 203 women admitted for acute pyelonephritis were reassessed 10 to 20 years after admission; renal scars were detected in 46% of these women on technetium (Tc) 99m-labeled dimercaptosuccinic acid scanning. However, the renal scars were not associated with hypertension or renal impairment.¹⁰⁹ The histologic finding of “chronic pyelonephritis” in patients with renal failure was formerly attributed to infection. However, this condition is now recognized as an end stage of many chronic inflammatory conditions of the kidney, and it is attributable to infection in only a few patients in whom there is a clear history of recurrent renal infection.

PATHOGENESIS

E. coli is isolated in 85% to 90% of women who present with acute uncomplicated pyelonephritis.^{2,110} Infecting strains are characterized by production of the P fimbria adhesin Gal(α1–4) Galβ disaccharide galabiose. This surface protein appears to have a direct role in the pathogenesis of pyelonephritis through induction of mucosal inflammation.^{38,111} A familial susceptibility to pyelonephritis has been reported and attributed to polymorphisms with decreased expression of the IL-8 receptor.¹¹² Other genetic and behavioral risk factors for pyelonephritis in healthy women are similar to those described for acute uncomplicated cystitis.¹¹⁰ For premenopausal women, these associations are frequency of sexual intercourse, history of urinary tract infection, history of urinary tract infection in the patient's mother, a new sexual partner, and recent spermicide use. The strongest association is with recent sexual intercourse. Women who experience pyelonephritis as children remain at increased risk of pyelonephritis as adults but experience a lower frequency of episodes.¹¹³ Diabetes is also an independent risk factor for pyelonephritis. Young diabetic women are 15 times more likely to be hospitalized for pyelonephritis than age-matched nondiabetic women. Behavioral risk factors associated with pyelonephritis in postmenopausal women have not yet been identified.

DIAGNOSIS

The classic clinical manifestation of renal infection is costovertebral angle pain or tenderness, often accompanied by fever and variable lower urinary tract symptoms. There is a wide spectrum of severity, however, from mild irritative symptoms with minimal costovertebral angle tenderness to severe symptoms that may include high fever, nausea and vomiting, and severe pain. Acute cholecystitis, renal colic, and pelvic inflammatory disease are occasionally confused with pyelonephritis. When patients present with severe symptoms, underlying complicating factors such as obstruction and abscess must be excluded through urgent imaging. A urine specimen for culture should be obtained before initiation of antimicrobial therapy in every case of suspected pyelonephritis. The culture will confirm the diagnosis of urinary tract infection and identify the specific infecting organism and susceptibilities so that antimicrobial therapy can be optimized. In 95% of women with pyelonephritis, $\geq 10^5$ CFU/mL of organisms are isolated from the urine culture.

Bacteremia is identified in 10% to 25% of women presenting with acute pyelonephritis if blood culture specimens are

collected routinely. However, the clinical utility of routine blood cultures is limited because bacteremia does not alter therapy, nor is it predictive of outcome.^{114,115} Thus blood cultures should be obtained selectively, usually only if the diagnosis is uncertain or the clinical presentation is severe. Growth of the same organism from both blood and urine usually confirms a urinary source for the infection. However, bacteria isolated from the urine occasionally originate from bacteremia from a source outside the urinary tract. This finding may also reflect hematogenous seeding with development of renal microabscesses, which is well described for *Staphylococcus aureus* in particular.¹¹⁶ The proportion of cases in which *S. aureus* bacteriuria has a nonurinary source, however, is controversial.^{116,117}

Additional investigations recommended for most patients presenting with acute pyelonephritis are measurements of peripheral leukocyte count and serum creatinine level. The leukocyte count is usually elevated and may be useful as a parameter to monitor the response to therapy. C-reactive protein and procalcitonin values are elevated in most women with acute pyelonephritis and tend to correlate with severity of presentation.^{2,118,119} The serum procalcitonin value at presentation is a marker for bacteremia¹²⁰ and septic shock,¹²¹ but it does not reliably discriminate between patients requiring inpatient therapy and those who can undergo outpatient therapy,¹²² and is not predictive of outcome.¹¹⁸ An elevated C-reactive protein value at discharge has been associated with prolonged hospitalization and postdischarge recurrence.¹¹⁹

DIAGNOSTIC IMAGING

When the clinical presentation of pyelonephritis is characteristic, symptoms are of mild or moderate severity, and the clinical response after initiation of antimicrobial therapy is prompt, routine diagnostic imaging is not indicated.^{48,123} Women whose clinical presentations are severe, in whom treatment fails, or who experience early posttreatment recurrent infection should undergo prompt imaging to rule out obstruction or abscesses and to determine whether intervention is necessary. Ultrasonography is often the initial imaging modality because it is safe and widely accessible.¹²⁴ The ultrasound examination in women with uncomplicated pyelonephritis usually yields normal results, but enlargement and edema in one or both kidneys are observed in 20% of patients.¹²⁵ Ultrasonography is less sensitive or specific for pyelonephritis than is either CT or MRI.

The optimal diagnostic imaging is contrast-enhanced CT, though this may be associated with a risk of developing contrast-induced nephropathy.¹²⁵ Abnormalities observed on CT are characterized as unilateral or bilateral, focal or diffuse, focal swelling or no focal swelling, and renal enlargement or no renal enlargement.¹²⁴ In addition to renal enlargement and edema, dilation of the collecting system in the absence of obstruction, wedge-shaped areas of decreased attenuation, and rounded low-attenuation masses with delayed enhancement may be observed. Obstruction of the renal tubules by inflammatory debris or impaired function with tubular ischemia may result in a “striated nephrogram.”¹²⁴ Abnormalities within the renal cortex and medulla, inflammatory changes in Gerota fascia or the renal sinuses, and thickening of the urothelium are sometimes observed. “Acute focal pyelonephritis” (also called acute lobar nephronia) is infection confined to a single lobe and may be more common in

women who have diabetes or are immunocompromised. The response to treatment is similar, however, for patients with or without this imaging finding.¹²⁵ In one study the presence of a focal lesion characterized by peripheral ring enhancement without central uptake of contrast material on CT or MRI at presentation was the only imaging finding that was correlated with subsequent development of renal scars.¹⁰⁸

TREATMENT

The majority of women with uncomplicated pyelonephritis receive treatment as outpatients.^{48,70} Indications for hospitalization include pregnancy, hemodynamic instability, uncertain gastrointestinal absorption or compliance with oral therapy, the need to exclude complicating factors such as obstruction and abscess, and the necessity of monitoring or treatment of associated medical illnesses. Appropriate supportive management for hypotension, nausea and vomiting, and pain should be initiated promptly. When oral tolerance is uncertain because of nausea and vomiting, a strategy frequently used in emergency department management is to provide a single parenteral dose of ceftriaxone, 1 g, or of gentamicin, 120 mg, followed by oral therapy once gastrointestinal symptoms are controlled.

Many parenteral antimicrobial regimens are effective for pyelonephritis (Table 36.6). Options for empiric parenteral therapy include aminoglycosides, extended-spectrum cephalosporins such as cefotaxime and ceftriaxone, and fluoroquinolones such as ciprofloxacin or levofloxacin.^{2,70} Aminoglycosides have unique efficacy for the treatment of renal infection as they are bound in high concentrations in the renal cortex.² Ceftriaxone therapy is the preferred empirical regimen for pregnant women. Although it is suggested that gentamicin be avoided in pregnancy because of potential fetal ototoxicity, excess otologic impairment has not been reported in large cohorts of newborn infants stratified by gentamicin exposure in utero.¹²⁶ Thus when cephalosporins cannot be used because of antimicrobial resistance or patient intolerance, gentamicin remains an alternate antimicrobial for treatment of pregnant women. For ESBL-producing *E. coli* resistant to all first-line therapies, a carbapenem (meropenem, ertapenem, doripenem) is recommended.

Oral therapy selected on the basis of urine culture results can then be prescribed to complete the antimicrobial course. In most young, nonpregnant women, acute pyelonephritis is effectively managed with outpatient oral therapy (Table 36.6).^{48,70} The recommended empirical antimicrobial choice is either ciprofloxacin or levofloxacin.⁷⁰ Oral TMP/SMX is effective, but because of the high prevalence of TMP/SMX-resistant *E. coli* in many communities, this agent is recommended only when the infecting organism is known to be susceptible. Other oral regimens are also effective and may be appropriate, depending on organism susceptibility and the patient's tolerance. The usual duration of treatment was previously 10 to 14 days, but ciprofloxacin 500 mg twice daily given for 5 or 7 days,^{127,128} levofloxacin 750 mg daily for 5 days, and a course of a single dose of ceftriaxone 1 g and cefixime 400 mg daily for 6 days¹²⁹ are all effective shorter regimens.¹³⁰

A satisfactory clinical response is usually observed by 48 to 72 hours after initiation of antimicrobial therapy. Risk factors predictive of a poor outcome are hospitalization, isolation of a resistant organism, concurrent diabetes mellitus,

Table 36.6 Antimicrobial Regimens for Treatment of Acute Uncomplicated Pyelonephritis in Women With Normal Renal Function and Susceptible Organisms

First-Line Therapy	Other Therapy
Oral	
Ciprofloxacin, 500 mg bid or 1000 mg extended-release preparation, qd × 7 days	TMP/SMX, 160/800 mg bid × 7–14 days
Levofloxacin, 750 mg qd × 5 days	Amoxicillin, 500 mg PO tid × 14 days ^a
	Amoxicillin/clavulanic acid, 500 mg tid or 875 mg PO bid × 14 days ^a
	Cephalexin, 500 mg qid × 14 days ^a
	Cefuroxime axetil, 500 mg bid × 14 days ^a
	Cefixime, 400 mg qd × 14 days ^a
Parenteral^b	
Ciprofloxacin, 400 mg q12h × 7 days	Ertapenem, 1 g qd
Levofloxacin, 750 mg qd × 5 days	Meropenem, 500 mg q6h
Gentamicin or tobramycin, 3–5 mg/kg qd, ± ampicillin, 1 g q4–6h	Piperacillin/tazobactam, 3.375 g q6h
Ceftriaxone, 1–2 g qd ^a	Doripenem, 500 mg q8h
Cefotaxime, 1 g q8h ^a	Ceftolozane/tazobactam, 1 g q8h
	Ceftazidime/avibactam, 2 g q8h

^aRecommended for pregnant women.^bChange to oral therapy to complete course once condition is clinically stable.

bid, Twice a day; PO, by mouth; qd, once a day; qid, four times a day; tid, twice a day; TMP/SMX, trimethoprim/sulfamethoxazole.

and history of renal stones.¹³¹ All these variables are suggestive of greater severity of illness at presentation or of a complicated infection. Prophylactic antimicrobial strategies similar to those for recurrent cystitis are effective for prevention of recurrent uncomplicated pyelonephritis.

COMPLICATED URINARY TRACT INFECTION

EPIDEMIOLOGY

The frequency of complicated urinary tract infection is highly variable, depending on the underlying genitourinary abnormality (Table 36.1).^{1,5} Some individuals with a transient abnormality, such as a ureteric stone complicated by pyelonephritis, may experience only a single infection. Other patients, including those with indwelling devices or persistent obstruction, may experience frequent recurrent infections. For instance, in men with spinal cord injury in whom voiding is managed with an indwelling catheter, the incidence is 2.72 infections/1000 days; when voiding is managed with intermittent catheterization, the infection incidence is 0.41/1000 days.¹³² In residents of long-term care facilities with long-term indwelling urethral catheters, the incidence of symptomatic infection is 3.2/1000 catheter days.¹³³

Complicated urinary tract infection is a frequent cause of hospitalization. The urinary tract is the most common source of community-acquired bacteremia,^{134,135} and most bacteremic episodes of urinary tract infection are attributable to complicated infection.¹³⁶ Patients who have obstruction, an indwelling catheter, urologic tumors, or who have undergone recent manipulation of the urinary tract with mucosal bleeding are at increased risk of bacteremia and severe sepsis. The genitourinary tract is the source of infection in about 10% of patients admitted to critical care units with septic shock.¹³⁷

These patients are also at risk for local suppurative complications, such as renal or perinephric abscess, or metastatic infection after bacteremia, such as septic arthritis, osteomyelitis, or endocarditis. Serious complications are more common in patients who are diabetic, are immunocompromised, or have long-term urologic devices or obstruction. Renal functional impairment in patients with complicated urinary tract infection is usually attributable to the underlying abnormality or to organ failure complicating septic shock, rather than being a direct consequence of infection. For instance, introduction of voiding strategies to maintain low bladder pressure and to prevent reflux have almost eliminated the complication of chronic renal failure in persons with spinal cord injury, despite a continued high incidence of urinary tract infection in these patients.¹³⁸

PATHOGENESIS

Microbiology

Host impairment rather than organism virulence is the major determinant of infection. *E. coli* remains the organism most frequently isolated in complicated urinary tract infection.^{4,5} The *E. coli* strains are characterized by a low frequency of expression of virulence factors in comparison with strains isolated in acute uncomplicated infection.⁴⁰ Many other bacteria and yeast species are also isolated.^{5,139,140} Enterobacteriaceae such as *Klebsiella* species, *Enterobacter* species, *Serratia* species, *Citrobacter* species, *Proteus mirabilis*, *Morganella morganii*, and *Providencia stuartii* are common. Other gram-negative organisms that may be isolated include *Pseudomonas aeruginosa*, *Stenotrophomonas maltophilia*, and *Acinetobacter* species. Gram-positive organisms are also frequently isolated, including group B streptococci, *Enterococcus* species, and coagulase-negative staphylococci. *S. aureus* is less commonly isolated. *Candida* species may be isolated, usually from patients who have diabetes or indwelling urologic devices or who are undergoing broad-spectrum antimicrobial therapy.¹⁴¹ Organisms isolated from patients with complicated urinary tract infection often have increased antimicrobial resistance.¹⁴² Risk factors for isolation of a resistant organism include a history of recent antimicrobial therapy or hospitalization, indwelling urethral catheter, invasive urologic procedures, nursing home residence, and some comorbidities.

Biofilm formation is universal in patients with chronic indwelling urinary devices.¹⁴³ Following insertion, a conditioning layer composed of proteins and other host components immediately coats the device. This conditioning layer provides a surface for subsequent attachment of bacteria or yeast that originate from the periurethral flora or drainage bags or are introduced after disruption of the closed drainage system. Organisms grow along the device, elaborating an extracellular polysaccharide substance, and colonies of microorganisms

persist within this relatively protected environment. Urine components such as Tamm–Horsfall protein and magnesium or calcium ions are also incorporated into the biofilm. Organisms ascend in the biofilm along the interior and exterior surfaces of the device and reach the bladder within days. The initial infection is usually with a single organism, but a polymicrobial flora is invariably present on long-term indwelling devices.^{26,143} Urease-producing organisms such as *P. mirabilis*, *K. pneumoniae*, *M. morganii*, and *P. stuartii* are isolated more frequently from individuals with long-term indwelling catheters and persist longer than other organisms, such as *Enterococcus* species. Complications attributed to biofilm formation with urease-producing organisms include development of renal or bladder stones and obstruction of the device.¹⁴⁴

Although the most common device is the urethral catheter, the process of biofilm formation is similar with other indwelling devices, such as ureteric stents and nephrostomy tubes.^{145–147} From 34% to 42% of ureteral stents are found at removal to be colonized with bacteria or yeast species, often with multiple organisms. More than 50% of organisms identified by stent biofilm analysis are not isolated from culture of urine specimens collected at the same time.

Uncommon bacteria are occasionally a cause of infection. Some of these organisms may not be identified with standard laboratory procedures for processing urine specimens. *Corynebacterium urealyticum* is a urease-producing gram-positive rod associated with the unique clinical manifestations of encrusted cystitis or pyelonephritis.¹⁴⁸ This infection is characterized by ulcerative inflammation and struvite encrustations on the bladder or renal pelvis wall. Pyelitis, if untreated, may lead to destruction of the kidney. *U. urealyticum* is another urease-producing bacterium that may cause cystitis or pyelonephritis, often with urolithiasis. Case reports suggest a predisposition in immunocompromised individuals, particularly those with hypogammaglobulinemia; healthy persons may also become infected. *Aerococcus sanguinicola* is a rare cause of complicated urinary tract infection; the diagnosis is usually made by isolation of the organism in the blood culture.¹⁴⁹ *Aerococcus urinae* was isolated in 0.3% to 0.8% of urine specimens in one clinical microbiology laboratory, usually from older persons with underlying abnormalities and bacteremia.¹⁵⁰ Anaerobic organisms are seldom identified in the absence of suppurative complications such as abscess.¹⁵¹

Host Factors

Genitourinary abnormalities facilitate infection through increased entry of organisms into the bladder (e.g., by intermittent catheterization, urologic procedures) and subsequent persistence due to incomplete voiding (e.g., as a result of obstruction, urolithiasis, diverticula, reflux), or in biofilm on urologic devices.^{4,5} Asymptomatic bacteriuria is common in patients with persistent abnormalities and universal in patients with long-term indwelling catheters.¹⁵² The determinants that lead to symptomatic infection in chronically bacteriuric individuals are not well characterized. However, obstruction and mucosal trauma with bleeding are well-recognized risk factors for bacteremia and sepsis in patients with preexisting bacteriuria.

CLINICAL PRESENTATIONS

Complicated urinary tract infection manifests across a wide clinical spectrum of signs and symptoms, from mild, irritative

symptoms of lower tract infection to pyelonephritis and bacteremia, including septic shock.^{4,5,153} Localizing signs and symptoms consistent with cystitis or pyelonephritis are usually present. Patients with indwelling urethral catheters or other indwelling devices usually present with fever alone, although costovertebral angle pain or tenderness, hematuria, or catheter obstruction, if present, identifies a genitourinary source. Patients with chronic neurologic impairment sometimes report symptoms that are not classic for urinary tract infection. For instance, spinal cord-injured patients experience increased bladder and leg spasms or autonomic dysreflexia,¹³⁸ whereas patients with multiple sclerosis may present with fatigue or deterioration in neurologic function.⁵

The clinical diagnosis of symptomatic infection is often problematic in older populations with cognitive impairment.^{154–156} These patients frequently have chronic genitourinary symptoms and impaired communication, which limits the assessment of signs and symptoms. Because bacteriuria is very common in elderly individuals with functional impairment, nonlocalizing clinical deterioration is frequently attributed to urinary tract infection because the urine culture is positive.¹⁵⁷ However, nonlocalizing clinical manifestations, including fever, are unlikely to have a urinary source in elderly persons without a long-term indwelling catheter.^{25,155} Changes in character of the urine, such as cloudiness and odor, are also frequently interpreted as symptoms of urinary tract infection. Cloudiness may be attributed to pyuria, which usually accompanies bacteriuria, and an unpleasant odor is suggestive of production of polyamines by bacteria in the urine. However, alterations in characteristics of the urine are neither sensitive nor specific for the diagnosis of infection. They may be attributable to other causes, such as precipitation of crystals and dehydration. Thus changes in character of the urine should not be interpreted as symptoms of urinary tract infection.

LABORATORY DIAGNOSIS

A urine specimen for culture should be obtained before initiation of antimicrobial therapy for every patient with suspected complicated urinary tract infection. Because of the wide variety of potential infecting organisms and increased likelihood of resistant strains, definitive microbiologic characterization is necessary to optimize antimicrobial management.

The presence of biofilm on devices such as indwelling catheters and stents within the urinary tract complicates interpretation of the urine culture in some patients with complicated infection.^{7,143–145} In patients with a long-term indwelling catheter, the catheter should be replaced, and the new catheter should be used to sample bladder urine and avoid contamination by organisms present in the biofilm of the old catheter.^{25,26} When the intestine is interposed into the urinary tract through creation of an ileal conduit, continent cutaneous diversion, or neobladder, urine collected through the conduit or reservoir is often bacteriuric, regardless of symptoms.¹⁵⁸ The organisms isolated represent mixed gram-positive flora, including streptococcal species and *Staphylococcus epidermidis*, but uropathogenic strains such as *E. coli*, *P. mirabilis*, *P. aeruginosa*, and *Enterococcus faecalis* may also be present. Similar urine culture findings are reported in 30% to 60% of patients with orthoptic bladder substitution or augmentation cystoplasty¹⁵⁹; individuals with these reservoirs who practice clean intermittent catheterization are more

likely to have positive culture findings.¹⁵⁸ Thus when symptomatic urinary tract infection is suspected, results of a urine culture in such a patient must be interpreted in the context of this usual bacteriuria.

Infection with a fastidious organism should be considered when the clinical presentation suggests symptomatic urinary tract infection but urine culture results are repeatedly negative, particularly when pyuria is present. These organisms may include *C. urealyticum*, *U. urealyticum*, and *Haemophilus* species. A persistently alkaline pH with pyuria but a negative urine culture suggests a urease-producing organism such as *C. urealyticum* or *U. urealyticum*. The laboratory should be consulted if a fastidious organism is considered, and appropriate specimens should be collected for additional laboratory evaluation to maximize the likelihood of isolating potential infecting organisms.

Patients with genitourinary abnormalities frequently have pyuria, whether or not they have bacteriuria or symptomatic infection, and so pyuria by itself is not diagnostic for urinary tract infection.⁵ The absence of pyuria, however, has a high negative predictive value for ruling out urinary tract infection in some populations, such as elderly patients.²⁵ The severity of clinical manifestations determines whether additional investigation, such as blood culture or a peripheral leukocyte count, is indicated. Renal function should be assessed in every patient with complicated urinary tract infection. A second urine culture after antimicrobial therapy is not recommended if the patient remains asymptomatic.

ANTIMICROBIAL TREATMENT

The principles of management for individuals with complicated urinary tract infection include prompt specimen collection to identify the specific infecting organism, characterization of renal function and underlying abnormalities, and early institution of appropriate antimicrobial therapy. The antimicrobial regimen selected is individualized on the basis of site of infection, severity of manifestations, known or presumed infecting organism and susceptibility, and the patient's tolerance.⁵ When the presenting symptoms are mild, it is preferable to delay initiation of antimicrobial therapy until results of the urine culture are available. This approach allows selection of a narrow-spectrum agent specific for the infecting organism and minimizes antimicrobial pressure, which promotes resistance.

When patients present with severe symptoms, empirical antimicrobial therapy is initiated pending urine culture results. Previous urine culture results, if available, and recent antimicrobial therapy received by the patient should be considered in the selection of the empiric regimen.⁵ Parenteral therapy is indicated for patients who present with hemodynamic instability, who cannot tolerate or absorb oral medications, or who are known or suspected to have an infecting organism resistant to available oral options. Patients who present with severe sepsis, including septic shock, should receive initial empirical antimicrobial therapy that provides broad coverage for both gram-positive and gram-negative bacteria, including resistant organisms.¹⁵³ Appropriate regimens for these patients may include an aminoglycoside, with or without ampicillin; a cephalosporin with an aminoglycoside; a carbapenem (ertapenem, imipenem, meropenem); an extended-spectrum cephalosporin, or a β -lactam/ β -lactamase inhibitor. The new β -lactam/ β -lactamase inhibitors—ceftazidime/avibactam and

ceftolozane/tazobactam—are effective for the treatment of complicated infection with resistant organisms.^{160–164} Intravenous fosfomycin has also been used for some highly resistant organisms, usually in combination with another broad-spectrum antimicrobial.¹⁶⁵

Fluoroquinolones with good urinary excretion and broad gram-negative coverage—norfloxacin, ciprofloxacin, and levofloxacin—are indicated for empirical oral therapy.⁵ Many other oral agents are effective and may be appropriate, depending on patient tolerance and the specific infecting organism.⁵ These include TMP/SMX, amoxicillin, amoxicillin/clavulanic acid, oral cephalosporins, and doxycycline. Oral fosfomycin has been used for some resistant organisms, with success rates of 50%–85% reported.^{166–168} Nitrofurantoin is effective for treatment of some episodes of bladder infection, but it is not effective for renal or prostate infection. It is contraindicated for treatment of patients with renal failure because peripheral neuropathy has been reported to occur with accumulation of toxic metabolites. *K. pneumoniae*, *P. aeruginosa*, and *P. mirabilis* strains are uniformly resistant, but nitrofurantoin remains effective for many resistant organisms such as vancomycin-resistant *Enterococcus* and most ESBL-producing *E. coli*.

Substantial clinical improvement is expected by 48 to 72 hours after initiation of effective antimicrobial therapy. Empirical therapy is reassessed at this time, considering the clinical response and urine culture results. Therapy is usually modified to an appropriate narrow-spectrum parenteral or oral agent to complete a 7- to 10-day course. If an organism isolated in the pretherapy urine culture specimen is resistant to the empirical antimicrobial, therapy should be altered to an antimicrobial agent to which the infecting organism is susceptible, even if clinical improvement has occurred. Shorter courses of therapy with treatment durations of 5 days are effective in some patients.¹⁶⁹

OTHER INTERVENTIONS

The optimal management of complicated urinary tract infection requires characterization of the underlying genitourinary abnormality and appropriate urologic or other interventions to assist in resolution of the current infection and prevention of subsequent infections. Urgent diagnostic imaging or urologic investigation is indicated for patients who have severe systemic symptoms or whose symptoms do not respond to appropriate antimicrobial therapy despite isolation of a susceptible organism. The goal of early imaging is to identify obstruction or abscesses, for which immediate drainage may be necessary for source control.¹³² When the underlying abnormality is already well characterized—as in the case of a patient with an indwelling catheter or a neurogenic bladder managed with intermittent catheterization—further investigations may still be appropriate if the patient has experienced a recent change in frequency or severity of infections. Such a patient remains at risk for development of urolithiasis, tumors, and suppurative complications.

The approach to diagnostic imaging and urologic investigation is determined by the clinical presentation and the patient's previous history. A plain radiograph of the abdomen may identify emphysematous infections and some stones. CT is the imaging modality of choice. It identifies calculi, gas, hemorrhage, calcification, obstruction, renal enlargement, and inflammatory masses. Contrast media-enhanced scans

are recommended (though are associated with risk of contrast-induced nephropathy, which should be considered in clinical decision making), with helical and multislice CT to study different phases of contrast excretion.^{123–125} Ultrasound examination is less sensitive and specific than other imaging modalities such as spiral CT and MRI, but it may be more accessible.¹²⁵

Appropriate supportive care must be initiated promptly. The removal and replacement of a long-term indwelling catheter before institution of antimicrobial therapy are associated with a more rapid defervescence and a lower risk of early relapse after therapy, as well as facilitating collection of a more valid urine culture specimen.²⁶ The clinical benefits of catheter replacement are presumed to result from removal of the high concentration of organisms in the biofilm, which are often not eradicated by antimicrobial therapy and remain a source for relapse. Urologic investigations such as cystoscopy, retrograde pyelography, and urodynamic studies should be obtained as appropriate. If encrusted *C. urealyticum* infection is present, surgical resection of the encrustations is required together with antimicrobial therapy. *C. urealyticum* strains are generally susceptible to vancomycin, tetracyclines, and fluoroquinolones.¹⁴⁸

MANAGEMENT OF RECURRENT INFECTION

When symptoms recur early after antimicrobial treatment, the susceptibility of the pretherapy infecting organism should be reviewed to confirm that the antimicrobial agent prescribed is effective. If the organism is susceptible, underlying genitourinary abnormalities should be reviewed and further evaluation obtained, if appropriate, to identify abnormalities such as abscesses that may necessitate drainage. Even if the abnormalities cannot be fully corrected, interventions to improve urine drainage may decrease the frequency of episodes of symptomatic infection.^{4,5} Indwelling devices should be removed whenever possible. Evidence-based infection control guidelines provide recommendations for prevention of catheter-acquired urinary tract infection.^{170–172} and multifaceted programs are effective in decreasing catheter use and infection rates.^{173,174} Specific practices include catheter use for restricted indications, limiting duration of use, safe practices for insertion and care, and appropriate catheter size. Interventions that have been shown not to be effective and that are not recommended include use of different catheter materials, antimicrobial coating of catheters, antiseptic or antimicrobial meatal care, and instillation of antiseptics into the drainage bag. Patients with long-term indwelling catheters and other indwelling devices experience infection because of biofilm formation on these devices; therefore the prevention of catheter-acquired infection will ultimately require development of biofilm-resistant biomaterials.^{175,176}

Long-term prophylactic antimicrobial therapy is not recommended.^{5,138,177} When patients with impaired voiding or indwelling devices are given prophylactic antimicrobial agents, there is little, if any, decrease in symptomatic infection, and rapid reinfection with resistant organisms is uniformly observed.⁵ For selected patients with a persistent abnormality and who experience symptomatic relapsing infection with an organism that cannot be eradicated, suppressive therapy may be considered. The goal of suppressive therapy is not to prevent reinfection but either to control symptomatic episodes when infecting bacteria cannot be eradicated or to

prevent stone enlargement when inoperable infection stones are present. The antimicrobial regimen is selected on the basis of the infecting organism and is prescribed initially at a full therapeutic dose. If the patient remains clinically stable and the urine is sterile, this dose is usually decreased to about half after 4 to 6 weeks. Suppressive therapy is not appropriate for patients with indwelling devices because biofilm formation facilitates rapid reinfection and emergence of resistant organisms. However, short-term use of such therapy (several weeks or months) for patients with complex urologic abnormalities and indwelling devices may occasionally be considered as part of a palliative care strategy.

A novel approach proposed for control of recurrent infection in patients with impaired bladder emptying is “bacterial interference.”¹⁷⁸ This strategy establishes asymptomatic bacteriuria with a nonpathogenic *E. coli* strain in patients with impaired voiding. The avirulent strain in the bladder then prevents infection by other, potentially more virulent, strains. The proposed mechanisms for this protective effect include blocking of bacterial receptors present on uroepithelial cells, competition for nutrients in the urine, and toxin production. Preliminary clinical trials have demonstrated some efficacy of this approach in a small number of highly selected patients.¹⁷⁹

ASYMPTOMATIC BACTERIURIA

EPIDEMIOLOGY

Asymptomatic bacteriuria is a common finding, particularly in women, older persons, and some patients with persistent genitourinary abnormalities (Table 36.7).¹⁵² The prevalence of bacteriuria among sexually active young women ranges from 3% to 5% but is <1% among age-matched controls who are not sexually active. Bacteriuria is present in 5% to 10% of healthy postmenopausal women¹⁵² and in 20% of women older than 80 years living in the community.¹⁸⁰ Asymptomatic bacteriuria is uncommon in younger men, but the prevalence increases in men older than 65 years, presumably concurrently with age-related prostate hypertrophy. Bacteriuria occurs in 10% of healthy men older than 80 years living in the community.¹⁸⁰ Among residents of nursing homes who do not have indwelling catheters, 20% to 50% of women and 15% to 40% of men are bacteriuric. The prevalence among persons with long-term indwelling catheters is 100%.¹⁵⁵ Among spinal cord-injured patients with impaired voiding and no indwelling catheter, the prevalence of bacteriuria is 50%, regardless of the method used for bladder emptying.¹³⁸

Patients with an increased prevalence of asymptomatic bacteriuria also have a higher incidence of symptomatic urinary tract infection. However, this higher frequency of symptomatic infection is not attributable to bacteriuria. The same biologic determinants promote both asymptomatic and symptomatic infection. Bacteriuria in healthy young women is usually transient, but up to 8% have acute cystitis within 1 week of initial identification of a positive urine culture result.¹⁸¹ In women with diabetes¹⁸² and female or male residents of nursing homes,¹⁵⁵ persistent bacteriuria for months or years, frequently with the same strain, is often observed. No long-term negative outcomes have been attributed to asymptomatic bacteriuria.^{152,182} Bacteriuric individuals are not at increased risk of hypertension or chronic renal failure, and survival is similar to that for persons without bacteriuria.

Table 36.7 Asymptomatic Bacteriuria in Normal Populations and in Selected Patients With Underlying Genitourinary Abnormalities

Population/Patients	Prevalence of Bacteriuria (%)
Healthy Women	
Aged 20–50 years:	
Sexually active	3–5
Not sexually active	<1
Aged 50–70 years	3–9
Aged ≥80 years	14–22
Healthy Men	
Aged <65 years	<1
Aged ≥80 years	6–10
Patients With Complicated Genitourinary Abnormalities	
Spinal Cord Injury	
Bladder retrained	25
Intermittent catheterization	23–89
Sphincterotomy/condom	58
Ileal neobladder	30–60
Residence in Long-Term Care Facility	
Women	25–57
Men	19–37
Indwelling Urethral Catheter	
Short term	5%–7% acquisition/day
Long term	100
Urethral Stents	
Temporary	45
Permanent	100

Data from Nicolle LE, Bradley S, Colgan R, et al. Infectious Diseases Society of America guidelines for the diagnosis and treatment of asymptomatic bacteriuria in adults. *Clin Infect Dis*. 2005;40:643–654; and Rodhe N, Mölstad S, Englund L, et al. Asymptomatic bacteriuria in a population of elderly residents living in a community setting: prevalence, characteristics, and associated factors. *Fam Pract*. 2006;23:303–307.

Harmful short-term outcomes attributable to asymptomatic bacteriuria are recognized in only two distinct populations: pregnant women and patients who undergo traumatic genitourinary procedures.¹⁵² The prevalence of bacteriuria during early pregnancy is 3% to 7%, similar to that among age-matched nonpregnant women. The physiologic changes that accompany the increased progesterone levels in pregnancy include smooth muscle relaxation and decreased peristalsis, which result in dilation of the renal pelvis and ureters. In later pregnancy, ureteric obstruction may result from pressure of the uterus at the pelvic brim. In 20% to 35% of pregnant women with bacteriuria that is not treated, acute pyelonephritis develops later in pregnancy, usually at the end of the second trimester or early in the third trimester. This incidence of pyelonephritis is twentyfold to thirtyfold higher among

untreated women with bacteriuria than among women whose initial screening urine cultures yielded negative results or in whom bacteriuria was treated. Acute pyelonephritis in later pregnancy is associated with premature delivery and poorer fetal outcomes. The second group of bacteriuric patients at risk are those who undergo traumatic urologic procedures. If bacteriuria remains untreated, bacteremia develops in as many as 60% after the procedure, and in 5% to 10% progresses to severe sepsis or septic shock.

PATHOGENESIS

Microbiology

E. coli is isolated from 80% of healthy women with asymptomatic bacteriuria.¹⁸¹ Most of the remaining bacterial strains are *K. pneumoniae*, *Enterococcus* species, and coagulase-negative staphylococci. For men older than 65 years of age, coagulase-negative staphylococci are isolated most frequently, followed by *E. coli* and *Enterococcus* species. In patients with underlying genitourinary abnormalities, a wider variety of organisms are isolated. *E. coli* and other organisms associated with asymptomatic bacteriuria are characterized by the relative absence of or lack of expression of recognized virulence factors.¹⁸³ Some strains that are strongly adherent but unable to stimulate an epithelial cell IL-6 cytokine response have been described. These relatively avirulent *E. coli* strains may originate from nonvirulent commensal strains or may evolve from virulent strains by attenuation of virulence genes.^{184,185} Other organisms that are frequently isolated, such as coagulase-negative staphylococci and *Enterococcus* species, are relatively nonpathogenic and seldom associated with symptomatic infection. Biofilm formation results in polymicrobial bacteriuria in patients with chronic indwelling catheters.¹⁴³

Host Factors

The genetic and behavioral risk factors and genitourinary abnormalities associated with asymptomatic bacteriuria are similar to those described for uncomplicated and complicated symptomatic urinary tract infection. For younger women, behavioral risks are sexual activity and spermicide use.¹⁸¹ Bacteriuric women older than 80 years who reside in the community are characterized by reduced mobility, urinary incontinence, and receiving estrogen treatment; and men older than 80 years with bacteriuria are characterized by prostate disease, history of stroke, and residence in supervised housing.¹⁸⁰ Functional impairment is the major risk factor associated with asymptomatic bacteriuria in the long-term care facility population without indwelling catheters.¹⁵⁵ Higher residual urine volume is not associated with increased bacteriuria in elderly populations but has been associated with bacteriuria in patients referred to an ambulatory urology clinic.¹⁵⁵

DIAGNOSIS

The criterion for identification of asymptomatic bacteriuria is an organism quantitative count of $\geq 10^5$ CFU/mL in a urine culture. For women, two consecutive urine specimens with similar culture results are recommended, but a single specimen is sufficient for men.¹⁵² When an initial urine culture in a young woman yields positive results, a second positive result in a specimen obtained within 2 weeks confirms bacteriuria in 85% to 90% of women. If the second specimen result is negative, the initial positive culture result may have

been contaminated; however, bacteriuria may also have resolved spontaneously. If a voided specimen has a low quantitative count of a single potential uropathogen, and if it is essential to rule out bacteriuria, a second urine specimen should be collected as a first morning void.

Pyuria usually accompanies bacteriuria, but there is variability in the frequency of pyuria observed in different populations. Pyuria is present in only 50% of bacteriuric pregnant women; therefore screening for pyuria is not a reliable method to rule out bacteriuria in pregnancy.¹⁸⁶ Pyuria is present in about 75% of diabetic women with bacteriuria, in 90% of bacteriuric patients undergoing hemodialysis, and in >90% of elderly persons with bacteriuria.¹⁵² Pyuria also occurs in 30% to 70% of bacteriuric patients with short-term indwelling catheters and in 100% of those with long-term indwelling catheters. Some other biomarkers have been explored to be used in differentiating asymptomatic from symptomatic infection, particularly in the nursing home population. Urinary IL-6 has some association with symptomatic infection, but has not been shown, to date, to have clinical utility.^{187,188}

TREATMENT

Screening for and treatment of asymptomatic bacteriuria is recommended only for pregnant women or for patients who will undergo a traumatic genitourinary tract procedure.¹⁵² For other patients, treatment does not improve short- or long-term clinical outcomes, whereas negative consequences such as reinfection with organisms of increased antimicrobial resistance and adverse drug effects occur.^{152,189} In girls and women, an increased frequency of symptomatic infection has been reported after treatment of asymptomatic bacteriuria.^{152,190} This finding may be attributable to alteration of vaginal flora by antimicrobial therapy or to replacement of a benign organism in the urine by a more virulent organism. Studies suggest no benefits of treatment of asymptomatic bacteriuria for patients with complicated infection, including renal transplant recipients.^{191–193} Treatment of patients with asymptomatic bacteriuria is identified as a major contributor to inappropriate antimicrobial use, and an important target for intervention in antimicrobial stewardship programs.^{194,195}

Identification and treatment of asymptomatic bacteriuria early in pregnancy reduces the risk of pyelonephritis from between 25% and 30% to between 1% and 2%.¹⁵² A recent trial in the Netherlands enrolling only low-risk pregnant women reported a much lower rate of pyelonephritis, but a significant association of pyelonephritis and untreated asymptomatic bacteriuria was still observed.^{196,197} Current recommendations are that all pregnant women should be screened for bacteriuria by urine culture at the end of the first trimester and treated if bacteriuria is found.¹⁸⁶ Preferred regimens include a 5- or 7-day course of nitrofurantoin¹⁹⁸ or a 7-day course of amoxicillin, amoxicillin/clavulanic acid, or a cephalosporin. The specific regimen is chosen on the basis of the organism isolated and the patient's tolerance. Shorter antimicrobial courses are sometimes prescribed, but these abbreviated courses are likely less effective.¹⁹⁹ TMP/SMX should be avoided, especially in the first trimester, and fluoroquinolones are contraindicated during pregnancy. After treatment for asymptomatic or symptomatic urinary tract infection, urine cultures should be performed at least monthly. If infection recurs, prophylactic antimicrobial therapy should

be initiated and continued for the duration of the pregnancy. Nitrofurantoin and cephalexin are the preferred prophylactic regimens because these are considered safe for the fetus.

Initiation of effective antimicrobial treatment immediately before a traumatic urologic procedure prevents bacteremia and sepsis in a bacteriuric patient.^{152,200} Conceptually, this approach is surgical prophylaxis rather than treatment of asymptomatic bacteriuria. A single dose of an antimicrobial agent is usually adequate, although some guidelines recommend that the antimicrobial agent be continued after transurethral resection of the prostate until the indwelling catheter is removed.¹⁵² Antimicrobial agents are not recommended for minor urologic procedures such as cystoscopy and urodynamic studies or before replacement of a long-term urethral catheter, as the risk of bacteremia and sepsis with these interventions is low and clinical outcomes are not improved with prophylactic antimicrobial agents.^{5,171,201}

PROSTATITIS

The development of the National Institutes of Health classification of prostatitis syndromes (Table 36.8) and subsequent application of this classification in clinical trials and patient care have greatly advanced the understanding and appropriate management of this common problem.²⁰² Only acute or chronic bacterial prostatitis is considered attributable to infection and has indications for antimicrobial therapy.^{28,203,204}

Acute bacterial prostatitis is a severe infection, which is a urologic emergency.²⁸ This syndrome is usually community acquired, although healthcare-associated infection may occur after prostate biopsy.²⁰⁵ Affected men present with severe systemic manifestations including fever and marked urinary symptoms of dysuria and frequency. Urinary obstruction and intense suprapubic pain are often present. Bacteremia is reported in 27% of episodes. A digital rectal examination is not recommended because it may precipitate bacteremia.²⁸

E. coli is isolated in 70% of episodes. These strains are characterized by the presence of multiple virulence factors.²⁰⁶ *Proteus* species, *Klebsiella* species, *Enterococcus* species, *P.*

Table 36.8 National Institutes of Health Classification of Prostatitis Syndromes

Class	Description
I	Acute bacterial prostatitis
II	Chronic bacterial prostatitis
III	Chronic pelvic pain syndrome (CPPS)
IIIa	Inflammatory CPPS Leukocytes present in semen, in urine after prostate massage, or in expressed prostate secretions
IIIb	Noninflammatory CPPS Absence of leukocytes in specimens
IV	Asymptomatic inflammatory prostatitis Leukocytes in specimens similar to those in inflammatory CPPS, but no symptoms

Data from Krieger JN, Nyberg L, Nickel JC. NIH consensus definition and classification of prostatitis. *JAMA*. 1999;281:236–237.

aeruginosa, and *S. aureus* are each isolated in fewer than 10% of cases.²⁰⁵ Management includes bladder drainage by insertion of a urethral or suprapubic catheter, blood and urine cultures to characterize the infecting organism, and initiation of empirical parenteral antimicrobial therapy. Most antimicrobial agents are active in the acutely inflamed prostate. A combination of a β -lactam and an aminoglycoside is considered first-line therapy, although other broad-spectrum parenteral antibiotics, such as piperacillin/tazobactam and carbapenems, are also effective.²⁹ Once the infecting organism and susceptibility are confirmed and there has been an adequate clinical response to parenteral therapy, the antibiotic is modified to oral therapy to complete a 6-week course. A fluoroquinolone, either ciprofloxacin or levofloxacin, is recommended as oral therapy if the infecting organism is susceptible. When there is not a prompt clinical response following bladder drainage and initiation of effective antimicrobial therapy, CT or MRI is indicated to search for the uncommon complication of prostate abscess, which occurs in about 5% of cases. When an abscess is identified, transrectal ultrasonography-guided aspiration is usually effective for drainage. A small proportion of patients, 10% to 15%, develop chronic bacterial prostatitis following acute prostatitis.²⁰⁶

Chronic bacterial prostatitis occurs in men with persistent prostate infection.^{28,203,207–209} Bacteria enter the prostate from the urethra and persist because of limited antimicrobial diffusion or activity in the gland as well as the frequent presence of infected prostate stones in older men. A common clinical presentation is recurrent acute cystitis because bacteria in the prostate intermittently enter the bladder. The same organism is often repeatedly isolated, but the intervals between symptomatic episodes may last months or even years. Other symptoms are generally mild, such as irritative voiding symptoms and discomfort localized to the testicles, lower back, or perineum. Results of the prostate examination are usually normal, but tenderness may occasionally be elicited.

Chronic bacterial prostatitis is documented microbiologically in only 10% of men who present with the clinical syndrome of chronic prostatitis or chronic pelvic pain syndrome. The diagnosis requires paired cultures of midstream and post-prostatic massage urine specimens.²⁹ The midstream specimen confirms negative results of a urine culture, and the post-prostatic massage specimen reveals pyuria and organisms of presumed prostate origin.^{28,207} Gram-negative organisms, including Enterobacteriaceae and *P. aeruginosa*, and gram-positive *Enterococcus* species, *S. aureus*, coagulase-negative staphylococci, and group B streptococci are the bacteria most commonly isolated.²⁰⁶ Sexually transmitted organisms such as *Chlamydia trachomatis*, *U. urealyticum*, *Mycoplasma genitalium*, and *Trichomonas vaginalis* are uncommon and, when present, are usually identified in younger men.²⁹ The clinical relevance of post-prostatic massage cultures, however, remains controversial. In one study of 463 patients and 121 age-matched controls, 70% of patients were found to harbor at least one organism in a post-prostatic massage specimen, and uropathogens such as *E. coli* were isolated from 8% of patients and 8.3% of controls.²¹⁰ In another study, gram-positive organisms were isolated from 6% of 470 men after prostatic massage, but 97% of these organisms were not confirmed on second culture.²¹¹

Despite this uncertainty, chronic bacterial prostatitis does respond to appropriate antimicrobial treatment, although

relapse after treatment is common. Ciprofloxacin and levofloxacin are the first choices for antimicrobial treatment of chronic bacterial prostatitis when susceptible organisms are isolated. These agents penetrate well into the prostate and seminal fluid, and they remain active in the acidic environment of the prostate. Cure rates at 6 months after a 4-week course are 75% to 89%,^{29,203} although late relapses may still occur. Doxycycline and macrolides are considered second-line drugs but are preferred for gram-positive infections.²⁰³ Men who present for the first time with chronic pelvic pain syndrome and with evidence of inflammation (i.e., leukocytes) in expressed prostatic secretions, but in whom cultures are negative, should be prescribed a 4-week trial of antimicrobial therapy if they have not previously received a prolonged antimicrobial course.^{203,212} In reported case series, as many as 10% of men with such findings show response to antimicrobial therapy despite negative culture results; however, comparative clinical trials in treatment-naïve men with negative culture results have not been reported. If symptoms persist or recur after this 4-week antimicrobial trial, and if results of post-prostatic massage cultures remain negative, further antimicrobial therapy is not indicated.^{207,213,214}

URINARY TRACT INFECTION IN UNIQUE PATIENT POPULATIONS

RENAL TRANSPLANT RECIPIENTS

Urinary tract infection accounts for 45% to 60% of infections that occur in patients after renal transplantation.^{215–217} By 6 months after transplantation, 17% of recipients have experienced at least one urinary tract infection, and by 3 years, posttransplantation infection is reported in 60% of female recipients and 47% of male recipients.²¹⁸ The incidence is highest in the early posttransplant period, when immunosuppressive therapy is most intense and infection complicates surgical intervention and the use of urologic devices such as urinary catheters and ureteric stents.^{219–221} The clinical presentation is usually asymptomatic bacteriuria, but both cystitis and pyelonephritis are also common; 3% to 14% of episodes are associated with bacteremia.^{217,220,222} Early posttransplantation infections tend to be more severe and to manifest more frequently as pyelonephritis or with bacteremia.²²³

Risk factors for urinary tract infection are (1) patient specific, such as female gender, diabetes mellitus, pretransplantation urinary tract infections, prolonged prior dialysis, polycystic kidney disease, and diabetes; or (2) transplant procedure related, such as allograft trauma, microbial contamination of cadaveric kidneys, technical complications related to ureteral anastomosis, the presence of urinary catheters and ureteric stents, immunosuppression, reimplantation, and vesicoureteric reflux.²²¹ Independent risk factors for any symptomatic or asymptomatic infection in the first year after transplantation are reported to be older age, female gender, higher number of days of indwelling catheter after transplant, anatomic genitourinary abnormalities, and urinary infection within 1 month prior to the procedure.²²⁴ For acute pyelonephritis at any time after transplantation, independent risk factors are female gender, experiencing acute rejection episodes, higher number of urinary infections, and receiving

mycophenolate mofetil.²²⁵ Following transplantation, the risk for development of urinary tract infection correlates with the duration of perioperative catheterization or ureteral stent.^{216,226} The routine use of ureteric stents at transplantation decreases the overall transplantation complication rate, but stent implantation is associated with a small, but significant, increased risk of urinary tract infection.

Asymptomatic bacteriuria is associated with symptomatic urinary infection but is not a risk factor for graft loss or impaired graft function.¹⁵² Transplant recipients with asymptomatic bacteriuria who progress to graft failure usually also experience symptomatic infection. Graft failure in these patients is usually attributable to urologic abnormalities that promote infection rather than a direct consequence of infection. Symptomatic urinary tract infection has also not been independently associated with graft survival.^{222,227–230} However, case reports have described transplant recipients receiving stable immunosuppressive regimens who experience deterioration in graft function that is coincident with an episode of acute pyelonephritis.²³¹ This occurrence may be attributable to activation of the immune system by the infection.

The principles of management are similar to those for any patient with complicated urinary tract infection. They include prompt clinical diagnosis and initiation of antimicrobial therapy, obtaining appropriate specimens for culture, and urologic evaluation for underlying genitourinary abnormalities that may promote infection. Enterobacteriaceae, particularly *E. coli*, are the most common infecting organisms, but a variety of other bacteria or yeasts may be isolated. The choice of antimicrobial agent is determined by the susceptibility of the infecting organism and the patient's tolerance. A 1-week course of antimicrobial agent for cystitis and 2 weeks for pyelonephritis is recommended. The effect of type and intensity of immunosuppressive therapy on antimicrobial efficacy has not been reported. Bacteriuria does not compromise renal function^{193,228} and treatment of bacteriuria does not improve clinical outcomes, including the frequency of symptomatic episodes or graft function.^{191–193} Current guidelines therefore do not recommend screening for bacteriuria.²³² Treatment of asymptomatic candiduria in renal transplant recipients has also not been shown to be beneficial and is not recommended.²³³

Urinary tract infections after transplantation are limited by ensuring optimal surgical technique, which includes minimizing the perioperative duration of indwelling urinary devices. TMP/SMX prophylaxis given for the first 6 months after transplantation decreases the risk of both symptomatic and asymptomatic urinary tract infection, as well as other infections, but the effectiveness is compromised by increasing antimicrobial resistance.^{234,235} When urinary tract infection is diagnosed in patients receiving TMP/SMX prophylaxis, the organisms isolated are invariably resistant to TMP/SMX. Frequent recurrent symptomatic infection after transplantation may be attributable to an inadequate duration of antimicrobial treatment, resistance of the infecting organism, the presence of urologic abnormalities such as obstruction and stones, or infection of native kidneys. When infection recurs shortly after treatment, previous urine culture results should be reviewed to establish whether reinfection or relapse is occurring and to confirm that the infecting organism is susceptible to the antimicrobial therapy given. If relapse with

a susceptible organism occurs after a 2-week course of therapy, retreatment with a 4- to 6-week course is recommended, although the effectiveness of such prolonged retreatment has not been critically evaluated.²²² Repeated relapse after prolonged antimicrobial therapy in patients without identified urologic abnormalities is often caused by infection localized to the native kidneys. Such patients usually have a history of recurrent urinary tract infection before transplantation. The organisms frequently cannot be eradicated, presumably because of failure of antibiotics to achieve effective levels in the nonfunctioning kidney. Long-term suppressive therapy may be necessary to prevent further symptomatic episodes in these patients.

PERSONS WITH RENAL FAILURE

Urinary tract infections in patients with mild-to-moderate renal failure usually respond adequately to antimicrobial therapy.²³⁶ When renal impairment is severe, however, adequate urine or renal levels of antimicrobial agents may not be achieved because of limited kidney perfusion. If renal failure occurs with acute pyelonephritis, the response following initiation of antimicrobial therapy may be delayed or the risk of relapse after therapy may be increased. Systematic evaluation of antimicrobial efficacy for treatment of urinary tract infection in patients with renal failure has been limited.⁵ Aminoglycosides have little penetration into nonfunctioning kidneys and are not recommended for treatment. Fluoroquinolones, TMP/SMX, ampicillin, and cephalosporins have all been effective for treatment in individual case reports. Ertapenem therapy for patients with cystitis or catheter-acquired urinary tract infection is effective but associated with a more prolonged time to negative cultures for patients with impaired renal function.²³⁷ For patients with urinary infection with multiple drug-resistant organisms treated with fosfomycin, chronic kidney disease was associated with a significantly increased risk of treatment failure.¹⁶⁶

Another therapeutic problem arises when infection is localized to a unilateral nonfunctioning kidney. Antimicrobial therapy sterilizes the urine and often ameliorates symptoms because high urine levels are achieved by antimicrobial excretion through the functioning kidney. However, organisms persist in the kidney with impaired function because antimicrobial penetration to the site of infection is inadequate. When antimicrobial therapy is discontinued, a prompt relapse of infection occurs. Localization of infection to one kidney can be documented by culture of urine collected directly from the ureter following bladder irrigation with normal saline to remove infected bladder urine before ureteric catheterization. For symptomatic relapsing infection when a nonfunctioning kidney is known or suspected to be infected, management options include a trial of a more prolonged course of antimicrobial therapy, continuous suppressive therapy, and surgical removal of the nonfunctioning kidney.

PERSONS WITH URINARY STONES

Urinary tract infection complicates urolithiasis in several ways; infection may be the cause of stone formation, noninfected metabolic stones may become colonized with bacteria which persist in biofilm, and obstructing noninfected stones may precipitate infection proximal to the obstruction.²³⁸ In any

patient with urolithiasis and infection, the infection should be controlled with appropriate antimicrobial therapy before urologic manipulations.²³⁹

Infection stones, also called “struvite stones,” are a complication of infection with urease-producing organisms such as *P. mirabilis*.²⁴⁰ Urease catalyzes the hydrolysis of urea in the urine. This process produces ammonia and alkaline urine, favoring precipitation of magnesium ammonium phosphate, carbonate apatite, and monoammonium urate.²⁴¹ These crystals are incorporated into bacterial biofilm, creating the infection stone.¹⁴³ Infection stones continue to enlarge, sometimes rapidly, and ultimately cause obstruction and renal failure if not adequately treated. Patients at increased risk for development of infection stones are those with long-term indwelling catheters, urinary tract obstruction, neurogenic voiding dysfunction, distal renal tubular acidosis, and medullary sponge kidney (see also Chapter 38: Urinary Stone Disease).

Management of infection stones requires complete removal of the stone, together with sterilization of the urine.²³⁹ Percutaneous ultrasonic lithotripsy is effective in removing 60% to 90% of these stones, whereas extracorporeal shock wave lithotripsy is 30% to 60% effective. Multiple stone fragments are passed after lithotripsy. Antimicrobial therapy, selected on the basis of urine culture results, is continued until all fragments are passed. The optimal duration of antimicrobial therapy after lithotripsy is controversial. Currently, 4 weeks is recommended, although some writers suggest a shorter duration. When percutaneous lithotripsy is not effective or is contraindicated, open surgery for stone removal or nephrectomy may be necessary. For selected elderly or debilitated patients with complex urologic or medical problems, stone removal may not be possible. In such patients, continuous suppressive antimicrobial therapy is recommended to limit stone enlargement and preserve renal function.²⁴² The antimicrobial agent is selected on the basis of urine culture results, and therapy is continued indefinitely if effective and tolerated.

Bacteria may adhere to the surface of a stone that is not initially infected and may subsequently persist in biofilm.¹⁴³ Culture of a voided urine specimen in such cases often does not reflect the bacteriology of the stone. In a series of 75 patients with renal stones that were not infection stones, 36 (49%) of the stones were colonized, but only 19 (53%) of these patients also had bacteriuria.²⁴³ Organisms isolated from the colonized stones included *E. coli* (75%), *Enterococcus* species (100%), *P. aeruginosa* (19%), *Klebsiella* species (31%), *P. mirabilis* (8.3%), *Streptococcus* species (31%), *Citrobacter* species (8.3%), *S. saprophyticus* (19%), *M. Morganii* (8.3%), and *Gemella* species (8.3%). Larger stones were more likely to be colonized. Patients with stones colonized with bacteria are at increased risk for postprocedure sepsis, even when the voided urine specimen culture has negative results.²⁴⁴ Perioperative antimicrobial agents are recommended for all patients with stones who have indwelling catheters or stents, because they are at increased risk of stone colonization. Even with antimicrobial prophylaxis, 10% of patients undergoing percutaneous nephrolithotomy experience postoperative fever. Positive urine culture, diabetes, staghorn calculi, and preoperative nephrostomy are risk factors for postprocedure fever.²⁴⁵

A noninfected stone causing ureteric obstruction may be complicated by infection of undrained urine proximal to

the stone. Complete obstruction is associated with a high risk for bacteremia and sepsis, and prompt drainage is essential for these patients.²⁴⁶ When a ureter is obstructed, the microbiology of the voided urine specimen may not be consistent with that of urine sampled directly from the renal pelvis. Among patients with obstruction and a positive culture for urine sampled from the renal pelvis, the voided urine specimen culture had positive results in only 16% of cases in one report, and the organisms isolated were concordant between the two specimens in only 23% of patients.²⁴⁷ Thus a urine specimen should be obtained for culture by percutaneous aspiration of the renal pelvis in patients with obstruction, whenever possible.

OTHER INFECTIONS OF THE URINARY TRACT

RENAL AND PERINEPHRIC ABSCESES

Renal and perinephric abscesses are uncommon suppurative complications associated with substantial morbidity and mortality. Renal abscesses are located entirely within the renal parenchyma, whereas perinephric abscesses occupy the retroperitoneal fat and fascia surrounding the kidney. Abscesses may involve both the renal and perinephric tissues: 25% to 39% of abscesses are intranephric, 19% to 25% are both intranephric and perinephric, and 42% to 51% are perinephric alone.^{248,249} These abscesses develop following ascending urinary tract infection and pyelonephritis, or after hematogenous spread to the renal cortex or retroperitoneum following bacteremia from another site. Multiple bilateral cortical microabscesses suggest hematogenous spread. Complicating factors such as diabetes, urolithiasis, and obstruction are present in most cases. A population-based study from Taiwan reported an incidence of hospitalization for renal and perinephric abscess of 4.6/10,000 person-years for diabetic patients and 1.1/10,000 for nondiabetic patients.²⁵⁰

The organisms most commonly isolated from abscesses are *E. coli*, *K. pneumoniae*, *P. mirabilis*, *S. aureus*, and anaerobes. *S. aureus* abscesses are most likely to have originated with hematogenous spread from another site of infection. The clinical manifestations of both renal and perinephric abscesses mimic those of acute pyelonephritis. The characteristic findings are fever and costovertebral angle pain or tenderness. The clinical course in patients treated initially for pyelonephritis is characterized by delayed clinical response or early symptomatic relapse posttherapy. In a multivariable analysis, patients subsequently found to have renal abscess rather than pyelonephritis were more likely to have diabetes mellitus, and to present with hypotension, acute renal impairment, and a peripheral leukocyte (white blood cell [WBC]) count >20,000 WBC/mL.²⁵¹

CT is the preferred imaging modality for diagnosing the presence of an abscess; it also characterizes the abscess size and extent and may identify the potential source.¹²⁴ Ultrasonography identifies most perinephric abscesses, but may not be able to distinguish an inflammatory lobar mass from a true renal abscess within the kidney.¹²⁵

The management goals for renal and perinephric abscesses include prompt diagnosis, early institution of effective antimicrobial therapy, and, in selected patients, abscess

drainage for both therapeutic and diagnostic purposes. Culture of the abscess fluid identifies the specific infecting organisms and directs antimicrobial choice. If abscess fluid cannot be sampled for culture, organisms isolated from blood or urine culture should be considered in the selection of antimicrobial therapy. Initial antimicrobial administration is usually intravenous. Once the patient's condition is stabilized, therapy can be continued with an appropriate oral antimicrobial agent that has good bioavailability. Small renal abscesses (up to 5 cm in diameter) may be managed medically but larger abscesses usually require drainage.^{248,249,252–254} About 70% of renal abscesses resolve with medical therapy alone.²⁴⁹ Perinephric or mixed abscesses are larger, and drainage is usually required. The current initial approach for these larger abscesses is to attempt percutaneous drainage and, if this is not effective, to proceed to open drainage or nephrectomy. Resolution of the abscess is monitored by repeated imaging studies. Antimicrobial therapy is continued until the abscess has completely resolved or until there is only a residual, stable scar.

INFECTED RENAL CYSTS

The frequency of occurrence and risk factors for development of cyst infection are not well described. A French series reported that 8.4% of 389 patients with autosomal dominant polycystic kidney disease admitted to the nephrology department over a 10-year period had definite or likely cyst infection²⁵⁵; 24% of the episodes were associated with bacteremia, and two patients had more than one episode. Infected cysts complicating polycystic kidney disease may be difficult to diagnose both clinically and microbiologically.²⁵⁶ Fever and abdominal pain or tenderness, often with bacteremia, are the usual clinical manifestations. However, the differential diagnosis includes pyelonephritis, infected kidney stones, perinephric abscess, cyst hemorrhage, and other intra-abdominal pathology. Most patients with infected cysts also have peripheral leukocytosis and elevated C-reactive protein values. *E. coli*, *K. pneumoniae*, *Enterococcus* species, and group B streptococci are the most common infecting organisms, but *Salmonella* spp., *P. aeruginosa*, *Clostridium* species, *Candida* species, and *Aspergillus* species have also been reported. In view of the wide spectrum of potential pathogens, the specific infecting organism should be confirmed by cyst aspiration whenever possible. Imaging studies are usually necessary to confirm the diagnosis and identify the implicated cyst. Ultrasonography and CT are not useful.²⁵⁵ MRI and WBC-labeled scans^{257,258} were successful in localizing the infected cyst in individual case reports. Positron emission tomography with fludeoxyglucose F-18 is reported to be most effective in localizing the infected cyst, but access to this technology is limited.^{256,259–261}

Once the potential infected cyst is identified, a cyst aspiration, if possible, can confirm the presence of infection and provide therapeutic drainage.²⁶² Antimicrobial penetration into the cyst is presumed to be transepithelial. Effective concentrations of TMP/SMX, chloramphenicol, and fluoroquinolones are achieved in the cyst.²³⁶ Cyst and serum levels of levofloxacin are similar, whereas cyst levels reported with ciprofloxacin are only 40% of serum levels.²⁶³ Therapeutic cyst antimicrobial levels are also achieved with penicillins, cephalosporins, aminoglycosides,²³⁶ and amphotericin B.²⁶⁴

Prolonged antimicrobial therapy, for at least 4 weeks, is recommended, although clinical trials defining the optimal duration of therapy have not been reported. In the French case series described previously, the mean duration of antibiotic therapy was 5 weeks, and the initial course of therapy was successful for 71% of cases.²⁵⁴ Nephrectomy is occasionally necessary when cyst drainage and appropriate antimicrobial therapy are not successful in controlling the infection (see also Chapter 45: Cystic Diseases of the Kidney).

EMPHYSEMATOUS CYSTITIS AND PYELONEPHRITIS

Emphysematous cystitis and pyelonephritis are acute necrotizing infections characterized by gas formation. Gas within the urinary collecting system can be seen after many interventional procedures, but emphysematous infection is characterized by gas in the tissues.¹²⁵ Gas is localized to the bladder wall and lumen in cystitis²⁶⁵ and in and around the kidney in pyelonephritis.²⁶⁶ Emphysematous pyelitis is a gas-forming infection restricted to the collecting system; the renal parenchyma is spared. Affected patients usually have diabetes with poor glucose control. Obstruction is another common predisposing factor for emphysematous pyelonephritis. *E. coli* and *K. pneumoniae* are the organisms most commonly isolated. High levels of glucose in the urine serve as a substrate for these bacteria, and large amounts of gas are generated through natural fermentation. In affected patients who are not diabetic, protein fermentation is a proposed source of gas formation.²⁶⁵ CT is considered the optimal imaging technique for confirming emphysematous infection and characterizing the extent of involvement.^{124,125}

A review of 135 cases of emphysematous cystitis identified over a period of 50 years reported that the median patient age was 66 years (range, <1 to 90 years); 64% of patients were women, and 67% had diabetes.²⁶⁵ Presenting symptoms ranged from pneumaturia alone through irritative lower tract voiding symptoms to severe illness suggestive of acute abdominal disease or sepsis; 7% of cases were asymptomatic. *E. coli* was isolated from 58%, *K. pneumoniae* from 21%, and *Clostridium* species and *Enterobacter aerogenes* each from 7% of cases. The diagnosis was apparent on a plain radiograph of the abdomen in 84% of cases. Medical management—including antimicrobial therapy, bladder drainage, and glycemic control—was usually effective. Surgical intervention was required for only 10% of cases and included cystectomy, partial cystectomy, and surgical debridement when concomitant emphysematous pyelonephritis was present. The overall mortality rate was only 7%.

Emphysematous pyelonephritis is a more serious infection with a substantially higher mortality rate.²⁶⁷ In case series of patients with emphysematous pyelonephritis, 62% to 100% were diabetic, and the majority were women.^{268–270} Serum glucose levels are often very high in the diabetic patients, and serum C-reactive protein was generally elevated. *E. coli* was isolated from 45% to 54% of episodes, and bacteremia occurred in 20% to 50% of subjects in whom blood cultures were performed. Plain abdominal radiographs identified gas in the kidney in only 50% of cases. Ultrasonography of the kidneys usually showed some abnormalities but was less accurate than CT in identifying emphysematous infection. Management includes antimicrobial therapy together with

percutaneous or open drainage of abscesses and correction of obstruction.^{266,267,269–271} Aggressive glucose control and supportive care are also necessary. The mortality rate ranges from 7% to 20% in patients with less severe manifestations to 70% in patients with a fulminant course characterized by necrosis, intravascular thrombosis, and microabscess formation.

Emergency nephrectomy was previously considered necessary for any patient presenting with emphysematous pyelonephritis. Currently, percutaneous drainage is the recommended initial approach. This is associated with similar or lower mortality rates than either emergency nephrectomy or medical management alone.^{267,271–274} In a retrospective case series describing a relatively small number of patients, the mortality rates were 50% with medical management alone, 25% with medical management and emergency nephrectomy, and 13.5% with medical management combined with initial percutaneous drainage.²⁶⁸ Delayed elective nephrectomy may subsequently be required for some patients.

XANTHOGRANULOMATOUS PYELONEPHRITIS

Xanthogranulomatous pyelonephritis is an uncommon, severe, subacute, or chronic suppurative process characterized by destruction and replacement of the renal parenchyma by granulomatous tissue containing histocytes and foamy cells.^{275,276} The inflammatory process may extend into perinephric structures such as Gerota fascia, the posterior perirenal space, the psoas muscle, the diaphragm, and the spleen. Less than 1% to 8% of kidneys removed or subjected to biopsy for inflammatory conditions are reported to show evidence of xanthogranulomatous pyelonephritis.²⁷⁶ The pathogenesis of this process is unknown, but potential contributing factors include chronic urinary tract infection, abnormal lipid metabolism, lymphatic obstruction, impaired leukocyte function, and vascular occlusion. Urine culture results are positive in 62% to 89% of cases. The organisms most commonly isolated from renal tissue are *P. mirabilis* (38%), *E. coli* (33%), *Klebsiella/Enterobacter* species (8%), *P. aeruginosa* (8%), and *S. aureus* (10%).

In a single-center experience, 35 (85%) of 41 cases presenting from 1994 to 2005 were women.²⁷⁷ Xanthogranulomatous pyelonephritis was responsible for 19% of 214 nephrectomies performed at this facility during the review period. The most common presenting symptoms were fever, flank or abdominal pain, weight loss, lower urinary tract symptoms, and gross hematuria. All patients had renal calculi. Of 17 cases in which urine culture was obtained, *E. coli* was isolated in 35% and *P. mirabilis* in 18%; in 35% there was no growth. A review from Greece of 39 cases occurring between 1980 and 1999 reported a female/male ratio of 2:1.²⁷⁵ The presenting history included urolithiasis or renal colic, recurrent urinary tract infections, and previous urologic procedures. All patients were symptomatic at presentation; symptoms included complaints of fever, flank or abdominal pain, chills, and malaise. Anorexia, weight loss, lower tract symptoms, and gross hematuria were also reported. *E. coli* was isolated in 15 cases and *P. mirabilis* in 12. Plain radiographs or intravenous pyelography (IVP) showed a nonfunctioning kidney in 63% of cases, single or multiple calculi in 52%, staghorn calculus in 48%, calyceal deformity in 26%, and hydronephrosis in 23%. In 35 New Zealand patients presenting from 2001 to 2013,²⁷⁸ 91% were female, staghorn calculi occurred in 51%,

and obstructing ureteric calculi in 23%. The most common organisms isolated were *E. coli* (46%) and *P. mirabilis* (20%).

CT is considered the optimal imaging modality; it identifies the abnormality in 74% to 90% of cases.^{124,125,279} Because of the current widespread access to CT, the diagnosis is now usually made preoperatively. Characteristic findings include an enlarged kidney, frequently with replacement of renal parenchyma and multiple fluid-filled cavities, together with urolithiasis. Ultrasonography identifies nonspecific abnormalities, including renal enlargement with relative preservation of renal contour and multiple hypoechoic round masses.¹²⁵ MRI shows abnormalities similar to those observed with CT.²⁷⁹ The differential diagnosis includes malignancy and tuberculosis. The usual management is nephrectomy; antimicrobial therapy has only a secondary role.²⁷⁶ If the diagnosis is made early, when there is only focal renal involvement, partial nephrectomy may be curative.

PYOCYSTIS

Pyocystis, also called “vesicle empyema,” is an infection characterized by a purulent fluid collection in the bladder of a patient with a nonfunctioning bladder. In effect, the bladder becomes an undrained abscess. This is a rare complication diagnosed in patients with anuric renal failure or surgically bypassed bladders. The clinical presentation includes suprapubic pain or distention, abdominal pain, foul-smelling urethral discharge, and fever or sepsis.^{280,281} Organisms isolated have included *E. coli*, *P. mirabilis*, *P. aeruginosa*, *Serratia marcescens*, *Streptococcus* species, *Enterococcus* species, and *Candida* species. Mixed cultures are common. It is not known whether anaerobic organisms may also contribute.

When the condition is suspected, a specimen of bladder fluid should be obtained for diagnosis and culture. The laboratory should be requested to identify all organisms isolated, so that the specimen is processed as abscess fluid rather than urine. The treatment approach involves systemic antimicrobial agents and urethral catheterization for bladder drainage. Antimicrobial therapy is directed by the specific organisms isolated. Bladder irrigation with either saline or an antibiotic solution is sometimes recommended, but whether irrigation provides an additional therapeutic benefit is not clear.²⁸² Surgical intervention to achieve adequate drainage is necessary, rarely, for recalcitrant cases.

UNCOMMON ORGANISMS

GENITOURINARY TUBERCULOSIS

Genitourinary tuberculosis is diagnosed in 1.1% to 1.5% of all tuberculosis cases, and 5% to 6% of cases of extrapulmonary tuberculosis.^{283–285} This infection is usually a consequence of local reactivation following hematogenous dissemination of *Mycobacterium tuberculosis* to the renal cortex during primary pulmonary infection. The renal cortex is also frequently involved with miliary disease, with multiple granulomas usually present. The high oxygen tension of the renal cortex is favorable for renal localization. Men are infected twice as often as women. The latent period, from the time of initial pulmonary infection to diagnosis of clinical urogenital tuberculosis, is 22 years on average with a range of 1 to 46

years.²⁸⁶ Reactivation of tuberculosis usually occurs in only one kidney; thus disease is characteristically unilateral. Contiguous involvement of the collecting system leads to *M. tuberculosis* bacilluria with subsequent ureteric and bladder infection. Prostate and epididymis infection may result directly from hematogenous dissemination rather than contiguous spread. The kidneys are affected in 60% to 100% of cases, the ureters in 19% to 41%, the bladder in 15% to 20%, and the prostate or epididymis in 20% to 50% of men.²⁸⁶⁻²⁸⁸

Substantial morbidity is attributed to urogenital tuberculosis. Severe calyceal clubbing and dilation of the renal pelvis and ureters leading to total destruction of the kidney and autonephrectomy occur in 23% to 33% of cases, and renal failure occurs in 1% to 10%.²⁸⁶ A summary of cases reported worldwide reported that 27% of patients had a nonfunctioning unilateral kidney at presentation, but this proportion varied from 8% to 72% in different countries, presumably reflecting the timeliness of diagnosis.²⁸⁹

Presenting genitourinary symptoms are often vague or nonspecific and may include back or flank pain, dysuria, and urinary frequency and, in women, infertility. As many as 50% of patients have no localizing genitourinary symptoms. About 25% to 33% of patients have systemic complaints, usually pulmonary symptoms, fever, and weight loss.^{286,290} The subacute presentation and results of the initial evaluation can mimic those of xanthogranulomatous pyelonephritis. In men, additional features suggestive of tuberculosis include an enlarged, hard, and nontender epididymis; thickened or beaded vas deferens; indurated or nodular prostate; and nontender testicular mass.²⁸⁴ Tuberculous granulomatous prostatitis may manifest as a nodular prostate with an elevation

of prostate-specific antigen that is clinically indistinguishable from that in prostate carcinoma. Most patients with renal tuberculosis have evidence of concomitant extragenital disease. The most common site is the lungs, but the pulmonary disease is usually inactive. The chest radiograph is abnormal in 67% to 75% of patients, and the tuberculin skin test positive in 60% to 90%.

Urinalysis findings are abnormal in >90% of patients with genitourinary tuberculosis.²⁸³ Sterile pyuria with hematuria is most common, but sterile pyuria alone and gross or microscopic hematuria may also occur.^{283,287,288} For patients with HIV infection, an algorithm that incorporates negative results of routine urine culture and the presence of pyuria, albuminuria, or hematuria has good predictive value for detecting genitourinary tuberculosis.^{291,292} As many as half of HIV-positive patients, however, have concomitant bacteriuria with other organisms at presentation, and this finding may obscure the initial assessment of the urinalysis.

IVP was traditionally the standard imaging approach, but CT is now preferred.²⁸⁶ Imaging studies identify unilateral disease in 75% of cases of renal tuberculosis (Fig. 36.1). The characteristic early finding is erosions of the renal calyx; the erosions subsequently progress to papillary necrosis, hydro-nephrosis, renal parenchymal cavitation, and dilated calyces. Ureteric tuberculosis is characterized by a thickened ureteric wall and strictures. Lesions are most common in the distal third of the ureter. Bladder tuberculosis may manifest as reduced bladder volume with wall thickening, ulceration, and filling defects resulting from granulomatous involvement. In advanced disease, scarring results in permanent loss of volume and a residual small, irregular, calcified bladder. The

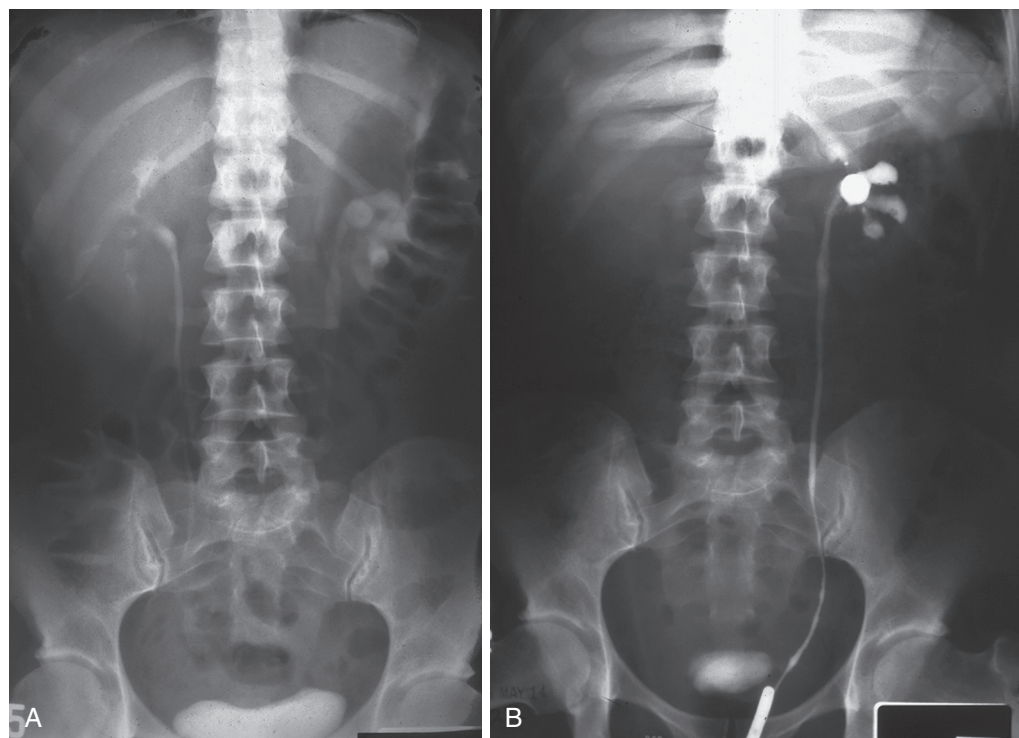


Fig. 36.1 Renal tuberculosis. (A) Intravenous pyelogram, showing unilateral hydronephrosis and calyceal distention. (B) Retrograde pyelogram in the same patient, showing distal ureteric narrowing.

most common finding on CT is renal calcification, present in 50% of cases.²⁹³ Other characteristic findings are hydrocalyx secondary to infundibular stenosis and cavity formation. In advanced disease, cortical loss, uroepithelial thickening, and dystrophic calcification are present.

The diagnosis is confirmed by growth of *M. tuberculosis* in urine or tissue culture. Appropriate specimens for mycobacterial culture should be obtained whenever this diagnosis is considered. Three sequential early morning urine specimens for mycobacterial culture are recommended. Urine cultures have positive results in 75% to 90% of affected patients. A positive acid-fast bacillus smear of urine usually signifies *M. tuberculosis*, but culture confirmation is essential to rule out colonization with nonpathogenic mycobacteria and for susceptibility testing of the infecting strain.^{294,295} If a suggestive renal abnormality is present but urine culture results are negative, tissue biopsy may be necessary to confirm the diagnosis. Polymerase chain reaction (PCR) nucleic acid antigen testing of urine specimens has been reported to be more sensitive than culture and, if available, provides a more rapid diagnosis.^{285,292} However, culture is still necessary to isolate the organism for susceptibility testing.

Antimycobacterial treatment should be provided under the supervision of a physician with expertise in tuberculosis management. The treatment of genitourinary tuberculosis is similar to that of extrapulmonary tuberculosis at other sites. The initial regimen consists of four drugs (isoniazid, rifampin, pyrazinamide, and ethambutol) for 2 months, followed by two drugs (isoniazid and rifampin) for 4 months if the isolate is susceptible to first-line therapy. Follow-up IVP every 6 months has been recommended to identify ureteral scarring or obstruction, which may develop or progress during therapy as part of the healing process. Corticosteroid therapy does not prevent this complication.²⁸³ Ureteral reimplantation, endoscopic balloon dilation, or implantation of ureteral stents may be necessary if progressive obstruction develops.²⁹⁶ Nephrectomy is rarely required but may be indicated for intractable pain, untreatable infection proximal to a stricture, uncontrollable hematuria or hypertension, or drug resistance. Bladder augmentation surgery may be required if the bladder is scarred and contracted following treated tuberculosis infection.

BACILLUS CALMETTE–GUÉRIN INFECTION

Intravesical vaccine instillation of bacillus Calmette–Guérin (BCG) is considered first-line treatment for superficial bladder tumors and carcinoma in situ. Treatment with this biologic therapy is complicated by systemic or local BCG infection in <5% of cases.^{297–299} BCG bladder instillation is followed by local irritative symptoms such as dysuria in 80% of cases, but these symptoms do not usually persist beyond 48 hours.³⁰⁰ If symptoms do persist, isoniazid for 14 days is recommended. When symptoms continue despite isoniazid therapy, a full course of antituberculous therapy is recommended.³⁰¹ Less common genitourinary infections that accompany BCG instillation include prostatitis in 1% to 3% of patients, epididymitis in 0.2%, and, in rare cases, testicular abscesses, bladder ulcers, local skin infections, or renal infection. Reflux may be a risk factor for the rare complication of BCG pyelonephritis.³⁰² Localized genitourinary infection tends to have a delayed onset and is usually not apparent clinically

until more than 3 months after BCG treatment.^{296,297} BCG may not be isolated from urine or tissue cultures.²⁹⁷ Tissue biopsy usually shows necrotizing granulomatous inflammation, and this finding in the context of prior BCG therapy is sufficient for diagnosis even if subsequent culture results are negative.²⁹⁹

FUNGAL URINARY TRACT INFECTION

Fungal urinary tract infection is usually caused by *Candida* species. These organisms have an extensive array of virulence factors that may facilitate successful colonization and invasion of the urinary tract.³⁰³ Infection may be either antegrade, following candidemia, or retrograde via the urethra.³⁰³ Clinical manifestations are asymptomatic candiduria, cystitis, and pyelonephritis. Bladder or renal fungus balls and systemic fungemia are rare complications.³⁰⁴

Candida albicans is isolated from more than 50% of episodes, followed by *Candida glabrata*, *Candida tropicalis*, and *Candida parapsilosis*.^{305,306} Candiduria is usually identified in patients who are seriously ill with multiple comorbid conditions, and most infections are asymptomatic.³⁰⁴ The most important risk factors for candiduria are the presence of an indwelling catheter or other indwelling urologic device, diabetes mellitus, and exposure to broad-spectrum antimicrobial agents.³⁰⁶ Treatment of asymptomatic candiduria is not beneficial and is currently recommended only for selected patients with high-risk neutropenia (<100 WBC/mm³) or before a traumatic urologic procedure.^{307–309} Indwelling devices should be removed whenever possible to facilitate resolution of candiduria. Imaging studies to exclude fungus balls should be considered in patients who have recurrent symptomatic infection or urinary obstruction.³¹⁰ When fungus balls are present, surgical removal is required for cure of the infection.

Fluconazole is the treatment of choice for *Candida* urinary tract infection because it is excreted in the urine in active form and high urinary levels are achieved.^{308,309} A 2-week course of fluconazole, 200 to 400 mg daily, is recommended (Table 36.9). Amphotericin B deoxycholate, 0.3 to 0.6 mg/

Table 36.9 Treatment of Candiduria

	Cystitis	Pyelonephritis
Fluconazole (if isolate is susceptible)	200–400 mg daily, 14 days	200–400 mg daily, 14 days
Amphotericin B deoxycholate	0.3–0.6 mg/kg daily, 1–7 days	0.5–0.7 mg/kg daily, 14 days
Amphotericin B bladder irrigation	5–50 mg/L continuous irrigation for 2–7 days	Not indicated
Flucytosine	25 mg/kg qid, 7–10 days	25 mg/kg qid, 14 days ^a

^aMay be used in combination with amphotericin B.

qid, Four times a day.

From Pappas PG, Kauffman CA, Andes D, et al. Clinical practice guidelines for the management of candidiasis 2009: update by the Infectious Diseases Society of America. *Clin Infect Dis*. 2009;48:503–535.

kg daily, is the alternative treatment recommended for fluconazole-resistant strains, including most *C. glabrata* organisms, as well as in patients who are allergic to fluconazole or in whom treatment fails despite optimal fluconazole therapy and urologic management. Treatment duration with amphotericin B is 1 to 7 days for cystitis and 2 weeks for pyelonephritis. Amphotericin B may also be effective for treatment of fungal cystitis when administered as a bladder washout, but this approach is not generally recommended because it requires several days of urethral catheterization and the optimal dose, frequency, and duration are not well established.³¹¹ With the lipid formulations of amphotericin B, concentrations of active drug achieved in renal tissue and urine are very low, so these formulations are not recommended.

The antifungal 5-flucytosine is excreted in the urine and is indicated for treatment of *Candida* urinary tract infection as a single agent or in combination with amphotericin B for renal infection.^{308,309} Resistance to 5-flucytosine develops rapidly when this drug is used as a single agent, and the frequent adverse effects of bone marrow suppression and enterocolitis also limit its use, particularly in patients with renal failure. Echinocandins such as caspofungin, micafungin, and anidulafungin and other azoles such as itraconazole, voriconazole, and posaconazole, which are not excreted into the urine, are not recommended for treatment of urinary tract infection.^{308,309} There are, however, case reports of successful treatment of *C. glabrata* urinary tract infection with caspofungin³¹² and a case series of selected patients treated with micafungin reported a success rate of 75%.³¹³

VIRAL INFECTIONS

Viral urinary tract infections are uncommon in adults and occur largely in immunocompromised patients.^{314–317} Clinical manifestations of viral infection generally follow reactivation of latent infection in immunocompromised patients, although de novo infection may occur. The usual clinical manifestation is hemorrhagic cystitis, but nephropathy has also been described.^{314,318–320} The most common viruses are adenovirus, parvovirus B19, and cytomegalovirus (CMV). More than one viral infection may coexist. Most adult cases occur in recipients of hematopoietic stem cell transplants, particularly those with severe graft-versus-host disease, and in renal transplant recipients. Infection has also been reported in other immunosuppressed patients, such as HIV-infected patients with low CD4⁺ cell counts.^{316,321} HIV-associated nephropathy as a distinct clinical entity, including consideration of the role of host genetic background, glomerular cell viral entry, and replication, is considered in Chapter 32. Management of parvovirus B19 infection in the transplanted kidney is discussed in Chapter 70. Adenovirus or CMV infection is diagnosed by viral culture or PCR of the urine. Management includes minimization of immunosuppressive therapy if possible. For HIV-infected patients, antiretroviral therapy to increase the CD4⁺ cell count should be initiated. CMV infection responds to treatment with ganciclovir or foscarnet.³¹⁴ Treatment of adenovirus infection uses cidofovir, although some efficacy has been reported with vidarabine.³¹⁴ There are also case reports of successful treatment of adenovirus with ganciclovir and ribavirin.^{314,322}

PARASITIC INFESTATIONS OF THE URINARY TRACT

The most common and important parasitic infestation of the urinary tract is with *Schistosoma haematobium*.^{323–325} This parasite is acquired after exposure to contaminated water. *Schistosoma* larvae penetrate the intact skin and migrate in the blood to the liver, where they transform into young worms (schistosomulae) that mature in 4 to 6 weeks and then further migrate to the perivesical venules. The life span of the adult worm in the venule is usually 3 to 5 years but can be longer. Most eggs produced by the adult worms enter the bladder lumen and are removed in the urine. However, some eggs are retained locally in the bladder wall, where they incite an eosinophilic inflammatory and granulomatous immune response that causes progressive fibrosis. The functional abnormality early in the disease is obstruction of the bladder neck. Late complications include recurrent bacterial urinary tract infection, bladder or ureteric stone formation, renal functional abnormalities, and, ultimately, kidney failure.^{324,326} *Schistosoma* infestation is also a risk factor for squamous cell carcinoma of the bladder.³²⁷ The relative risk of cancer for persons with schistosomiasis varies from 1.8 to 23.5; the incidence is highest in the 30- to 50-year age group.³²⁸

The prevalence of infection in endemic areas is high. Surveys from rural Zimbabwe revealed that 60% of women younger than 20 years of age and 29% of those aged 45 to 49 years have eggs in the urine; HIV-infected women older than 35 years had a significantly higher prevalence of infestation.³²⁹ Travelers to endemic areas may acquire infection with only minimal exposure to contaminated water^{330–332} (see also Chapters 76–78 for discussion of schistosomiasis in Latin America, Africa, and the Near and Middle East).

Acute genitourinary symptoms occur in up to 50% of cases following acquisition of infection and include hematuria, which is often terminal, together with dysuria and urinary frequency. Microhematuria is reported in 41% to 100% and gross hematuria in 0% to 97% of patients with chronic schistosomal infestation. Radiologic abnormalities are present in the upper urinary tract in 2% to 62% of chronic cases.³²⁴ Ultrasonography of the urinary tract demonstrates thickening of the bladder wall, granulomatous changes, hydronephrosis, and, on occasion, bladder or ureteric calcification. The diagnosis is established by identification of parasite eggs in the urine or biopsy specimens or by serologic findings.³²⁵ Urine specimens collected for identification of eggs should be obtained on consecutive days between 1100 and 1300 hours because egg passage is maximal at this time. Sedimentation or filtration of the urine before examination increases the sensitivity of egg detection.^{324,333}

Treatment with one dose of praziquantel, 40 mg/kg body weight, will cure 80% of cases.³²⁵ Follow-up urine specimens for parasite examination are recommended 3 months after treatment to identify patients in whom treatment has failed and must be repeated. Bladder wall thickening and hydro-ureters may be reversed in the majority of affected patients if treatment is given early in the course of infestation.³³⁴ However, when chronic disease is established and fibrotic lesions are present, changes may not be reversible, and corrective surgery or management of end-stage renal disease is required.³³³

The protozoal parasite *T. vaginalis* is commonly transmitted sexually and is occasionally identified on microscopy with routine urinalysis. In women, the parasite may originate from contamination of the urine by vaginal secretions, but the organism is a well-described cause of urethritis for both men and women. Whenever *T. vaginalis* is identified, treatment of the patient and his or her sexual partners is indicated, regardless of symptoms. The recommended treatment is a single dose of metronidazole, 2 g, or tinidazole, 2 g.³³⁵

Echinococcus granulosus infestation occasionally involves the kidneys.³²⁰ Renal cysts are reported in 2% to 3% of cases of hydatid disease.^{336,337} The diagnosis is usually made after an incidental finding of a cyst in the kidneys, ureters, bladder, or testes by imaging or in the investigation of nonspecific symptoms.³³⁸ On occasion, flank pain or a mass is present. Hydatid cysts are not excreted in the urine. Treatment consists of surgical cyst removal or marsupialization; nephrectomy is occasionally necessary.³³⁹ Perioperative albendazole therapy is also usually recommended for patients with hydatid disease. A less common helminthic infestation is *Wuchereria bancrofti* (filariasis), which may cause lymphatic obstruction and rupture into the urinary collecting system, producing chyluria.³²³

 Complete reference list available at [ExpertConsult.com](https://www.expertconsult.com).

KEY REFERENCES

- Hooton TM. Uncomplicated urinary tract infection. *N Engl J Med*. 2012;366:1028–1037.
- Grigoryan L, Trautner BW, Gupta K. Diagnosis and management of urinary tract infections in the outpatient setting. *JAMA*. 2014;312:1677–1684.
- Abraham SN, Miao Y. The nature of immune responses to urinary tract infections. *Nat Rev Immunol*. 2015;15:655–663.
- Ragnarsdottir B, Svanborg C. Susceptibility to acute pyelonephritis or asymptomatic bacteriuria: host-pathogen interaction in urinary tract infections. *Pediatr Nephrol*. 2012;27:2017–2029.
- High KP, Bradley SF, Gravenstein S, et al. Clinical practice guideline for the evaluation of fever and infection in older adult residents of long term care facilities. *Clin Infect Dis*. 2009;48:149–171.
- Lipsky BA, Byren I, Hoey CT. Treatment of bacterial prostatitis. *Clin Infect Dis*. 2010;50:1641–1652.
- Bermingham SL, Ashe JF. Systematic review of the impact of urinary tract infections on health-related quality of life. *BJU Int*. 2012;110:e830–e836.
- Hooton TM, Roberts PL, Cox ME, et al. Voided midstream urine culture and acute cystitis in premenopausal women. *N Engl J Med*. 1983;369:1891–1913.
- Perrotta C, Aznar M, Mejia R, et al. Oestrogens for preventing recurrent urinary tract infection in postmenopausal women. *Cochrane Database Syst Rev*. 2008;(2):CD005131.
- Bent S, Saint S. The optimal use of diagnostic testing in women with acute uncomplicated cystitis. *Am J Med*. 2002;113(suppl 1A):20S–28S.
- Gupta K, Hooton TM, Naver KG, et al. International clinical practice guidelines for the treatment of acute uncomplicated cystitis and pyelonephritis in women: a 2010 update by the Infectious Diseases Society of America and the European Society for Microbiology and Infectious Diseases. *Clin Infect Dis*. 2011;52:e103–e120.
- Falagas ME, Rafailidis PI. Fosfomycin: the current status of the drug. *Clin Infect Dis*. 2015;61:114–1146.
- Hutner A, Verhaegh EM, Harbarth S, et al. Nitrofurantoin revisited: a systematic review and analysis of controlled trials. *J Antimicrob Chemother*. 2015;70:2456–2464.
- Beerepoot MAJ, Geerlings SE, van Haarst EP, et al. Nonantibiotic prophylaxis for recurrent urinary tract infections: a systematic review and meta-analysis of randomized controlled trials. *J Urol*. 2013;190:1981–1989.
- Jepson RG, Craig JC. Cranberries for preventing urinary tract infections. *Cochrane Database Syst Rev*. 2008;(1):CD001321.
- O'Brien VP, Hannan TJ, Nielsen HV, et al. Drug and vaccine development for the treatment and prevention of urinary tract infections. *Microbiol Spectr*. 2016;4:1–42.
- Luthje P, Brauner A. Novel strategies in the prevention and treatment of urinary tract infections. *Pathogens*. 2016;5(13):1–14.
- Demetris J, Menias CD. State of the art: imaging of renal infections. *Emerg Radiol*. 2007;14:13–22.
- Sandberg T, Skoog G, Hermansson AB, et al. Ciprofloxacin for 7 days versus 14 days in women with acute pyelonephritis: a randomized, open-label and double-blind, placebo controlled non-inferiority trial. *Lancet*. 2012;380:484–490.
- D'Hondt F, Everaert K. Urinary tract infections in patients with spinal cord injuries. *Curr Infect Dis Rep*. 2011;13:544–551.
- Marcus RJ, Post JC, Stoodley P, et al. Biofilms in nephrology. *Expert Opin Biol Ther*. 2008;8:1159–1166.
- Nicolic LE, Bradley S, Colgan R, et al. Infectious Diseases Society of America guidelines for the diagnosis and treatment of asymptomatic bacteriuria in adults. *Clin Infect Dis*. 2005;40:643–654.
- Nicolic LE. Urinary tract infection. *Crit Care Clin*. 2013;3:699–715.
- Mody L, Juthani-Mehta M. Urinary tract infections in older women: a clinical review. *JAMA*. 2014;311:844–854.
- Wullt B, Agace W, Mansson W. Bladder, bowel and bugs—bacteriuria in patients with intestinal urinary diversion. *World J Urol*. 2004;22:186–195.
- Hooton TM, Bradley SF, Cardenas DD, et al. Diagnosis, prevention, and treatment of catheter-associated urinary tract infections in adults: 2009 international clinical practice guidelines from the Infectious Diseases Society of America. *Clin Infect Dis*. 2010;50(5):625–663.
- Gould CV, Umscheid CA, Agarwal RK, et al. Health Care Infection Control Practices Advisory Committee: guidelines for prevention of catheter-associated urinary tract infections 2009. *Infect Control Hosp Epidemiol*. 2010;31(4):319–326.
- Siddiq DM, Darouiche RO. New strategies to prevent catheter-associated urinary tract infections. *Nat Rev Urol*. 2012;9:305–314.
- Green H, Rahamimov R, Golbert E, et al. Consequences of treated versus untreated asymptomatic bacteriuria in the first year following kidney transplantation: retrospective observational study. *Eur J Clin Microbiol Infect Dis*. 2012;32:127–131.
- Origen J, Lopez-Medrano F, Fernandez-Ruiz M, et al. Should asymptomatic bacteriuria be systematically treated in kidney transplant recipients? Results from a randomized controlled trial. *Am J Transplant*. 2016;16(10):2943–2953.
- Widmer M, Gulmezoglu AM, Mignini L, et al. Duration of treatment of asymptomatic bacteriuria during pregnancy. *Cochrane Database Syst Rev*. 2011;(12):CD000491.
- Schaeffer AJ, Nicolic LE. Urinary tract infections in older men. *N Engl J Med*. 2016;374:562–571.
- Rees J, Abrahams M, Doble A, et al. Diagnosis and treatment of chronic bacterial prostatitis and chronic prostatitis/chronic pelvic pain syndrome: a consensus guideline. *BJU Int*. 2015;116:509–525.
- Gilbert DN. Urinary tract infections in patients with chronic renal failure. *Clin J Am Soc Nephrol*. 2006;1:327–331.
- Flannigan R, Choy WH, Chew B, et al. Renal struvite stones—pathogenesis, microbiology, and management strategies. *Nat Rev Urol*. 2014;11:333–341.
- Coelho RF, Schneider-Montero ED, Mesquita JLB, et al. Renal and perinephric abscess. Analysis of 65 consecutive cases. *World J Surg*. 2007;31:431–436.
- Hung C-H, Liou J-D, Yan M-Y, et al. Immediate percutaneous drainage compared with surgical drainage of renal abscess. *Int Urol Nephrol*. 2007;39:51–55.
- Saller M, Rafat C, Zahar J-R, et al. Cyst infections in patients with autosomal dominant polycystic kidney disease. *Clin J Am Soc Nephrol*. 2009;4:1183–1189.
- Jouret F, Lhommel R, Devuyt O, et al. Diagnosis of cyst infection in patients with autosomal dominant polycystic kidney disease: attributes and limitations of the current modalities. *Nephrol Dial Transplant*. 2012;27:3746–3756.
- Thomas AA, Lane BR, Thomas AZ, et al. Emphysematous cystitis: a review of 135 cases. *BJU Int*. 2007;100:17–20.
- Pontin AR, Barnes RD. Current management of emphysematous pyelonephritis. *Nat Rev Urol*. 2009;6:272–279.
- Bjurlin MA, Hurley SD, Kim DY, et al. Clinical outcomes of nonoperative management in emphysematous urinary tract infections. *Urol*. 2012;79:1281–1285.

286. Figuerido AA, Lucon AM. Urogenital tuberculosis: update and review of 8961 cases from the world literature. *Rev Urol.* 2008;10:207–217.
299. Perez-Jacoiste Asin MA, Fernandez-Ruiz M, Lopez-Medrano F, et al. Bacillus Calmette-Guerin (BCG) infection following intravesical BCG administration as adjunctive therapy for bladder cancer. *Medicine (Baltimore).* 2014;93:236–254.
304. Kauffman CA, Fisher JF, Sobel JD, et al. *Candida* urinary tract infections: diagnosis. *Clin Infect Dis.* 2011;52(suppl 6):S452–S456.
308. Pappas PG, Kauffman CA, Andes D, et al. Clinical practice guidelines for the management of candidiasis 2009: update by the Infectious Diseases Society of America. *Clin Infect Dis.* 2009;48:503–535.
310. Sadegi BJ, Patel BK, Wilbur AC, et al. Primary renal candidiasis: importance of imaging and clinical history in diagnosis and management. *J Ultrasound Med.* 2009;28:507–514.
314. Paduch DA. Viral lower urinary tract infections. *Curr Urol Rep.* 2007;8:324–335.
325. Colley DG, Bustinduy AL, Secor WE, et al. Human schistosomiasis. *Lancet.* 2014;383:2253–2264.

REFERENCES

- Hooton TM. Uncomplicated urinary tract infection. *N Engl J Med*. 2012;366:1028–1037.
- Nicolle LE. Uncomplicated urinary tract infections in adults including uncomplicated pyelonephritis. *Urol Clin North Am*. 2008;35:1–12.
- Grigoryan L, Trautner BW, Gupta K. Diagnosis and management of urinary tract infections in the outpatient setting. *JAMA*. 2014;312:1677–1684.
- Neal DE. Complicated urinary tract infections. *Urol Clin North Am*. 2008;35:13–22.
- Nicolle LE, AMMI-Canada Guidelines Committee. Complicated urinary tract infection in adults. *Can J Infect Dis Med Microbiol*. 2005;16:349–360.
- Montagnini Spaine D, Mamizuka EM, Pereira Cedenho A, et al. Microbiologic aerobic studies on normal male urethra. *Urology*. 2000;56:207–210.
- Kunin CM, Evans C, Bartholomew D, et al. The antimicrobial defense mechanism of the female urethra: a reassessment. *J Urol*. 2002;168:413–419.
- Riemersma WA, van der Schee CJ, van der Meijden W, et al. Microbial population diversity in the urethras of healthy males and males suffering from nonchlamydial, nongonococcal urethritis. *J Clin Microbiol*. 2003;41:1977–1986.
- Schneeweiss J, Koch M, Umek W. The human urinary microbiome and how it relates to urogynecology. *Int Urogynecol J*. 2016;27:1307–1312.
- Sobel JD. Pathogenesis of urinary tract infection: role of host defenses. *Infect Dis Clin North Am*. 1997;3:531–549.
- Weichhart T, Haidinger M, Horl WH, et al. Current concepts of molecular defense mechanisms operative during urinary tract infection. *Eur J Clin Invest*. 2008;38:29–38.
- Abraham SN, Miao Y. The nature of immune responses to urinary tract infections. *Nat Rev Immunol*. 2015;15:655–663.
- Raffi HS, Bates JM Jr, Laszik Z, et al. Tamm-Horsfall protein acts as a general host-defense factor against bacterial cystitis. *Am J Nephrol*. 2005;25:570–578.
- Finer G, Landau D. Pathogenesis of urinary tract infections with normal female anatomy. *Lancet Infect Dis*. 2004;4:631–635.
- Ragnarsdottir B, Svanborg C. Susceptibility to acute pyelonephritis or asymptomatic bacteriuria: host-pathogen interaction in urinary tract infections. *Pediatr Nephrol*. 2012;27:2017–2029.
- Wullt B, Bergsten G, Fischer H, et al. The host response to urinary tract infection. *Infect Dis Clin North Am*. 2003;17:279–302.
- Ragnarsdottir B, Fischer H, Godaly G, et al. TLR- and CXCR1-dependent innate immunity: insights into the genetics of urinary tract infections. *Eur J Clin Invest*. 2008;38:12–20.
- Billips BK, Forrestal SG, Rycyk M, et al. Modulation of host innate immune response in the bladder by uropathogenic *Escherichia coli*. *Infect Immun*. 2007;75:5353–5360.
- Loughman JA, Hunstad DA. Induction of indoleamine 2, 3-dioxygenase by uropathogenic bacteria attenuates innate responses to epithelial infection. *J Infect Dis*. 2012;205:1830–1839.
- Hernandez JG, Sunden F, Connolly J, et al. Genetic control of the variable innate immune response to asymptomatic bacteriuria. *PLoS ONE*. 2011;6:1–11.
- Godaly G, Otto G, Burdick MD, et al. Fimbrial lectins influence the chemokine repertoire in the urinary tract mucosa. *Kidney Int*. 2007;71:778–786.
- Thumbikat P, Waltenbaugh C, Schaeffer AJ, et al. Antigen specific responses accelerate bacterial clearance in the bladder. *J Immunol*. 2006;176:3080–3086.
- Kantek A, Mottonen T, Al-Kaila K, et al. P fimbria-specific B cell responses in patients with urinary tract infection. *J Infect Dis*. 2003;188:1885–1891.
- Nicolle LE. Urinary tract infections in special populations: diabetes, renal transplantation, HIV infection and spinal cord injury. *Infect Dis Clin North Am*. 2014;28:91–104.
- High KP, Bradley SF, Gravenstein S, et al. Clinical practice guideline for the evaluation of fever and infection in older adult residents of long term care facilities: 2008 update by the Infectious Diseases Society of America. *Clin Infect Dis*. 2009;48:149–171.
- Raz R, Schiller D, Nicolle LE. Chronic indwelling catheter replacement before antimicrobial therapy for symptomatic urinary tract infection. *J Urol*. 2000;164:1254–1258.
- Lipsky BA. Urinary tract infections in men. *Ann Intern Med*. 1989;110:138–150.
- Frimodt-Moller N. Correlation between pharmacokinetic/pharmacodynamic parameters and efficacy for antibiotics in the treatment of urinary tract infection. *Int J Antimicrob Agents*. 2002;19:546–553.
- Benway BM, Moon TD. Bacterial prostatitis. *Urol Clin North Am*. 2008;35:23–32.
- Lipsky BA, Byren I, Hoey CT. Treatment of bacterial prostatitis. *Clin Infect Dis*. 2010;50:1641–1652.
- Wagenlehner FME, Weidner W, Naber KG. Pharmacokinetic characteristics of antimicrobials and optimal treatment of urosepsis. *Clin Pharmacokinet*. 2007;46:291–305.
- Wagenlehner FME, Wagenlehner C, Redman R, et al. Urinary bactericidal activity of doripenem versus that of levofloxacin in patients with complicated urinary tract infections or pyelonephritis. *Antimicrob Agents Chemother*. 2009;53:1567–1573.
- Nicolle L, Duckworth H, Sitar D, et al. Pharmacokinetics/pharmacodynamics of levofloxacin 750 mg once daily in young women with acute uncomplicated pyelonephritis. *Int J Antimicrob Agents*. 2008;31:287–289.
- Foxman B, Gillespie B, Koopman J, et al. Risk factors for second urinary tract infection among college women. *Am J Epidemiol*. 2000;151:1194–1205.
- Jackson SL, Boyko EJ, Scholes D, et al. Predictors of urinary tract infection after menopause. *Am J Med*. 2004;117:903–911.
- Birmingham SL, Ashe JF. Systematic review of the impact of urinary tract infections on health-related quality of life. *BJU Int*. 2012;110:e830–e836.
- Nickel JC, Lee JC, Grantmyre JE, et al. Natural history of urinary tract infection in a primary care environment in Canada. *Can J Urol*. 2005;12:2718–2737.
- Johnson JR, Owens K, Gajewski A, et al. Bacterial characteristics in relation to clinical source of *Escherichia coli* isolated from women with acute cystitis or pyelonephritis and uninfected women. *J Clin Microbiol*. 2005;43:6064–6072.
- Johnson JR. Microbial virulence determinants and the pathogenesis of urinary tract infection. *Infect Dis Clin North Am*. 2003;17:261–278.
- Gunther NW, Lockett J, Johnson DE, et al. In vivo dynamics of type 1 fimbria regulation in uropathogenic *Escherichia coli* during experimental urinary tract infection. *Infect Immun*. 2001;69:2838–2846.
- Takahashi A, Kanamaru S, Kurazono H, et al. *Escherichia coli* isolates associated with uncomplicated and complicated cystitis and asymptomatic bacteriuria possess similar phylogenies, virulence genes, and O-serogroup profiles. *J Clin Microbiol*. 2006;44:4589–4592.
- Foxman B, Manning SD, Tallman P, et al. Uropathogenic *Escherichia coli* are more likely than commensal *E. coli* to be shared between heterosexual sex partners. *Am J Epidemiol*. 2002;156:1133–1140.
- Johnson JR, Clabots C. Sharing of virulent *Escherichia coli* clones among household members of a woman with acute cystitis. *Clin Infect Dis*. 2006;43:c101–c108.
- Manges AR, Johnson JR, Foxman B, et al. Widespread distribution of urinary tract infections caused by a multidrug-resistant *Escherichia coli* clonal group. *N Engl J Med*. 2001;345:1007–1013.
- Banerjee R, Johnson JR. A new clone sweeps clean: the enigmatic emergence of *Escherichia coli* sequence type 131. *Antimicrob Agents Chemother*. 2014;58:4997–5004.
- Kuroda M, Yamashita A, Hirakawa H, et al. Whole genome sequence of *Staphylococcus saprophyticus* reveals the pathogenesis of uncomplicated urinary tract infection. *Proc Natl Acad Sci U. S. A*. 2005;102:13272–13277.
- Vogel T, Verreault R, Gourdeau M, et al. Optimal duration of antimicrobial therapy for uncomplicated urinary tract infection in older women: a double blind randomized controlled trial. *Can Med Assoc J*. 2004;170:469–473.
- Hooton TM, Roberts PL, Cox ME, et al. Voided midstream urine culture and acute cystitis in premenopausal women. *N Engl J Med*. 1983;369:1891–1913.
- Hooton TM. The current management strategies for community-acquired urinary tract infection. *Infect Dis Clin North Am*. 2003;17:303–332.
- Silverman JA, Schreiber HL IV, Hooton TM, et al. From physiology to pharmacy: developments in the pathogenesis and treatment of recurrent urinary tract infections. *Curr Urol Rep*. 2013;14:448–456.
- Hannan TJ, Totsika M, Mansfield KJ, et al. Host-pathogen checkpoints and population bottlenecks in persistent and intracellular uropathogenic *Escherichia coli* bladder infection. *FEMS Microbiol Rev*. 2012;36:616–648.

51. Czaja CA, Stamm WE, Stapleton AE, et al. Prospective cohort study of microbial and inflammatory events immediately preceding *Escherichia coli* recurrent urinary tract infection in women. *J Infect Dis*. 2009;200:528–536.
52. Raz R, Gennesin Y, Wasser J, et al. Recurrent urinary tract infections in postmenopausal women. *Clin Infect Dis*. 2000;30:152–156.
53. Scholes D, Hooton TM, Roberts PL, et al. Risk factors for recurrent urinary tract infection in young women. *J Infect Dis*. 2000;182:1177–1182.
54. Vincent CR, Thomas TL, Reyes L, et al. Symptoms and risk factors associated with first urinary tract infection in college age women: a prospective cohort study. *J Urol*. 2013;189:904–910.
55. Brown J, Vittinghoff E, Kanaya AM, et al. Urinary tract infections in post-menopausal women. Effects of hormone therapy and risk factors. *Obstet Gynecol*. 2001;98:1045–1052.
56. Foxman B, Somsel P, Tallman P, et al. Urinary tract infection among women aged 40 to 65: behavioral and sexual risk factors. *J Clin Epidemiol*. 2001;54:710–718.
57. Hu KK, Boyko EJ, Scholes D, et al. Risk factors for urinary tract infections in post-menopausal women. *Arch Intern Med*. 2004;164:989–993.
58. Moore EE, Howes SE, Scholes D, et al. Sexual intercourse and risk of symptomatic urinary tract infection in post-menopausal women. *J Gen Intern Med*. 2008;23:595–599.
59. Perrotta C, Aznar M, Mejia R, et al. Oestrogens for preventing recurrent urinary tract infection in postmenopausal women. *Cochrane Database Syst Rev*. 2008;(2):CD005131.
60. Lipsky BA. Urinary tract infections in men. *Ann Intern Med*. 1989;110:138–150.
61. Tomas ME, Getman D, Donskey CJ, et al. Overdiagnosis of urinary tract infection in adult women presenting to an emergency department. *J Clin Microbiol*. 2015;53:2685–2692.
62. Bent S, Nallamothu BK, Simel DL, et al. Does this woman have an acute uncomplicated urinary tract infection? *JAMA*. 2002;287:2701–2710.
63. Bent S, Saint S. The optimal use of diagnostic testing in women with acute uncomplicated cystitis. *Am J Med*. 2002;113(suppl 1A):20S–28S.
64. Nicolle LE, Madsen KS, Debeeck GO, et al. Three days of pivmecillinam or norfloxacin for treatment of acute uncomplicated urinary infection in women. *Scand J Infect Dis*. 2002;34:487–492.
65. Nys S, van Merode T, Bartelds AIM, et al. Urinary tract infection in general practice patients: diagnostic tests versus bacteriological culture. *J Antimicrob Chemother*. 2006;57:955–958.
66. St. John A, Boyd JC, Lowes AJ, et al. The use of urinary dipstick tests to exclude urinary tract infection. *Am J Clin Pathol*. 2006;126:428–436.
67. Ferry SA, Holm SE, Stenlund H, et al. The material course of uncomplicated lower urinary tract infection in women illustrated by a randomized placebo controlled study. *Scand J Infect Dis*. 2004;36:296–301.
68. Christiaens TC, DeMeyere M, Verschraegen G, et al. Randomized controlled trial of nitrofurantoin versus placebo in the treatment of uncomplicated urinary tract infection in adult women. *Br J Gen Pract*. 2002;52:729–734.
69. Gagyori I, Bleidorn J, Kochen MM, et al. Ibuprofen versus fosfomycin for uncomplicated urinary tract infection in women: randomized controlled trial. *BMJ*. 2015;351:1–10.
70. Gupta K, Hooton TM, Naver KG, et al. International clinical practice guidelines for the treatment of acute uncomplicated cystitis and pyelonephritis in women: a 2010 update by the Infectious Diseases Society of America and the European Society for Microbiology and Infectious Diseases. *Clin Infect Dis*. 2011;52:e103–e120.
71. Katchman EA, Milo G, Paul M, et al. Three day versus long duration of antibiotic treatment for cystitis in women: systematic review and meta-analysis. *Am J Med*. 2005;118:1196–1207.
72. Kahlmeter G, Ahman J, Matuschek E. Antimicrobial resistance of *Escherichia coli* causing uncomplicated urinary tract infection: a European update for 2014 and comparison with 2000 and 2008. *Infect Dis Ther*. 2015;4:417–423.
73. Stewardson AJ, Gaia N, Francois P, et al. Collateral damage from oral ciprofloxacin versus nitrofurantoin in outpatients with urinary tract infections: a culture-free analysis of gut microbiota. *Clin Microbiol Infect*. 2015;21:344.e1–344.e11.
74. Falagas ME, Rafailidis PI. Fosfomycin: the current status of the drug. *Clin Infect Dis*. 2015;61:114–1146.
75. Hooton TM, Roberts PL, Stapleton AE. Cefpodoxime vs ciprofloxacin for short course treatment of acute uncomplicated cystitis. *JAMA*. 2012;307:583–589.
76. Gupta K, Hooton TM, Roberts PL, et al. Short-course nitrofurantoin for the treatment of acute uncomplicated cystitis in women. *Arch Intern Med*. 2007;167:2207–2212.
77. Huttner A, Verhaegh EM, Harbarth S, et al. Nitrofurantoin revisited: a systematic review and meta-analysis of controlled trials. *J Antimicrob Chemother*. 2015;70:2456–2464.
78. Gupta K, Sahm DF, Mayfield D, et al. Antimicrobial resistance among uropathogens that cause community-acquired urinary tract infections in women: a nationwide analysis. *Clin Infect Dis*. 2001;33:89–94.
79. Naber KG, Schito G, Botto H, et al. Surveillance study in Europe and Brazil on clinical aspects and antimicrobial resistance epidemiology in females with cystitis (ARESC): implications for empiric therapy. *Eur Urol*. 2008;54:1164–1175.
80. Anesi JA, Lautenbach E, Nachamkin I, et al. Clinical and molecular characterization of community-onset urinary tract infections due to extended-spectrum cephalosporin-resistant enterobacteriaceae. *Infect Control Hosp Epidemiol*. 2016;37:1433–1439.
81. Raz R, Chazan B, Kennes Y, et al. Israel Urinary Tract Infection Group: empiric use of trimethoprim-sulfamethoxazole in the treatment of women with uncomplicated urinary tract infections in a geographical area with a high prevalence of TMP-SMX resistant organisms. *Clin Infect Dis*. 2002;34:1165–1169.
82. McNulty CAM, Richards J, Livermore DM, et al. Clinical evidence of laboratory-reported antibiotic resistance in acute uncomplicated urinary tract infection in primary care. *J Antimicrob Chemother*. 2006;58:1000–1008.
83. Pitout JD, Gregson DB, Church PL, et al. Community-wide outbreaks of clonally related CTX-M-14 beta-lactamase-producing *Escherichia coli* strains in the Calgary health region. *J Clin Microbiol*. 2005;43:2844–2849.
84. Walker E, Lyman A, Gupta K, et al. Clinical management of an increasing threat: outpatient urinary tract infections due to multidrug-resistant uropathogens. *Clin Infect Dis*. 2016;63:960–965.
85. Lawrentschuk N, Ooi J, Pang A, et al. Cystoscopy in women with recurrent urinary tract infection. *Int J Urol*. 2006;13:350–353.
86. Muller AE, Verhaegh EM, Harbarth S, et al. Nitrofurantoin's efficacy and safety as prophylaxis for urinary tract infections: a systematic review of the literature and meta-analysis of controlled trials. *Clin Microbiol Infect*. 2016;1:e1–e8.
87. Duenas-Garcia OF, Sullivan G, Hall CD, et al. Pharmacological agents to decrease new episodes of recurrent lower urinary tract infections in postmenopausal women: a systematic review. *Female Pelvic Med Reconstr Surg*. 2016;22(Mar/Apr):63–69.
88. Stapleton A. Novel approaches to prevention of urinary tract infection. *Infect Dis Clin North Am*. 2003;17:457–472.
89. Beerepoot MA, Geerlings SE, van Haarst EP, et al. Nonantibiotic prophylaxis for recurrent urinary tract infections: a systematic review and meta-analysis of randomized controlled trials. *J Urol*. 2013;190:1981–1989.
90. Jepson RG, Craig JC. Cranberries for preventing urinary tract infections. *Cochrane Database Syst Rev*. 2008;(1):CD001321.
91. Barbosa-Cenik C, Brown MB, Buxton M, et al. Cranberry juice fails to prevent recurrent urinary tract infection: results from a randomized placebo-controlled trial. *Clin Infect Dis*. 2011;52:23–30.
92. Beerepoot MA, ter Riet G, Nys S, et al. Cranberries vs antibiotics to prevent urinary tract infections. A randomized, double blind noninferiority trial in premenopausal women. *Arch Intern Med*. 2011;25:1270–1278.
93. Kontiokari T, Sundquist K, Nuutinen M, et al. Randomized trial of cranberry lingonberry juice or *Lactobacillus GG* drink for the prevention of urinary tract infections in women. *BMJ*. 1571;322(7302):2001.
94. Beerepoot MA, ter Riet G, Nys S, et al. Lactobacilli vs antibiotics to prevent urinary tract infections: a randomized, double-blind, noninferiority trial of postmenopausal women. *Arch Intern Med*. 2012;172:704–712.
95. Schwenger EM, Tejani AM, Loewen PS. Probiotics for preventing urinary tract infections in adults and children (review). *Cochrane Database Syst Rev*. 2015;(12):Art. No.: CD008772.
96. Hannan TJ, Hooton TM, Hultgren SJ. Estrogen and recurrent UTI. What are the facts? *Sci Transl Med*. 2013;5(190):fs23.
97. Raz R, Colodner R, Rohana Y, et al. Effectiveness of estriol-containing vaginal pessaries and nitrofurantoin macrocrystal therapy in the prevention of recurrent urinary tract infection in postmenopausal women. *Clin Infect Dis*. 2003;36:1362–1368.

98. Barber AE, Norton JP, Spivack AM, et al. Urinary tract infections: current and emerging management strategies. *Clin Infect Dis*. 2013;57:719–724.
99. O'Brien VP, Hannan TJ, Nielsen HV, et al. Drug and vaccine development for the treatment and prevention of urinary tract infections. *Microbiol Spectr*. 2016;4:1–42.
100. Ciani O, Arendsen E, Romancik M, et al. Intravesical administration of combined hyaluronic acid (HA) and chondroitin sulfate (CS) for the treatment of female recurrent urinary tract infections: a European multicenter nested case-control study. *BMJ*. 2016;6:e009669.
101. Torella M, Del Deo F, Grimaldi A, et al. Efficacy of an orally administered combination of hyaluronic acid, chondroitin sulfate, curcumin and quercetin for the prevention of recurrent urinary tract infections in postmenopausal women. *Eur J Obstet Gynecol Reprod Biol*. 2016;207:125–128.
102. Luthje P, Brauner A. Novel strategies in the prevention and treatment of urinary tract infections. *Pathogens*. 2016;5(13):1–14.
103. Wagenlehner FM, Ballarini S, Pilatz A, et al. A randomized, double-blind, parallel group multicenter clinical study of *Escherichia coli* lyophilized lysate for the prophylaxis of recurrent uncomplicated urinary tract infections. *Urol Int*. 2015;95:167–176.
104. Brown P, Ki M, Foxman B. Acute pyelonephritis among adults: cost of illness and considerations for the economic evaluation of therapy. *Pharmacoeconomics*. 2005;23:1123–1142.
105. Hill JB, Sheffield JS, McIntire DD, et al. Acute pyelonephritis in pregnancy. *Obstet Gynecol*. 2005;105:18–23.
106. Kooman JP, Barendregt JNM, van der Sande FM, et al. Acute pyelonephritis: a cause of renal failure? *Neth J Med*. 2000;57:185–189.
107. Nahar A, Akon M, Hanes D, et al. Pyelonephritis and acute renal failure. *Am J Med*. 2004;328:121–123.
108. Piccoli B, Colla L, Burdese M, et al. Development of kidney scars after acute uncomplicated pyelonephritis: relationship with clinical laboratory and imaging data at diagnosis. *World J Urol*. 2006;24:66–73.
109. Raz R, Sakran W, Chazan B, et al. Long-term follow-up of women hospitalized for acute pyelonephritis. *Clin Infect Dis*. 2003;37:1014–1020.
110. Scholes D, Hooton TM, Roberts PL, et al. Risk factors associated with acute pyelonephritis in healthy women. *Ann Intern Med*. 2005;142:20–27.
111. Lane MC, Mobley HLT. Role of P-fimbrial-mediated adherence in pyelonephritis and persistence of uropathogenic *Escherichia coli* (UPEC) in the mammalian kidney. *Kidney Int*. 2007;72:19–25.
112. Lundstedt A-C, Leijonhufvud I, Ragnarsdottir B, et al. Inherited susceptibility to acute pyelonephritis: a family study of urinary tract infection. *J Infect Dis*. 2007;195(8):1227–1234.
113. Geback C, Hansson S, Martinell J, et al. Urinary tract infection pattern in adult women followed from childhood. *Pediatr Nephrol*. 2016;31:1107–1111.
114. Velasco M, Martinez JA, Moreno-Martinez A, et al. Blood cultures for women with uncomplicated acute pyelonephritis: are they necessary? *Clin Infect Dis*. 2003;45:1127–1130.
115. Pasternack EL 3rd, Topinka MA. Blood cultures in pyelonephritis: do results change therapy? *Acad Emerg Med*. 2000;7:1170.
116. Paraboutis IG, Tsagalou EP, Lepinski JL, et al. Primary *Staphylococcus aureus* urinary tract infection: the role of undetected hematogenous seeding of the urinary tract. *Eur J Clin Microbiol Infect Dis*. 2010;29:1095–1101.
117. Huggan PJ, Murdoch DR, Gallagher K, et al. Concomitant *Staphylococcus aureus* bacteriuria is associated with poor clinical outcomes in adults with *S. aureus* bacteraemia. *J Hosp Infect*. 2008;69:345–349.
118. Lemiale V, Renaud B, Moutereau S, et al. A single procalcitonin level does not predict adverse outcomes of women with pyelonephritis. *Eur Urol*. 2007;51:1394–1401.
119. Yang WJ, Cho IR, Seong H, et al. Clinical implication of serum C reactive protein in patients with uncomplicated acute pyelonephritis as marker of prolonged hospitalization and recurrence. *Urology*. 2009;73:19–22.
120. Ha YE, Kang C-I, Wi YM, et al. Diagnostic usefulness of procalcitonin as a marker of bacteremia in patients with acute pyelonephritis. *Scand J Clin Lab Invest*. 2013;73:444–448.
121. Park JH, Wee JH, Choi SP, et al. Serum procalcitonin level for the prediction of severity in women with acute pyelonephritis in the ED: value of procalcitonin in acute pyelonephritis. *Am J Emerg Med*. 2013;31:1092–1097.
122. Claessens Y-E, Schmidt J, Batard E, et al. Can C-reactive protein, procalcitonin and mid-regional pro-atrial natriuretic peptide measurements guide choice of in-patient or out-patient care in acute pyelonephritis? Biomarkers in Sepsis (BIS) multicenter study. *Clin Microbiol Infect*. 2010;16:753–760.
123. Stunell H, Buckley O, Feeney J, et al. Imaging of acute pyelonephritis in the adult. *Eur Radiol*. 2002;17:1820–1828.
124. Demetris J, Menias CD. State of the art: imaging of renal infections. *Emerg Radiol*. 2007;14:13–22.
125. Vourganti S, Agarwal PK, Bodner DR, et al. Ultrasonographic evaluation of renal infections. *Radiol Clin North Am*. 2006;44:763–765.
126. Kirkwood A, Harris C, Timar N, et al. Is gentamicin ototoxic to the fetus? *J Obstet Gynaecol Can*. 2007;29:140–145.
127. Talan DA, Stamm WE, Hooton TM, et al. Comparison of ciprofloxacin (7 days) and trimethoprim/sulfamethoxazole (14 days) for acute uncomplicated pyelonephritis in women. *JAMA*. 2000;283:1583–1590.
128. Sandberg T, Skoog G, Hermansson AB, et al. Ciprofloxacin for 7 days versus 14 days in women with acute pyelonephritis: a randomized, open-label and double-blind, placebo controlled non-inferiority trial. *Lancet*. 2012;380:484–490.
129. Moustafa F, Nguyen G, Mathevon T, et al. Evaluation of the efficacy and tolerance of a short 7 day third-generation cephalosporin treatment in the management of acute pyelonephritis in young women in the emergency department. *J Antimicrob Chemother*. 2016;71:1660–1664.
130. Peterson J, Kaul S, Khashab M, et al. A double-blind randomized comparison of levofloxacin 750 mg once-daily for five days with ciprofloxacin 400/500 mg twice daily for 10 days for the treatment of complicated urinary tract infections and acute pyelonephritis. *Urology*. 2008;71:17–22.
131. Pertel PE, Haverstock D. Risk factors for a poor outcome after therapy for acute pyelonephritis. *BJU Int*. 2006;98:141–147.
132. Esclarin De Ruz A, Garcia Leoni E, Herruzo Cabrera R. Epidemiology and risk factors for urinary tract infection in patients with spinal cord injury. *J Urol*. 2000;164:1285–1289.
133. Stevenson KB, Moore J, Colwell H, et al. Standardized infection surveillance in long term care: interfacility surveillance from a regional cohort of facilities. *Infect Control Hosp Epidemiol*. 2005;26:231–238.
134. Sogaard M, Schonheyder HC, Riis A, et al. Short-term mortality in relation to age and co-morbidity in older adults with community-acquired bacteremia: a population-based cohort study. *J Am Geriatr Soc*. 2008;56:1593–1600.
135. Lee C-C, Chen S-Y, Chaing I-J, et al. Comparison of clinical manifestations and outcome of community-acquired bloodstream infections among the oldest old, elderly, and adult patients. *Medicine (Baltimore)*. 2007;86:138–144.
136. Bahagon Y, Rauch D, Schlesinger Y, et al. Prevalence and predictive features of bacteraemic urinary tract infection in emergency department patients. *Eur J Clin Microbiol*. 2007;26:349–352.
137. Kumar A, Roberts D, Wood KE, et al. Duration of hypotension before initiation of effective antimicrobial therapy is the critical determinant of survival in human septic shock. *Crit Care Med*. 2006;34:1589–1596.
138. D'Hondt F, Everaert K. Urinary tract infections in patients with spinal cord injuries. *Curr Infect Dis Rep*. 2011;13:544–551.
139. Naber KG, Llorens L, Kaniga K, et al. Intravenous doripenem at 500 milligrams versus levofloxacin at 250 milligrams, with an option to switch to oral therapy, for treatment of complicated lower urinary tract infection and pyelonephritis. *Antimicrob Agents Chemother*. 2009;53:3782–3792.
140. Tabiban JH, Gornbein J, Heidari A, et al. Uropathogens and host characteristics. *J Clin Microbiol*. 2008;46:3980–3986.
141. Kauffman CA, Vazquez JA, Sobel JD, et al. National Institute for Allergy and Infectious Diseases Mycoses Study Group: prospective, multicenter surveillance study of funguria in hospitalized patients. *Clin Infect Dis*. 2000;30:14–18.
142. Trandogdue Z, Bartolett R, Cai T, et al. Antimicrobial resistance in urosepsis: outcomes from the multinational, multicenter global prevalence of infections in urology (GPIU) study 2003-2013. *World J Urol*. 2016;34:1193–1200.
143. Marcus RJ, Post JC, Stoodley P, et al. Biofilms in nephrology. *Expert Opin Biol Ther*. 2008;8:1159–1166.
144. Stickler DJ, Feneley RCL. The encrustation and blockage of long-term indwelling bladder catheters: a way forward in prevention and control. *Spinal Cord*. 2010;48:784–790.
145. Akay AF, Aflay U, Gedik A, et al. Risk factors for lower urinary tract infection and bacterial stent colonization in patients with double J ureteral stent. *Int Urol Nephrol*. 2007;39:95–98.

146. Liaw A, Knudsen B. Urinary tract infections associated with urethral stents: a review. *Arch Esp Urol*. 2016;69:479.
147. Kehinde EO, Rotimi VO, Al-Hunayan A, et al. Bacteriology of urinary tract infection associated with indwelling J ureteral stents. *J Endourol*. 2004;18:891–896.
148. Soriano F, Tauch A. Microbiological and clinical features of *Corynebacterium urealyticum*: urinary tract stones and genomics as the Rosetta stone. *Clin Microbiol Infect*. 2008;14:632–643.
149. Ibler K, Jensen KT, Ostergaard C, et al. Six cases of *Aerococcus sanguinicola* infection. Clinical relevance and bacterial identification. *Scand J Infect Dis*. 2008;40:761–765.
150. Christensen JJ, Kilian M, Fussing V, et al. *Aerococcus urinae* polyphasic characterization of the species. *APMIS*. 2005;113:517–525.
151. Brook I. Urinary tract and genitourinary suppurative infections due to anaerobic bacteria. *Int J Urology*. 2004;11:133–141.
152. Nicolle LE, Bradley S, Colgan R, et al. Infectious Diseases Society of America guidelines for the diagnosis and treatment of asymptomatic bacteriuria in adults. *Clin Infect Dis*. 2005;40:643–654.
153. Nicolle LE. Urinary tract infection. *Crit Care Clin*. 2013;3:699–715.
154. Loeb M, Bentley DW, Bradley S, et al. Development of minimum criteria for the initiation of antibiotics in residents of long term care facilities: results of a consensus conference. *Infect Control Hosp Epidemiol*. 2001;22:120–124.
155. Nicolle LE. Urinary tract infections in the older adult. *Clin Geriatr Med*. 2016;32:523–538.
156. Mody L, Juthani-Mehta M. Urinary tract infections in older women: a clinical review. *JAMA*. 2014;311:844–854.
157. Juthani-Mehta M, Quagliarello V, Perrelli E, et al. Clinical features to identify urinary tract infection in nursing home residents: a cohort study. *J Am Geriatr Soc*. 2009;57:963–970.
158. Wullt B, Agace W, Mansson W. Bladder, bowel and bugs—bacteriuria in patients with intestinal urinary diversion. *World J Urol*. 2004;22:186–195.
159. Suriano F, Gallucci M, Flammia GP, et al. Bacteriuria in patients with an orthoptic ileal neobladder: urinary tract infection or asymptomatic bacteriuria? *BJU Int*. 2007;101:1576–1579.
160. van Duin D, Bonomo RA. Ceftazidime/avibactam and ceftolozane/tazobactam: second-generation β -lactam/ β -lactamase inhibitor combinations. *Clin Infect Dis*. 2016;63:234–241.
161. Huntington JA, Sakoulas G, Umeh O, et al. Efficacy of ceftolozane/tazobactam versus levofloxacin in the treatment of complicated urinary tract infections (cUTIs) caused by levofloxacin-resistant pathogens: results from the ASPECT-cUTI trial. *J Antimicrob Chemother*. 2016;71:2014–2021.
162. Carmeli Y, Armstrong J, Laud PJ, et al. Ceftazidime-avibactam or best available therapy in patients with ceftazidime-resistant Enterobacteriaceae and *Pseudomonas aeruginosa* complicated urinary tract infections or complicated intra-abdominal infections (REPRISE): a randomized, pathogen-directed, phase 3 study. *Lancet Infect Dis*. 2016;16:661–673.
163. Wagenlehner FM, Sobel JD, Newell P, et al. Ceftazidime-avibactam versus doripenem for the treatment of complicated urinary tract infections, including acute pyelonephritis: RECAPTURE, a phase 3 randomized trial program. *Clin Infect Dis*. 2016;63:754–762.
164. Popejoy MW, Paterson DL, Cloutier D, et al. Efficacy of ceftolozane/tazobactam against urinary tract and intra-abdominal infections caused by ESBL-producing *Escherichia coli* and *Klebsiella pneumoniae*: a pooled analysis of phase 3 clinical trials. *J Antimicrob Chemother*. 2017;72:268–272.
165. Grabein B, Graninger W, Rodriguez Barrio J, et al. Intravenous fosfomycin—back to the future. Systematic review and meta-analysis of the clinical literature. *Clin Microbiol Infect*. 2016;12:epub.
166. Seroy JT, Grim SA, Reid GE. Treatment of MDR urinary tract infections with oral fosfomycin: a retrospective analysis. *J Antimicrob Chemother*. 2016;71:2563–2568.
167. Veve MP, Wagner JL, Kenney RM. Comparison of fosfomycin to ertapenem for outpatient or step-down therapy of extended-spectrum β -lactamase urinary tract infection. *Int J Antimicrob Agents*. 2016;48:56–60.
168. Sastry S, Glarke LG, Alrowais H, et al. Clinical appraisal of fosfomycin in the era of antimicrobial resistance. *Antimicrob Agents Chemother*. 2016;59:7355–7361.
169. Jarrell AS, Wood GC, Ponnappa S, et al. Short-duration treatment of catheter-associated urinary tract infections in critically ill trauma patients. *J Trauma Acute Care Surg*. 2015;79:649–653.
170. Lo E, Nicolle L, Coffin SE, et al. Strategies to prevent catheter-associated urinary tract infections in acute care hospitals. *Infect Control Hosp Epidemiol*. 2014;35:464–479.
171. Hooton TM, Bradley SF, Cardenas DD, et al. Diagnosis, prevention, and treatment of catheter-associated urinary tract infections in adults: 2009 international clinical practice guidelines from the Infectious Diseases Society of America. *Clin Infect Dis*. 2010;50(5):625–663.
172. Gould CV, Umscheid CA, Agarwal RK, et al. Health Care Infection Control Practices Advisory Committee: guidelines for prevention of catheter-associated urinary tract infections 2009. *Infect Control Hosp Epidemiol*. 2010;31(4):319–326.
173. Saint S, Greene MT, Krein SL, et al. A program to prevent catheter-associated urinary tract infection in acute care. *N Engl J Med*. 2016;374:2111–2119.
174. Mullin KM, Kovacs CS, Fatica C, et al. A multifaceted approach to reduction of catheter-associated urinary tract infections in the intensive care unit with an emphasis on “Stewardship of Culturing.” *Infect Control Hosp Epidemiol*. 2016;[epub Nov 1-3].
175. Siddiq DM, Darouiche RO. New strategies to prevent catheter-associated urinary tract infections. *Nat Rev Urol*. 2012;9:305–314.
176. Singha P, Locklin J, Handa H. A review of the recent advances in antimicrobial coatings for urinary catheters. *Acta Biomater*. 2017;50:20–40.
177. Moltzahn F, Haeni K, Birkhauser FD, et al. Peri-interventional antibiotic prophylaxis only vs continuous low-dose antibiotic treatment in patients with JJ stents: a prospective randomized trial analyzing the effect on urinary tract infections and stent related symptoms. *BJU Int*. 2012;111:289–295.
178. Sundén F, Hakansson L, Ljunggren E, et al. Bacterial interference—is deliberate colonization with *Escherichia coli* 83972 an alternative treatment for patients with recurrent urinary tract infection? *Int J Antimicrob Agents*. 2006;28(suppl 1):S26–S29.
179. Prasad A, Cevallos ME, Riosa S, et al. A bacterial interference strategy for prevention of UTI in persons practicing intermittent catheterization. *Spinal Cord*. 2009;47(7):565–569.
180. Rodhe N, Mölstad S, Englund L, et al. Asymptomatic bacteriuria in a population of elderly residents living in a community setting: prevalence, characteristics, and associated factors. *Fam Pract*. 2006;23:303–307.
181. Hooton TM, Scholes D, Stapleton AE, et al. A prospective study of asymptomatic bacteriuria in sexually active young women. *N Engl J Med*. 2000;343:991–997.
182. Nicolle LE, Zhanell GG, Harding GK. Microbiological outcomes in women with diabetes and untreated asymptomatic bacteriuria. *World J Urol*. 2006;24:61–65.
183. Mabbett AN, Ulett GC, Watts RE, et al. Virulence properties of asymptomatic bacteriuria *Escherichia coli*. *Int J Med Microbiol*. 2009;299:53–63.
184. Zdziarski J, Svanborg C, Wullt B, et al. Molecular basis of commensalism in the urinary tract: low virulence or virulence attenuation. *Infect Immun*. 2008;76:695–703.
185. Klemm P, Hancock V, Schembri MA. Mellowing out: adaptation to commensalism by *Escherichia coli* strain 83972. *Infect Immun*. 2007;75:3688–3695.
186. Mignini L, Carroli M, Abalos E, et al. World Health Organization Asymptomatic Bacteriuria Trial Group: accuracy of diagnostic tests to detect asymptomatic bacteriuria during pregnancy. *Obstet Gynecol*. 2009;113(2 Pt 1):346–352.
187. Sundén F, Wullt B. Predictive value of urinary interleukin-6 for symptomatic urinary tract infections in a nursing home population. *Int J Urol*. 2016;23:168–174.
188. Sundvall P-D, Elm M, Ulleryd P, et al. Interleukin-6 concentrations in the urine and dipstick analyses were related to bacteriuria but not symptoms in the elderly: a cross sectional study of 421 nursing home residents. *BMC Geriatr*. 2014;14:88.
189. Leone M, Perrin A-S, Granier I, et al. A randomized trial of catheter change and short course of antibiotics for asymptomatic bacteriuria in catheterized ICU patients. *Intensive Care Med*. 2007;33:726–729.
190. Cai T, Mazzoli S, Mondaini N, et al. The role of asymptomatic bacteriuria in young women with recurrent urinary infections: to treat or not to treat? *Clin Infect Dis*. 2012;55:771–777.
191. Green H, Rahamimov R, Golbert E, et al. Consequences of treated versus untreated asymptomatic bacteriuria in the first year following kidney transplantation: retrospective observational study. *Eur J Clin Microbiol Infect Dis*. 2012;32:127–131.

192. Moradi M, Abbasi M, Moradi A, et al. Effect of antibiotic therapy on asymptomatic bacteriuria in kidney transplant recipients. *Urol J*. 2005;2:32–35.
193. Origen J, Lopez-Medrano F, Fernandez-Ruiz M, et al. Should asymptomatic bacteriuria be systematically treated in kidney transplant recipients? Results from a randomized controlled trial. *Am J Transplant*. 2016;16(10):2943–2953.
194. Trautner BW, Grigoryan L, Petersen NJ, et al. Effectiveness of an antimicrobial stewardship approach for urinary catheter-associated asymptomatic bacteriuria. *JAMA Intern Med*. 2015;175:1120–1127.
195. Hartley SE, Kuhn L, Valley S, et al. Evaluating a hospitalist-based intervention to decrease unnecessary antimicrobial use in patients with asymptomatic bacteriuria. *Infect Control Hosp Epidemiol*. 2016;37:1044–1051.
196. Kazemier BM, Koningstein FN, Schneeberger C, et al. Maternal and neonatal consequences of treated and untreated asymptomatic bacteriuria in pregnancy: a prospective cohort study with an embedded randomized controlled trial. *Lancet Infect Dis*. 2015;15:1324–1333.
197. Nicolle LE. Management of asymptomatic bacteriuria in pregnant women. *Lancet Infect Dis*. 2015;15:1252–1254.
198. Lumbiganon P, Villar J, Laopaiboon M, et al. One-day compared with 7-day nitrofurantoin for asymptomatic bacteriuria in pregnancy: a randomized controlled trial. *Obstet Gynecol*. 2009;113(2 Pt 1):339–345.
199. Widmer M, Gulmezoglu AM, Mignini L. Duration of treatment of asymptomatic bacteriuria during pregnancy. *Cochrane Database Syst Rev*. 2015;(11):CD000491.
200. European Association of Urology. *Guidelines on urological infections (version 2015)*. www.uroweb.org/guidelines/http://uroweb.org/wpcontent/uploads/EAU-Guidelines-Urological-Infections-v2pdf.
201. Garcia-Perdomo HA, Lopez H, Corbonell J, et al. Efficacy of antibiotic prophylaxis in patients undergoing cystoscopy: a randomized clinical trial. *World J Urol*. 2013;31:1433–1439.
202. Krieger JN, Nyberg L, Nickel JC. NIH consensus definition and classification of prostatitis. *JAMA*. 1999;281:236–237.
203. Schaeffer AJ. Clinical practice: chronic prostatitis and the chronic pelvic pain syndrome. *N Engl J Med*. 2006;355:1690–1698.
204. Bowen DK, DiLubanza E, Schaeffer AJ. Chronic bacterial prostatitis and chronic pelvic pain syndrome. *BMJ Clin Evid*. 1802;08:2015.
205. Etienne M, Chavanet P, Sibert L, et al. Acute bacterial prostatitis: heterogeneity in diagnostic criteria and management. Retrospective multicentric analysis of 371 patients diagnosed with acute prostatitis. *BMC Infect Dis*. 2008;8:12.
206. Krieger JN, Dobrindt U, Riley DE, et al. Acute *Escherichia coli* prostatitis in previously healthy young men: bacterial virulence factors, antimicrobial resistance, and clinical outcomes. *Urology*. 2011;77:1420–1425.
207. Touma NJ, Nickel JC. Prostatitis and chronic pelvic pain syndrome in men. *Med Clin North Am*. 2011;95:75–86.
208. Schaeffer AJ, Nicolle LE. Urinary tract infections in older men. *N Engl J Med*. 2016;374:562–571.
209. Krebs J, Bartel P, Pannek J. Bacterial persistence in the prostate after antibiotic treatment of chronic bacterial prostatitis in men with spinal cord injury. *Urology*. 2014;83:515–520.
210. Nickel JC, Alexander RB, Schaeffer AJ, et al. Chronic Prostatitis Collaborative Research Network Study Group: leukocytes and bacteria in men with chronic bacterial prostatitis/chronic pelvic pain syndrome compared to asymptomatic controls. *J Urol*. 2003;170:818–822.
211. Krieger JN, Ross SD, Limaye AP, et al. Inconsistent localization of gram positive bacteria to prostate specific specimens from patients with chronic prostatitis. *Urology*. 2005;66:721–725.
212. Pontari MA. Chronic prostatitis/chronic pelvic pain syndrome. *Urol Clin North Am*. 2008;35:81–89.
213. Alexander RB, Probert KJ, Schaeffer AJ, et al. Ciprofloxacin or tamsulosin in men with chronic prostatitis/chronic pelvic pain syndrome. *Ann Intern Med*. 2004;141:581–589.
214. Rees J, Abrahams M, Doble A, et al. Diagnosis and treatment of chronic bacterial prostatitis and chronic prostatitis/chronic pelvic pain syndrome: a consensus guideline. *BJU Int*. 2015;116:509–525.
215. de Souza RM, Olsburgh J. Urinary tract infection in the renal transplant patient. *Nat Clin Pract Nephrol*. 2008;4:252–264.
216. Dantas SR, Kuboyama RH, Mazzali M, et al. Nosocomial infections in renal transplant patients: risk factors and treatment implications associated with urinary tract and surgical site infections. *J Hosp Infect*. 2006;63:117–123.
217. Golebiewska J, Debska-Silzien A, Komarnicka J, et al. Urinary tract infections in renal transplant recipients. *Transplant Proc*. 2011;43:2985–2990.
218. Abbott KC, Swanson SJ, Richter ER, et al. Late urinary tract infection after renal transplantation in the United States. *Am J Kidney Dis*. 2004;44:353–362.
219. Saemann M, Horl WH. Urinary tract infection in renal transplant recipients. *Eur J Clin Invest*. 2008;38(suppl 2):58–65.
220. Valera B, Gentil MA, Cabello V, et al. Epidemiology of urinary infections in renal transplant recipients. *Transplant Proc*. 2006;38:2414–2415.
221. Schmaldienst S, Dittrich E, Horl WH. Urinary tract infections after renal transplantation. *Curr Opin Urol*. 2002;12:125–130.
222. Alangaden GJ, Thyagarajan R, Gruber SA, et al. Infectious complications after kidney transplantation: current epidemiology and associated risk factors. *Clin Transplant*. 2006;20:401–409.
223. Kamath NS, John GT, Neelakantan N, et al. Acute graft pyelonephritis following renal transplantation. *Transpl Infect Dis*. 2006;8:140–147.
224. Sorto R, Irizar SS, Delgadillo G, et al. Risk factors for urinary tract infections during the first year after kidney transplantation. *Transplant Proc*. 2010;42:280–281.
225. Rice JC, Safdar N. Urinary tract infections in solid organ transplant recipients. *Am J Transplant*. 2009;9(54):S267–S272.
226. Wilson CH, Bhatti AA, Rix DA, et al. Routine intraoperative ureteric stenting for kidney transplant recipients. *Cochrane Database Syst Rev*. 2005;(4):CD004925.
227. Saemann M, Hoel WH. Urinary tract infection in renal transplant recipients. *Eur J Clin Invest*. 2008;38(suppl 2):S267–S272.
228. Papasotiriou M, Savvidaki E, Kalliakmani P, et al. Predisposing factors for the development of urinary tract infections in renal transplant recipients and the impact on long term graft function. *Ren Fail*. 2011;33:404–410.
229. Fiorante S, Fernandez-Ruiz M, Lopez-Medrano F, et al. Acute graft pyelonephritis in renal transplant recipients: incidence, risk factors and long-term outcome. *Nephrol Dial Transplant*. 2011;26:1065–1073.
230. Valera B, Gentil MA, Cabello V, et al. Epidemiology of urinary infections in renal transplant recipients. *Transplant Proc*. 2006;38:2414–2415.
231. Audard V, Amor M, Desvieux D, et al. Acute graft pyelonephritis: a potential cause of acute rejection in renal transplant. *Transplantation*. 2005;80:1128–1130.
232. Kasiske BL, Zeier MG, Chapman JR, et al. KDIGO clinical practice guideline for the care of kidney transplant recipients: a summary. *Kidney Int*. 2010;77:299–311.
233. Safdar N, Slattery WR, Knasinski V, et al. Predictors and outcomes of candiduria in renal transplant recipients. *Clin Infect Dis*. 2005;40:1413.
234. Singh R, Bemelman FJ, Hodiament CJ, et al. The impact of trimethoprim-sulfamethoxazole as *Pneumocystis jirovecii* pneumonia prophylaxis on the occurrence of asymptomatic bacteriuria and urinary tract infections among renal allograft recipients: a retrospective before-after study. *BMC Infect Dis*. 2016;25(16):90.
235. Green H, Rahamimov R, Gafer U, et al. Antibiotic prophylaxis for urinary tract infections in renal transplant recipients: a systematic review and meta-analysis. *Transpl Infect Dis*. 2011;13:441–447.
236. Gilbert DN. Urinary tract infections in patients with chronic renal failure. *Clin J Am Soc Nephrol*. 2006;1:327–331.
237. Cunha BA, Giuga J, Gerson S. Predictors of ertapenem therapeutic efficacy in the treatment of urinary tract infections (UTIs) in hospitalized adults: the importance of renal insufficiency and urinary pH. *Eur J Clin Microbiol Infect Dis*. 2016;35:673–679.
238. Brown PD. Management of urinary tract infections associated with nephrolithiasis. *Curr Infect Dis Rep*. 2010;12:450–451.
239. Zanetta G, Paparella S, Trinchieri A, et al. Infections and urolithiasis: current clinical evidence in prophylaxis and antibiotic therapy. *Arch Ital Urol Androl*. 2008;80:5–12.
240. Flannigan R, Choy WH, Chew B. Renal struvite stones—pathogenesis, microbiology, and management strategies. *Nat Rev Urol*. 2014;11:333–341.
241. Miano R, Germani S, Vespasiani G. Stones and urinary tract infections. *Urol Int*. 2007;79(suppl 1):32–36.
242. Deutsch PG, Subramanian K. Conservative management of staghorn calculi: a single-centre experience. *BJU Int*. 2016;118:444–450.
243. Margel D, Ehrlich Y, Brown N, et al. Clinical implications of routine stone culture in percutaneous nephrolithotomy—a prospective culture. *Urology*. 2006;67:26–29.

244. Mariappan P, Smith G, Baraol SV, et al. Stone and pelvic urine culture and sensitivity are better than bladder urine as predictors of urosepsis following percutaneous nephrolithotomy. A prospective clinical study. *J Urol*. 2005;173:1610–1614.
245. Gutierrez J, Smith A, Geavlete P, et al. Urinary tract infections and post-operative fever in percutaneous nephrolithotomy. *World J Urol*. 2013;31:1135–1140.
246. Borofsky MS, Walter D, Shah O, et al. Surgical decompression is associated with decreased mortality in patients with sepsis and ureteral calculi. *World J Urol*. 2013;31:1135–1140.
247. Mariappan P, Loong CW. Midstream urine culture and sensitivity test is a poor predictor of infected urine proximal to the obstructing ureteral stone or infected stones: a prospective clinical study. *J Urol*. 2004;171:2142–2145.
248. Meng MV, Mario LA, McAninch JW. Current treatment and outcomes of perinephric abscesses. *J Urol*. 2002;168:1337–1340.
249. Coelho RF, Schneider-Montero ED, Mesquita JLB, et al. Renal and perinephric abscess. Analysis of 65 consecutive cases. *World J Surg*. 2007;31:431–436.
250. Ko M-C, Liu C-C, Liu C-K, et al. Incidence of renal and perinephric abscess in diabetic patients: a population-based national study. *Epidemiol Infect*. 2011;139:229–235.
251. Lim SK, Ng FC. Acute pyelonephritis and renal abscesses in adults—correlating clinical parameters with radiological (computer tomography) severity. *Ann Acad Med Singapore*. 2011;40:407–413.
252. Rubilotta E, Balzarro M, Lacola V, et al. Current clinical management of renal and perinephric abscesses. *Urologia*. 2014;81:144–147.
253. Hung C-H, Liou J-D, Yan M-Y, et al. Immediate percutaneous drainage compared with surgical drainage of renal abscess. *Int Urol Nephrol*. 2007;39:51–55.
254. Lee SH, Jung HJ, Mah SY, et al. Renal abscesses measuring 5 cm or less. Outcome of medical treatment without therapeutic drainage. *Yonsei Med J*. 2010;4:569–573.
255. Saller M, Rafat C, Zahar J-R, et al. Cyst infections in patients with autosomal dominant polycystic kidney disease. *Clin J Am Soc Nephrol*. 2009;4:1183–1189.
256. Joutet F, Lhomel R, Devuyst O, et al. Diagnosis of cyst infection in patients with autosomal dominant polycystic kidney disease: attributes and limitations of the current modalities. *Nephrol Dial Transplant*. 2012;27:3746–3756.
257. Chicoskie C, Chaoui A, Kuligowska E, et al. MRI isolation of infected renal cyst in autosomal dominant polycystic kidney disease. *Clin Imaging*. 2001;25:114–117.
258. Puliaiti C, Stephens MR, Kenche J, et al. Cyst infection in renal allograft recipients with adult polycystic kidney disease. The diagnostic value of labeled leukocyte scanning. Case reports. *Transplant Proc*. 2007;39:1841–1842.
259. Bleeker-Rovers CP, de Sévaux RGL, van Hamersvelt HW, et al. Diagnosis of renal and hepatic cyst infections by 18-F-fluorodeoxyglucose positron emission tomography in autosomal dominant polycystic kidney disease. *Am J Kidney Dis*. 2003;41(E22):1–4.
260. Soussan M, Sberro R, Wartski M, et al. Diagnosis and localization of renal cyst infection by 18F-fluorodeoxyglucose PET/CT in polycystic kidney disease. *Ann Nucl Med*. 2008;22:529–531.
261. Piccoli G, Arena V, Consiglio V, et al. Position emission tomography in the diagnostic pathway for intracystic infection in ADPKD and “cystic” kidneys: a case series. *Nephrol*. 2011;12:48–59.
262. Akinel D, Turkbey B, Yilmaz R, et al. Percutaneous treatment of pyocystitis in patients with autosomal dominant polycystic kidney disease. *Cardiovasc Intervent Radiol*. 2008;31:926–930.
263. Hiyama L, Tang A, Miller LG. Levofloxacin penetration into a renal cyst in a patient with autosomal dominant polycystic kidney disease. *Am J Kidney Dis*. 2006;47:E9–E12.
264. Hepburn MJ, Pennick GJ, Sutton DA, et al. *Candida krusei* renal cyst infection and measurement of amphotericin B levels in cystic fluid in a patient receiving AmBisome. *Med Mycol*. 2003;41:163–165.
265. Thomas AA, Lane BR, Thomas AZ, et al. Emphysematous cystitis: a review of 135 cases. *BJU Int*. 2007;100:17–20.
266. Mokabberi R, Ravakhab K. Emphysematous urinary tract infections: diagnosis, treatment and survival. *Am J Med Sci*. 2007;333:111–116.
267. Pontin AR, Barnes RD. Current management of emphysematous pyelonephritis. *Nat Rev Urol*. 2009;6:272–279.
268. Soo Park B, Lee SJ, Kim YW, et al. Outcome of nephrectomy and kidney-preserving procedure for the treatment of emphysematous pyelonephritis. *Scand J Urol Nephrol*. 2006;40:332–338.
269. Bjurlin MA, Hurley SD, Kim DY, et al. Clinical outcomes of nonoperative management in emphysematous urinary tract infections. *Urol*. 2012;79:1281–1285.
270. Lu Y-C, Chiang B-J, Pong Y-H, et al. Emphysematous pyelonephritis: clinical characteristics and prognostic factors. *Int J Urol*. 2013;21:277–282.
271. Ubee SS, McGlynn L, Fordham M. Emphysematous pyelonephritis. *BJU Int*. 2010;107:1474–1478.
272. Somani BK, Nabi G, Thorpe P, et al. Is percutaneous drainage the new gold standard in the management of emphysematous pyelonephritis? Evidence from a systematic review. *J Urol*. 2008;179:1844–1849.
273. Olvera-Posada D, Armengod-Fischer G, Vazquez-Lavista LG, et al. Emphysematous pyelonephritis: multicenter clinical and therapeutic experience in Mexico. *Urology*. 2014;83:1280–1284.
274. Lu Y-C, Chiang B-J, Pong Y-H, et al. Predictors of failure of conservative treatment among patients with emphysematous pyelonephritis. *BMC Infect Dis*. 2014;14:418.
275. Zorzos I, Moutzouris V, Korakianitis G, et al. Analysis of 39 cases of xanthogranulomatous pyelonephritis with emphasis on CT findings. *Scand J Urol Nephrol*. 2003;37:342–347.
276. Li L, Parwani AV. Xanthogranulomatous pyelonephritis. *Arch Pathol Lab Med*. 2011;135:671–674.
277. Korkes F, Favoretto RL, Broglio M, et al. Xanthogranulomatous pyelonephritis: clinical experience with 41 cases. *Urology*. 2008;71:178–180.
278. Addison B, Zargar H, Lilic N, et al. Analysis of 35 cases of xanthogranulomatous pyelonephritis. *ANZ J Surg*. 2015;85:150–153.
279. Loffroy R, Guiu B, Watfa J, et al. Xanthogranulomatous pyelonephritis in adults: clinical and radiological findings in diffuse and focal forms. *Clin Radiol*. 2007;62:884–890.
280. Remer EE, Peacock WF. Pyocystitis: two case reports of patients in renal failure. *J Emerg Med*. 2000;19:131–133.
281. Bibb JL, Servilla KS, Gibel LJ, et al. Pyocystitis in patients on chronic dialysis. A potentially misdiagnosed syndrome. *Int Urol Nephrol*. 2002;34:415–418.
282. Falagas ME, Vergidis PI. Irrigation with antibiotic containing solutions for the prevention and treatment of infection. *Clin Microbiol Infect*. 2005;11:862–867.
283. Wise GJ, Shteynshlyuger A. An update on lower urinary tract tuberculosis. *Curr Urol Rep*. 2008;9:305–313.
284. Jacob JT, Nguyen MLT, Ray SM. Male genital tuberculosis. *Lancet Infect Dis*. 2008;8:335–342.
285. Daher EF, da Silva GB, Barros EJG. Review: renal tuberculosis in the modern era. *Am J Trop Med Hyg*. 2013;88:54–64.
286. Figuerido AA, Lucon AM. Urogenital tuberculosis: update and review of 8961 cases from the world literature. *Rev Urol*. 2008;10:207–217.
287. Altıparmak MR, Trabulus S, Balkan II, et al. Urinary tuberculosis: a cohort of 79 adult cases. *Ren Fail*. 2015;37:1157–1163.
288. Singh JP, Priyadarshi V, Kundu AK, et al. Genito-urinary tuberculosis revisited—13 years’ experience of a single centre. *Indian J Tuberc*. 2013;60:15–21.
289. Figueiredo AA, Lucon AM, Júnior RF, et al. Epidemiology of urogenital tuberculosis worldwide. *Int J Urol*. 2008;15:827–832.
290. Altıntepe L, Tonbul Z, Ozbey I, et al. Urinary tuberculosis: ten years’ experience. *Ren Fail*. 2005;27:657–661.
291. Perez S, Andrade M, Bergel P, et al. A simple algorithm for the diagnosis of AIDS-associated genitourinary tuberculosis. *Clin Infect Dis*. 2006;42:1807–1808.
292. Hsieh HC, Lu PL, Chen YH, et al. Genitourinary tuberculosis in a medical center in southern Taiwan: an eleven year experience. *J Microbiol Immunol Infect*. 2006;39:408–413.
293. Burrill J, Williams CJ, Bain G, et al. Tuberculosis: a radiologic review. *Radiographics*. 2007;27:1255–1273.
294. Hillemann D, Richter E, Rusch-Gerdes S. Use of the BACTEC Mycobacteria Growth Indicator Tube 960 automated system for recovery of *Mycobacteria* from 9,558 extrapulmonary specimens, including urine samples. *J Clin Microbiol*. 2006;44:4014–4017.
295. Chan DSG, Choy MY, Wang S, et al. An evaluation of the recovery of *Mycobacteria* from urine specimens using the automated Mycobacteria Growth Indicator Tube System (BACTEC MGIT 960). *J Med Microbiol*. 2008;57:1220–1222.
296. Bansal P, Bansal N. The surgical management of urogenital tuberculosis our experience and long-term follow-up. *Urol Ann*. 2015;7(1):49–52.
297. Gonzalez DY, Musher DM, Brar I, et al. Spectrum of bacille Calmette-Guérin (BCG) infection after intravesical BCG immunotherapy. *Clin Infect Dis*. 2003;36:140–148.

298. Stock V, Dotevall L, Sandberg T, et al. Late bacilli Calmette-Guérin infection with a large focal urinary bladder ulceration as a complication of bladder cancer treatment. *BJU Int.* 2010;107:1592–1597.
299. Perez-Jacoiste Asin MA, Fernandez-Ruiz M, Lopez-Medrano F, et al. Bacillus Calmette-Guerin (BCG) infection following intravesical BCG administration as adjunctive therapy for bladder cancer. *Medicine (Baltimore).* 2014;93:236–254.
300. Rischmann P, Desgrandchamps F, Malavaud B, et al. BCG intravesical installations: recommendations for side-effects management. *Eur Urol.* 2000;37(suppl 1):33–36.
301. Nadasy KA, Patel RS, Emmett M, et al. Four cases of disseminated *Mycobacterium bovis* infection following intravesical BCG installation for treatment of bladder carcinoma. *South Med J.* 2008;101:91–95.
302. Garcia JE, Thiel DD, Broderick GA. BCG pyelonephritis following intravesical therapy for transitional cell carcinoma. *Can J Urol.* 2007;14:3523–3525.
303. Fisher JF, Kavanagh K, Sobel JD, et al. *Candida* urinary tract infection: pathogenesis. *Clin Infect Dis.* 2011;52(suppl 6):S437–S451.
304. Kauffman CA, Fisher JF, Sobel JD, et al. *Candida* urinary tract infections: diagnosis. *Clin Infect Dis.* 2011;52(suppl 6):S452–S456.
305. Kauffman CA, Vazquez JA, Sobel JD, et al. Prospective multicenter surveillance study of funguria in hospitalized patients. *Clin Infect Dis.* 2000;30:14–18.
306. Sobel JD, Fisher JF, Kauffman CA, et al. *Candida* urinary tract infections: epidemiology. *Clin Infect Dis.* 2011;52(suppl 6):S433–S436.
307. Sobel JD, Kauffman CA, McKinsey D, et al. Candiduria: a randomized double-blind study of treatment with fluconazole and placebo. *Clin Infect Dis.* 2000;30:19–24.
308. Pappas PG, Kauffman CA, Andes D, et al. Clinical practice guidelines for the management of candidiasis 2009: update by the Infectious Diseases Society of America. *Clin Infect Dis.* 2009;48:503–535.
309. Kauffman CA. Diagnosis and management of fungal urinary tract infection. *Infect Dis Clin North Am.* 2014;28:61–74.
310. Sadegi BJ, Patel BK, Wilbur AC, et al. Primary renal candidiasis: importance of imaging and clinical history in diagnosis and management. *J Ultrasound Med.* 2009;28:507–514.
311. Drew RH, Arthur RR, Perfect JR. Is it time to abandon the use of amphotericin B bladder irrigation? *Clin Infect Dis.* 2005;40:1465–1470.
312. Sobel JD, Bradshaw SK, Lipka CJ, et al. Caspofungin in the treatment of symptomatic candiduria. *Clin Infect Dis.* 2007;44:e46–e49.
313. Gabardi S, Martin S, Sura M, et al. Micafungin treatment and eradication of candiduria among hospitalized patients. *Int Urol Nephrol.* 2016;48:1881–1885.
314. Paduch DA. Viral lower urinary tract infections. *Curr Urol Rep.* 2007;8:324–335.
315. Runde V, Ross S, Trinschel R, et al. Adenoviral infection from allogeneic stem cell transplantation: report on 130 patients from 9 single SCT units involved in a prospective multicenter surveillance study. *Bone Marrow Transplant.* 2001;28:51–57.
316. Christensen LS, Madsen TV, Barfod T. Persistent erythrovirus B19 urinary tract infection in an HIV-positive patient. *Clin Microbiol Infect.* 2001;7(9):507–509.
317. Green WR, Greaves WL, Frederick WR, et al. Renal infection due to adenovirus in a patient with human immunodeficiency virus infection. *Clin Infect Dis.* 1994;18(6):989–991.
318. Bruno B, Zager RA, Boeckh MJ, et al. Adenovirus nephritis in hematopoietic stem-cell transplantation. *Transplant.* 2004;77:1049–1057.
319. Parasuraman RK, Zhang PL, Samarapungavan D, et al. Severe necrotizing adenovirus tubulointerstitial nephritis in a kidney transplant recipient. *Case Rep Transplant.* 2013;ID969186.
320. Waldman M, Kopp JB. Parvovirus B19-associated complications in renal transplant recipients. *Nat Clin Pract Nephrol.* 2007;3:540–550.
321. Mazoyer E, Douglas E, Verine J, et al. A case report of adenovirus-related acute interstitial nephritis in a patient with AIDS. *Am J Kidney Dis.* 2008;51:121–126.
322. Nakazawa Y, Suzuki T, Fukuyama T, et al. Urinary excretion of ganciclovir contributes to improvement of adenovirus-associated hemorrhagic cystitis after allogeneic bone marrow transplantation. *Pediatr Transplant.* 2009;13(5):632–635.
323. Kehinde ED, Anim JT, Hira PR. Parasites of urologic importance. *Urol Int.* 2008;81:1–13.
324. Gryseels B, Polman K, Clerinx J, et al. Human schistosomiasis. *Lancet.* 2006;368:1106–1118.
325. Colley DG, Bustinduy AL, Secor WE, et al. Human schistosomiasis. *Lancet.* 2014;383:2253–2264.
326. Khalaf I, Shokeir A, Shalaby M. Urologic complications of genitourinary schistosomiasis. *World J Urol.* 2012;30:31–38.
327. Honeycutt J, Hammam O, Fu C-L, et al. Controversies and challenges in research on urogenital schistosomiasis—associated bladder cancer. *Trends Parasitol.* 2014;30:324–332.
328. Parkin DM. The global health burden of infection-associated cancers in the year 2002. *Int J Cancer.* 2006;118:3030–3044.
329. Ndhlovu PD, Mduluzi T, Kjetland EF, et al. Prevalence of urinary schistosomiasis and HIV in females living in a rural community of Zimbabwe: does age matter? *Trans R Soc Trop Med Hyg.* 2007;101:433–438.
330. Salvana EMT, King CH. Schistosomiasis in travelers and immigrants. *Curr Infect Dis Rep.* 2008;10:42–49.
331. Nicolls DJ, Weld LH, Schwartz E, et al. Characteristics of schistosomiasis in travelers reported to the Geo Sentinel Surveillance Network 1997–2008. *Am J Trop Med Hyg.* 2008;79:729–734.
332. Coltart CEM, Chew A, Storror N, et al. Schistosomiasis presenting in travelers: a 15 year observational study at the Hospital for Tropical Diseases, London. *Trans R Soc Trop Med Hyg.* 2015;109:214–220.
333. Bichler K-H, Savatovsky I. UTI Working Group of the European Association of Urology: EAU guidelines for the management of urogenital schistosomiasis. *Eur Urol.* 2006;49:998–1003.
334. Magak P, Chang-Cojulun A, Kadzo H, et al. Case-control study of posttreatment regression of urinary tract morbidity among adults in *Schistosoma haematobium*-endemic communities in Kwale County, Kenya. *Am J Trop Med Hyg.* 2015;93:3371–3376.
335. Nanda N, Michel RG, Kurdgelashvili G, et al. Trichomoniasis and its treatment. *Expert Rev Anti Infect Ther.* 2006;4:125–135.
336. Biyabani SR, Abbas F, Ghaffer S, et al. Unusual presentations of hydatid disease of the urinary tract. *J Urol.* 2000;163:896–898.
337. Yilmaz Y, Kosem M, Ceylan K, et al. Our experience in eight cases with urinary hydatid disease. A series of 372 cases held in nine different clinics. *Int J Urol.* 2006;13:1162–1165.
338. Huang M, Zheng H. Clinical and demographic characteristics of patients with urinary tract hydatid disease. *PLoS ONE.* 2012;7:e47667.
339. Rexiati M, Mutalifu A, Azhah B, et al. Diagnosis and surgical treatment of renal hydatid disease: a retrospective analysis of 30 cases. *PLoS ONE.* 2014;9:e96602.

BOARD REVIEW QUESTIONS

1. A 45-year-old woman presents for routine assessment 6 months following renal transplant. She has no general or specific clinical complaints. A routine urine culture grows *Escherichia coli* $>10^5$ resistant to fluoroquinolones. The most appropriate management of this urine culture result for this woman is
 - a. Treat with a 7-day course of ertapenem
 - b. Repeat the urine culture
 - c. No interventions are necessary
 - d. Advise the patient to take cranberry capsules
 - e. Request an ultrasound examination

Answer: c

Rationale: This is asymptomatic bacteriuria. There is no indication for treatment of asymptomatic bacteriuria in otherwise well renal transplant patients who are 6 months following surgery. Treatment of asymptomatic bacteriuria does not decrease the frequency of symptomatic urinary tract infection or improve outcomes related to graft function, hospitalization, or other morbidity. Antimicrobial treatment of asymptomatic bacteriuria will encourage further emergence of more resistant bacteria, which could compromise future management of symptomatic infection, should it occur.

2. A 24-year-old sexually active university student presents with acute left-sided costovertebral angle pain associated with nausea and vomiting and is found to have a temperature of 39.2°C. A diagnosis of acute pyelonephritis is made. The patient has no history of recent travel or antimicrobial therapy within the previous 6 months. The appropriate initial management is
 - a. Admit to hospital for treatment with ertapenem
 - b. Provide a 7-day course of oral levofloxacin 750 mg daily
 - c. Obtain urine culture and manage symptomatically with antiemetics pending results of urine culture and bacterial susceptibilities
 - d. Give a single intravenous dose of ceftriaxone or gentamicin followed by an oral course of ciprofloxacin 500 mg bid for 7 days
 - e. Treat with gentamicin 180 mg once a day for 7 days

Answer: d

Rationale: The current recommended treatment of acute pyelonephritis is immediate empiric institution of antimicrobial therapy. The recommended therapy is a fluoroquinolone, either ciprofloxacin 500 mg bid for 7 days or levofloxacin 750 mg od for 7 days, with or without an initial single dose of parenteral ceftriaxone or aminoglycoside. The presence of vomiting in this case would support the use of an initial parenteral antimicrobial dose. With the increasing occurrence of extended-spectrum β -lactamase (ESBL)-producing *Escherichia coli* in

the community, which are also usually resistant to fluoroquinolones, initiation of ertapenem pending urine culture results may be considered when the patient is severely ill and considered at high risk for infection with an ESBL *E. coli*. High-risk patients would be those with recent travel to areas with a high prevalence of ESBL, who have received recent antimicrobial therapy, or who reside in a community where the local prevalence of ESBL is known to be high. In all cases, the urine culture results should be reviewed once available, usually at 48–72 hours, to ensure the empiric therapy initiated is effective for the organism isolated. If the organism isolated is resistant to the empiric therapy, the patient should be further treated with a full course of antimicrobial therapy to which the organism is susceptible, irrespective of whether there has been clinical improvement with the initial empiric regimen.

3. A 62-year-old gentleman who was born and raised in the Philippines presents with vague and nonspecific symptoms of discomfort in his left flank. Urine culture is negative, but urinalysis is positive for pyuria and microscopic hematuria. An infused computed tomography (CT) scan reveals a stricture of the left ureter and erosion of the left renal calyx. The appropriate microbiological specimen to confirm a diagnosis of genitourinary tuberculosis in this gentleman is
 - a. Urine for culture and sensitivity
 - b. Three consecutive early morning urine specimens sent for culture for *Mycobacterium tuberculosis*
 - c. Urine collected on three consecutive mornings for smear for acid-fast bacilli (AFB)
 - d. Urine polymerase chain reaction (PCR) for *M. tuberculosis*
 - e. Urine specimen collected for *M. tuberculosis*

Answer: b

Rationale: This gentleman comes from a highly endemic area for tuberculosis, and the nonspecific presentation together with the CT findings and urinalysis abnormalities would be consistent with genitourinary tuberculosis. Microbiological documentation is required to confirm the diagnosis before initiating treatment, and a culture method is preferred so antimicrobial susceptibility testing can be performed to optimize the antimycobacterial therapy. The preferred diagnostic test is three consecutive early morning urine specimens forwarded to the laboratory with a request for AFB culture. Urine for AFB staining is relatively unhelpful as atypical mycobacteria may be present as colonizers but lead to a positive AFB result. Urine PCR techniques are becoming more available but do not provide information with respect to the antimicrobial susceptibility of the organism.